

Tag-Friendly Lattice Sampler and Applications

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Abstract. The NIST lattice-based cryptographic standards are set to be widely adopted, offering solutions to the most common cryptographic needs, namely key establishment and authentication (signature). This shifted the attention to more advanced primitives such as threshold cryptography as well as privacy-enhanced technologies, where the transition is expected to be more complex. This is particularly true in the context of post-quantum anonymous authentication where the existing mechanisms may not match the performance requirements of industrial applications. An important avenue for improvement of said performances is the lattice sampler, which is at the center of these mechanisms. Despite recent progress, prior samplers neglected one component: the tag. The latter is not only necessary for security, but it also impacts the efficiency of the subsequent constructions if not handled properly.

In this paper, we introduce a new sampler with an enhanced tag management that yet retain the main features of current samplers, and can thus be used as a plug-in replacement. It offers a sampling quality independent of the tag, allowing for producing preimages that are both smaller and faster to generate than those from the very recent sampler of Jeudy and Sanders (Asiacrypt’25). Far from being anecdotal, plugging it into several advanced authentication mechanisms results in size improvements of up to 30%, while being 35% faster.

Keywords: Lattice-Based Cryptography · Tag-Based Trapdoors · Preimage Sampling · Trapdoor Switching · Advanced Signatures

1 Introduction

A preimage sampler is a fundamental tool for lattice-based cryptography, in particular for signature schemes where they allow for generating a short preimage $\mathbf{x}$ such that $\mathbf{C}\mathbf{x} = f(\mathbf{m})$, where $\mathbf{C}$ is the signer’s public key and $f(\mathbf{m})$ is some function of the message to be signed. Of course, the basic security properties of signature schemes require that this inversion cannot be done without knowing some specific information, called the trapdoor. Otherwise, anyone would be able to produce their own signatures from the public $\mathbf{C}$ alone. As signature schemes are generally expected to produce several signatures, one needs to further refine this requirement on $\mathbf{x}$: not only must it be produced with the trapdoor, but it must also leak no information on it. Arguably, achieving the latter property in a provable way is the hardest part of designing preimage samplers and explains

their intricacies. This also makes it difficult to compare samplers as many of them may share the same structure for the matrix $\mathbf{C}$ but may differ considerably in their security analyses and the distribution they produce for $\mathbf{x}$, which, in the end, determine their performance.

The Micciancio-Peikert Sampler. One prominent sampler of the state-of-the-art is the one introduced by Micciancio and Peikert [MP12] (hereafter denoted MP) with $\mathbf{C} = [\mathbf{I}_d | \mathbf{A} | \mathbf{T}\mathbf{G} - \mathbf{B}] \bmod q$, where $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2$, $\mathbf{R}_1$ and $\mathbf{R}_2$ constitute the trapdoor, $\mathbf{G} = [\mathbf{I}_d | b\mathbf{I}_d | \dots | b^{k-1}\mathbf{I}_d]$ is a gadget matrix for some integer b and $k = \lceil \log_b q \rceil$, and $\mathbf{T}$ is an invertible matrix called a *tag*. Reflecting our comment above, it is easy to sample a preimage $\mathbf{x}$ for such a matrix $\mathbf{C}$. By construction of $\mathbf{G}$, it is indeed easy to find some short $\mathbf{z}$ such that $\mathbf{G}\mathbf{z} = \mathbf{T}^{-1}f(\mathbf{m})$. Then, one can set $\mathbf{x} = [(\mathbf{R}_1\mathbf{z})^T | (\mathbf{R}_2\mathbf{z})^T | \mathbf{z}^T]^T$ as a valid preimage of $f(\mathbf{m})$ by $\mathbf{C}$. What is more difficult is to build preimages that do not leak the trapdoor as trivially as the $\mathbf{x}$ constructed this way. The solution devised in [MP12] is significantly more complex but manages to remove any leakage on $\mathbf{R}_1$ and $\mathbf{R}_2$.

Since its introduction, the MP sampler has been used by countless cryptographic constructions, in particular in the context of privacy-enhanced authentication that we discuss later. However, when plugged in these constructions, one faces some issues to reconcile their security properties with the very features of the sampler. As an example, let us consider the case of a standard digital signature where the signature on m is precisely a short preimage $\mathbf{x}$ generated by the sampler. A natural choice of computational assumption for proving security is ISIS (Inhomogeneous Short Integer Solution) that states the hardness of finding a short $\mathbf{x}'$ such that $\mathbf{C}'\mathbf{x}' = \mathbf{u}$ for some matrix $\mathbf{C}'$ and vector $\mathbf{u}$. Unfortunately, $\mathbf{C}$ has no chance of being a valid ISIS instance since it has been precisely designed for inverting any syndrome. One must then find a way, in the security proof, to replace $\mathbf{C}$ by $\mathbf{C}'$ while keeping the ability to answer signature queries. There are essentially two ways of doing that¹.

The first one consists in setting $f(\mathbf{m}) = H(\mathbf{m})$ for some hash function modelled as a random oracle. The latter then acts as some alternative trapdoor that will compensate the one of $\mathbf{C}$. It works, but in the process we have introduced a hash function that will create further issues if we plan to use this signature for privacy-preserving applications such as group signature [CvH91,BMW03], blind signature [Cha82], anonymous credentials [CL01], etc. Indeed, hash functions are notoriously hard to combine with Zero-Knowledge (ZK) protocols, a key component of these cryptographic mechanisms. As noted in [JS25b], this ROM-based approach thus precludes the main applications of the MP sampler, and we then do not consider it in this paper.

The second solution leverages the tag $\mathbf{T}$. Concretely, the signer will now use a different tag for each signature but set $f(\mathbf{m})$ as some ZK-friendly function of $\mathbf{m}$, typically $\mathbf{u} + \mathbf{D}\mathbf{m}$ for some public $\mathbf{u}$ and $\mathbf{D}$. To prove security, one will first guess the tag $\mathbf{T}^*$ used by the adversary in its forgery and then replace $\mathbf{B}$ by $\mathbf{B} + \mathbf{T}^*\mathbf{G}$. This implicitly defines the *effective* tag as $\mathbf{T} - \mathbf{T}^*$, which has

¹ We note it is also possible to design ad-hoc computational assumptions, often interactive [AKSY22,BLNS23], but the security assurances they provide are questionable.

the following consequences. When $\mathbf{T} \neq \mathbf{T}^*$, the signer retains a full gadget and is thus able to compute a valid signature as usual (provided the difference is invertible). When $\mathbf{T} = \mathbf{T}^*$, the gadget vanishes and the forgery readily gives a solution to the ISIS problem. Actually, far from being an artifact of the proof, one can note that the tag is here crucial for security. Without it, the verification equation would become linear in $\mathbf{x}$ and $\mathbf{m}$ and could easily be diverted to create forgeries.

However, it remains one subtlety in this tag-based proof. One must still ensure that $\mathbf{B}$ can be replaced by $\mathbf{B} + \mathbf{T}^* \mathbf{G}$ without being detected by the adversary. The structure of $\mathbf{B} = \mathbf{R}_1 + \mathbf{A} \mathbf{R}_2$ suggests to rely on the LWE (Learning With Errors) assumption but the latter requires an intermediate step where $\mathbf{B}$ would be replaced by a random matrix, which means that we lose, again, our trapdoor. This forced previous applications to either rely on statistical arguments (which led to much worse performance, e.g., [JRS23]) or to extend $\mathbf{C}$ with a matrix $\mathbf{A}_3$ hiding a second trapdoor whose only purpose was to correctly simulate the proof. The latter solution directly affects the dimension of the preimage $\mathbf{x}$ and hence the performance of all applications relying on this sampler.

1.1 Improvements of the Micciancio-Peikert Sampler

The central role of the MP sampler in many applications have made it a worthwhile target for improvements since any gain readily improves these applications as illustrated in, e.g., [JS25b]. In particular, an avenue for improving the MP sampler is to reduce the dimension of the preimage $\mathbf{x}$ it generates. Indeed, the very structure of $\mathbf{C}$ implies that it is at least $2d + dk$ (random oracle context), which can rise up to $2d + 2dk$ when one needs a second trapdoor (tag-based context). Of course, one can reduce this dimension by opting for smaller k , but this mechanically makes b , and hence the norm of $\mathbf{x}$, larger, then decreasing security.

Truncated Gadgets. First improvements [CGM19, YJW23, JRS24, JS25b] were based on the observation that one can write the gadget matrix $\mathbf{G}$ as $[\mathbf{G}_L | \mathbf{G}_H]$, where $\mathbf{G}_L = [\mathbf{I}_d | \dots | b^{\ell-1} \mathbf{I}_d]$ and $\mathbf{G}_H = [b^\ell \mathbf{I}_d | \dots | b^{k-1} \mathbf{I}_d]$. Therefore, for any small $\mathbf{z}$ such that $\mathbf{T} \mathbf{G} \mathbf{z} = f(\mathbf{m})$, one can parse $\mathbf{z}$ as $[\mathbf{z}_L^T | \mathbf{z}_H^T]^T$ and write $\mathbf{T} \mathbf{G}_H \mathbf{z}_H = f(\mathbf{m}) - \mathbf{T} \mathbf{G}_L \mathbf{z}_L$, which means that $\mathbf{z}_H$ is some approximate preimage of $f(\mathbf{m})$ for $\mathbf{T} \mathbf{G}_H$ as long as $\mathbf{T} \mathbf{G}_L \mathbf{z}_L$ remains small. If one now sets $\mathbf{C} = [\mathbf{I}_d | \mathbf{A} | \mathbf{T} \mathbf{G}_H - \mathbf{B}] \bmod q$, one then gets an approximate trapdoor preimage sampler generating short $\mathbf{x}$ such that $\mathbf{C} \mathbf{x} = f(\mathbf{m}) - \mathbf{e}$, where $\mathbf{e}$ is the truncation error, here equal to $\mathbf{T} \mathbf{G}_L \mathbf{z}_L$. Actually, if we let $\mathbf{x}_1$ be the component of $\mathbf{x}$ matching $\mathbf{I}_d$, one can even derive an exact sampler by setting $\mathbf{x}_1 = \mathbf{x}_1 + \mathbf{e}$.

Conceptually, one has thus packed the $d\ell$ components of $\mathbf{x}$ matching $\mathbf{G}_L$ into the error $\mathbf{e}$ that is merged with $\mathbf{x}_1$, resulting in a preimage with fewer dimensions. Note however that dropping more entries leads to a larger $\mathbf{e}$ which then deteriorates the security. The norm of $\mathbf{e}$ thus places an implicit limitation on the amount of entries we can truncate. Here again, we stress that the contribution of these works was not this mere rearrangement but the thorough security analyses and parameters identifications required for proving the security of their sampler. Actually, the samplers with truncated gadgets of [CGM19, YJW23, JRS24] are

only proven secure in the random-oracle context mentioned above, where the inverted syndrome is a reprogrammable ROM output. It is only very recently that Jeudy and Sanders [JS25b] provided a sampler with truncated gadgets that was proven secure in the tag-based context as well, i.e., without losing major properties of the MP sampler in the process, resulting in straightforward improvements of many advanced cryptographic constructions.

Limitations. Unfortunately, the sampler [JS25b] has an undesirable feature. Both the computational complexity and the norm of the preimage are affected by the tag $\mathbf{T}$. Regarding the norm, the reason is that the error $\mathbf{e}$, which is then incorporated in the preimage $\mathbf{x}$, depends on $\mathbf{T}$. The norm of the latter thus negatively affects the one of $\mathbf{e}$ and thus $\mathbf{x}$, and one must then keep the truncation parameter ℓ sufficiently low to compensate (since the norm of $\mathbf{G}_L \mathbf{z}_L$ grows with ℓ). Regarding the computational complexity, the problem is that the security analysis in [JS25b] requires to introduce some perturbation in the sampling process that does depend on the tag $\mathbf{T}$. Concretely, each fresh tag requires floating-point computations in the sampler of [JS25b] which complexifies and slows down the implementation. As the tag changes at each signing in the standard model approach (i.e., without random oracles), these computations cannot be done offline and thus directly affect the signing time.

If we sum up, the key issue here is the dependence on the tag and, more generally, the fact that this new truncated sampler poorly manages the tag. Securely sampling a preimage $\mathbf{x}$ with a quality that does not depend on $\mathbf{T}$ would enable choosing larger values of ℓ (and hence lower-dimensional $\mathbf{x}$) while having a more efficient sampling process.

Trapdoor Switching. A second improvement to the MP sampler was introduced in [AGJ+24] and specifically targets the double-trapdoor issue mentioned above. Their approach was based on the observation that, in the (standard) case where $\mathbf{T} = t\mathbf{I}_d$, instead of replacing $\mathbf{B}$ by $\mathbf{B} + t^*\mathbf{G}$ in one step, one can do it more progressively by first migrating from $\mathbf{B}$ to $\mathbf{B} + \text{diag}(t^*, 0, \dots, 0)\mathbf{G}$, and then to $\mathbf{B} + \text{diag}(t^*, t^*, 0, \dots, 0)\mathbf{G}$, etc. In other words, instead of affecting the dk columns of the gadget at once, they only affect k columns at each step, and thus only need a partial trapdoor (and hence a matrix $\mathbf{A}_3$ with fewer columns) to compensate. Concretely, their solution allows to reduce the dimension of $\mathbf{x}$ from $2d + 2dk$ to $2d + dk + k$, which is better, at least when $d > 1$. This partial trapdoor switching technique combines with the sampler of [JS25b], meaning that $\mathbf{A}_3$ can be reduced to $k - \ell$ columns.

Limitations. A natural question now is whether this idea could be further developed to affect only one column at once. This way, one would need only one extra column and not k . Unfortunately, this does not work well with the current sampling approach of [MP12, JS25b]. Indeed, this would essentially mean sampling with an heterogeneous gadget $(1, b, \dots, b^{i-1}, tb^i, b^{i+1}, \dots, b^{k-1})$, which would considerably deteriorate the quality of the preimage. Here again, the key issue is the poor interaction between the tag and the gadget (and hence the sampler). The latter indeed does not provide the flexibility to work with the more elaborate tags we need for efficiently proving security.

At this stage, we need to take a step back and recall that the limitations presented above are not only theoretical concerns. They have very practical consequences as each unnecessary dimension we add to the preimage significantly decreases the performance of the sampler and, in turn, the ones of all their applications, as evidenced in Section 4. We believe that these issues stem from the fact that all these improved samplers have retained the original MP approach for blending the tag and the gadget, that is, they continue to rely (implicitly or explicitly) on a matrix $\mathbf{TG}$. While this allows for simplifying the security analysis, this approach is not optimal and prevents the ideas underlying those new samplers to achieve their full potential.

1.2 Our Contributions

In this paper, we address the two limitations above while retaining the main features of [JS25b], in particular those related to its worst-case security. For the sake of clarity, we present our results in two steps so as to better identify the consequences of each modification. First, we design a sampler that removes any dependence on the tag $\mathbf{T}$ in the preimage. Then, we further improve it by changing some parts of the sampling process to better support trapdoor switching. We provide a thorough analysis to demonstrate security and compactness of our sampler resulting from these two steps. The latter retains a structure similar to prior gadget-based constructions and can then be used in place of that of [JS25b], leading to immediate improvements of many cryptographic constructions, as illustrated in Section 4.

Step 1: Tag-Independent Quality. As explained above, the key issue with the sampler of [JS25b] is that it still relies on the full gadget $\mathbf{G}$ and the associated sampling procedure, even in the truncated case. Indeed, the approximate preimage $\mathbf{z}_H$ is extracted from a full preimage $\mathbf{z}$, inherently yielding a *truncation error* $\mathbf{e} = \mathbf{TG}_L\mathbf{z}_L$. Conceptually, this monolithic approach for $\mathbf{G}$ has led previous approaches, including [MP12, JS25b], to work with what we call a *tagged gadget* $\mathbf{TG} = [\mathbf{TG}_L|\mathbf{TG}_H]$. This tagged gadget requires perfect symmetry in the application of the tag $\mathbf{T}$, a feature that is highly undesirable as both components now have completely different purposes and constraints.

Our solution is to handle the tag differently and work with a different tagged gadget. More concretely, in a first step, we slightly change it and aim at using $[\mathbf{G}_L|\mathbf{TG}_H]$ so that the error would only be $\mathbf{G}_L\mathbf{z}_L$, without any tag dependence. Actually, we go a little bit further in breaking the symmetry between the two components of the tagged gadget by allowing $\mathbf{G}_L$ and $\mathbf{G}_H$ to be built using two potentially different bases b_L and b_H . Of course, one can note that the corresponding lattice $\mathcal{L}_q^\perp([\mathbf{G}_L|\mathbf{TG}_H])$ has lost the very nice features of $\mathcal{L}_q^\perp(\mathbf{G})$, so replicating the usual gadget sampling for this tagged gadget would yield terrible performance (and still depend on $\mathbf{T}$). We therefore propose to follow an alternative approach that requires to add a mild constraint on the modulus q . We indeed assume $q = q_L q_H$ is composite and where the factors are so that every element modulo q_L (resp. q_H) can be decomposed in base b_L (resp. b_H) using $\mathbf{G}_L$ (resp. $\mathbf{G}_H$).

We first slightly change the definition of $\mathbf{G}_H$ to be more practical: we let $\mathbf{G}_L = [\mathbf{I}_d | b_L \mathbf{I}_d | \dots | b_L^{k_L-1} \mathbf{I}_d]$, $\mathbf{G}_H = [\mathbf{I}_d | b_H \mathbf{I}_d | \dots | b_H^{k_H-1} \mathbf{I}_d]$, and thus $\mathbf{G}_x$ has dk_x columns. We then redefine the matrix $\mathbf{C}$ as $[\mathbf{I} | \mathbf{A} | q_L \mathbf{T} \mathbf{G}_H - \mathbf{B}]$, and work with the tagged gadget $[\mathbf{G}_L | q_L \mathbf{T} \mathbf{G}_H]$. To sample $\mathbf{z}_L$ and $\mathbf{z}_H$ such that $\mathbf{G}_L \mathbf{z}_L + q_L \mathbf{T} \mathbf{G}_H \mathbf{z}_H = \mathbf{w} \bmod q$, we start by computing $\ell = \mathbf{w} \bmod q_L$ (recalling q_L divides q). We then sample a Gaussian $\mathbf{z}_L$ such that $\mathbf{G}_L \mathbf{z}_L = \ell \bmod q_L$, which can be done with usual samplers for gadget lattices. Then, when lifting modulo q , it holds that $\mathbf{w} - \mathbf{G}_L \mathbf{z}_L = q_L \mathbf{h} \bmod q$. After that, we can compute $\mathbf{T}^{-1} \mathbf{h} = \mathbf{T}^{-1}(\mathbf{w} - \mathbf{G}_L \mathbf{z}_L)/q_L \bmod q_H$ and sample a Gaussian $\mathbf{z}_H$ such that $\mathbf{G}_H \mathbf{z}_H = \mathbf{T}^{-1} \mathbf{h} \bmod q_H$, also with usual gadget lattice samplers. The last equation can be lifted modulo q as $q_L \mathbf{T} \mathbf{G}_H \mathbf{z}_H = \mathbf{w} - \mathbf{G}_L \mathbf{z}_L \bmod q$ giving the desired relation, with a truncation error $\mathbf{G}_L \mathbf{z}_L$ that no longer depends on the tag. It then remains to identify the right perturbation to get a secure sampler, which we defer to Section 3 as it is quite technical.

Step 2: Tag Management for Compact Trapdoor Switching. We can now focus our second problem, that of trapdoor switching. To answer it, we further change the $\mathbf{G}_H$ part of our tagged gadget. We indeed give more freedom in tag management so as to handle the trapdoor switching issue with a one-column matrix $\mathbf{A}_3$ (instead of k in the case of full gadgets and $k_H = k - \ell$ for truncated ones).

As we explained, the key problem with prior approaches was the inability to modify the gadget columns one by one since the sampling procedure required to homogeneously apply the tag on all the entries of the gadget component $(1, b, \dots, b^{k-1})$. Even in the truncated case after our first step, the tag applies homogeneously to $q_L(1, b_H, \dots, b_H^{k_H-1})$. This highlights the trapdoor switching issue as a tag management problem. Our new approach can be used to solve the latter by further refining our tagged gadget and adapting the sampling process accordingly. Concretely, we first re-write $\mathbf{G}_H$ differently as

$$\mathbf{G}_H = \left[\mathbf{e}_0 | \dots | \mathbf{e}_{d-1} \mid b_H \mathbf{e}_0 | \dots | b_H \mathbf{e}_{d-1} \mid \dots \mid b_H^{k_H-1} \mathbf{e}_0 | \dots | b_H^{k_H-1} \mathbf{e}_{d-1} \right]$$

where $\mathbf{e}_i$ is the i -th canonical vector, and even more compactly as $\mathbf{G}_H = [b_H^{[i/d]} \mathbf{e}_{(i \bmod d)}]_{i \in [0, dk_H-1]}$. We then define $\mathbf{G}_{H, \mathbf{t}} = [t_i b_H^{[i/d]} \mathbf{e}_{(i \bmod d)}]_{i \in [0, dk_H-1]}$ for any $\mathbf{t} = (t_0, \dots, t_{dk_H-1})$. Our new tagged gadget is now $[\mathbf{G}_L | q_L \mathbf{G}_{H, \mathbf{t}}]$, which provides the ability to apply different tag values for each column, a feature crucial to security proofs. Indeed, one can now move from $\mathbf{B} = \mathbf{R}_1 + \mathbf{A} \mathbf{R}_2$ to $\mathbf{B}' = \mathbf{R}_1 + \mathbf{A} \mathbf{R}_2 + q_L t^* \mathbf{G}_H$ in dk_H iterations, each involving a different $q_L \mathbf{G}_{H, \mathbf{t}_i}$, where $\mathbf{t}_i = (t^*, \dots, t^*, 0, \dots, 0)$ and t^* appears i times. As $q_L \mathbf{G}_{H, \mathbf{t}_i}$ differs from $q_L \mathbf{G}_{H, \mathbf{t}_{i+1}}$ in only one position, a single column $\mathbf{A}_3$ is sufficient to compensate.

So, this new tagged gadget addresses all the problems we have with trapdoor switching but it still remains to make the sampler work. Recall from our first step that after sampling $\mathbf{z}_L$, we were left with sampling $\mathbf{z}_H$ such that $\mathbf{T} \mathbf{G}_H \mathbf{z}_H = \mathbf{h} \bmod q_H$. With this modified tagged gadget, it has now become a problem of sampling a short $\mathbf{z}_H$ such that $\mathbf{G}_{H, \mathbf{t}} \mathbf{z}_H = \mathbf{h} = (\mathbf{w} - \mathbf{G}_L \mathbf{z}_L)/q_L \bmod q_H$. For that, we proceed in a similar way as when we separated $\mathbf{z}_L$ and $\mathbf{z}_H$, except that

now we separate all the entries of $\mathbf{z}_H$ as they now all are associated to a different tag t_i . First, the shape of $\mathbf{G}_{H,\mathbf{t}}$ allows us to deal with each entry of $\mathbf{h}$ separately. This means, we only need to describe how to sample $\mathbf{z}$ such that $\mathbf{g}_{H,\mathbf{t}}^T \mathbf{z} = h_i$ with $\mathbf{t} = (t_i, t_{i+d}, \dots, t_{i+d(k_H-1)})$, and where $\mathbf{g}_{H,\mathbf{t}}^T = [t_{i+jd} \cdot b_H^j]_j$. As in the first step, we sample the first component modulo b_H , lift the equation, and correct the syndrome for the next component. And we do that iteratively. For this approach to work, we need two requirements: (1) q_H must equal $b_H^{k_H}$ and (2) the tags, and the difference of any pair of tags, must be invertible modulo b_H . The former constraint is very mild in practice. The latter one, which is quite classical, can be managed by choosing b_H to be a very small prime with adequate splitting so that all ternary polynomials are invertible modulo b_H .

At the end of our iterative process, we get a vector $\mathbf{z}_H$ such that $\mathbf{G}_{H,\mathbf{t}} \mathbf{z}_H = \mathbf{h} \bmod b_H^{k_H}$, with the same gadget quality as before, which is in particular independent of the tag $\mathbf{t}$. The other steps remain unchanged so our preimage directly benefits from the reduction of the number of columns of $\mathbf{A}_3$ from k_H to 1.

Security Analysis. Another important part of our contributions is then to analyze the security and compactness of our resulting sampler. For security, we show its worst-case security, i.e., it can be used to securely invert syndromes that are arbitrary and possibly adversarially controlled, thus retaining the benefits of the sampler of [JS25b]. We also tailor the sampler’s perturbation and in particular identify the right distribution for the latter so that the convolution argument still works, and in a compact way. As we show, we manage to retain the same quality as [JS25b] but without the dependence on $\mathbf{T}$ and with a more compact trapdoor switching, leading to preimages that are both smaller and simpler to generate.

Performance and Applications. As lattice samplers are a key component of many cryptographic constructions, any improvement in their design has repercussions on the latter. To illustrate the concrete benefits of our new sampler, we focus on the area of privacy-preserving authentication that has been so far the main application of the MP sampler. This family of mechanisms encompasses, for example, blind signatures, anonymous credentials, group signatures that have been (or are currently being) standardized at ISO/IEC [ISO16,ISO25,ISO13] and used in many different contexts (e.g., [TCG15,Int16,App23]). It also allows us to provide a fair comparison with the very recent sampler in [JS25b]. Actually, we follow the [JS25b] approach, and first describe a standard model signature based on our sampler before presenting more elaborate cryptographic constructions. These practical examples translate the quite technical contributions we describe so far into very concrete performance improvements as shown in Table 1.1.

For example, we get a standard model signature that is 27% smaller than the state-of-the-art [JS25b], with signature sizes of 3607 bytes. Similarly, our sampler improves some key metrics of anonymous credentials and blind signatures by a similar factor. We stress that these improvements come at no cost regarding security as we still rely on the same well-known lattice assumptions.

By making the sampler quality tag-independent, we also removed the need for tag-dependent online computations. As we describe in the Section 3.5, all the

	Standard Model Signature			Blind Signature	Anonymous Credentials	
	Sig. Size	Sig. Generation	Sig. Verification	Blind Sig. Size	Cred. Size	Cred. Proof Size
[JS25b]	4.82 KB	84.26 ms	0.77 ms	36.28 KB	4.87 KB	53.98 KB
Ours	3.52 KB	55.31 ms	0.56 ms	26.19 KB	3.52 KB	46.89 KB

Table 1.1. Comparison with the state-of-the-art [JS25b] on some relevant metrics.

important sampling materials can be precomputed during the key generation and stored in the secret key. It in particular removes online sampling material computations and in turn the timing overhead of [JS25b] compared to [AGJ+24]. As these computations necessitate floating-point arithmetic, the sampler of [JS25b] would have needed enhanced implementation protection for the signing algorithm, while ours does not. To showcase these implementation improvements, we provide a proof-of-concept implementation in C of our new standard model signature based on our sampler. It shows enhanced timings compared to the sampler of [JS25b] due to the lack of online covariance computations. Actually, it even slightly outperforms the performance of the full gadget case [AGJ+24] on the sampling/signing times, while having a much more efficient verification due to the reduced dimension. Overall, for a similar security level, our sampler runs faster and yields preimages of smaller sizes, while being simpler to implement.

A full comparison for these applications can be found in Section 4. We expect similar improvements for all other constructions (e.g., group signature) that use the sampling framework of [MP12] or [JS25b]. We believe these improvements are particularly important in the context of privacy-preserving authentication that has been left aside by the numerous initiatives related to the post-quantum transition, leaving the industry with only few quantum-safe alternatives that are still relatively large.

2 Preliminaries

We use $\mathbb{N}, \mathbb{Z}, \mathbb{R}$ to respectively denote the set of natural integers, the ring of integers, and the field of reals. For two integers $a \leq b$, we define $[a, b] = \{a, \dots, b\}$ and $[b] = [1, b]$. For a positive integer q , we define $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$.

2.1 Linear Algebra

Vectors and matrices are written in bold lowercase and uppercase letters respectively. The canonical basis of $\mathbb{R}^d$ is denoted by $\mathbf{e}_1, \dots, \mathbf{e}_d$, the zero vector by $\mathbf{0}$ and the all one vector by $\mathbf{1}$, where the dimension is clear from the context. We define the regular ℓ^p norms of $\mathbb{R}^d$ by $\|\mathbf{a}\|_p = (\sum_{i \in [d]} |a_i|^p)^{1/p}$. For a set $S \subseteq \mathbb{R}^d$, we write $\text{Span}_{\mathbb{R}}(S)$ to be the subspace of $\mathbb{R}^d$ generated by S . For a matrix $\mathbf{A} \in \mathbb{R}^{d \times m}$, we define $\text{Span}_{\mathbb{R}}(\mathbf{A}) = \text{Span}_{\mathbb{R}}(\{\mathbf{A}\mathbf{e}_i; i \in [m]\}) \subseteq \mathbb{R}^d$. We also define $\ker(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^m : \mathbf{A}\mathbf{x} = \mathbf{0}\}$. The set of real unitary matrices of $\mathbb{R}^{d \times d}$ is denoted by $O_d(\mathbb{R})$. The transpose of a matrix $\mathbf{A}$ is denoted by $\mathbf{A}^T$.

We denote by $\mathcal{S}_d^+(\mathbb{R})$ (resp. $\mathcal{S}_d^{++}(\mathbb{R})$) the set of all $d \times d$ real symmetric positive semi-definite (resp. positive definite) matrices. These sets are partially ordered under the *Loewner order*, defined as follows: for $\mathbf{S}, \mathbf{R} \in \mathcal{S}_d^+(\mathbb{R})$, we write $\mathbf{S} \geq \mathbf{R}$ (resp. $\mathbf{S} > \mathbf{R}$) if and only if $\mathbf{S} - \mathbf{R} \in \mathcal{S}_d^+(\mathbb{R})$ (resp. $\mathcal{S}_d^{++}(\mathbb{R})$). An important property of the Loewner order on $\mathcal{S}_d^{++}(\mathbb{R})$ is its compatibility with matrix inversion: if $\mathbf{S} \geq \mathbf{R}$ for $\mathbf{S}, \mathbf{R} \in \mathcal{S}_d^{++}(\mathbb{R})$, then $\mathbf{R}^{-1} \geq \mathbf{S}^{-1}$, and the same implication holds for the strict inequality. However, this inversion compatibility does not generally extend to $\mathcal{S}_d^+(\mathbb{R})$ when using pseudoinverses. Lastly, for any matrix $\mathbf{S} \in \mathcal{S}_d^+(\mathbb{R})$, we denote by $\sqrt{\mathbf{S}}$ any full-rank matrix (not necessarily square) satisfying $\mathbf{S} = \sqrt{\mathbf{S}}\sqrt{\mathbf{S}}^T$.

For any real matrix $\mathbf{A} \in \mathbb{R}^{d \times m}$, there exists a unique matrix $\mathbf{A}^+ \in \mathbb{R}^{m \times d}$ satisfying the four Moore–Penrose conditions: (1) $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$, (2) $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$, (3) $(\mathbf{A}\mathbf{A}^+)^T = \mathbf{A}\mathbf{A}^+$, (4) $(\mathbf{A}^+\mathbf{A})^T = \mathbf{A}^+\mathbf{A}$. This matrix $\mathbf{A}^+$ is called the *Moore–Penrose pseudoinverse* of $\mathbf{A}$, or simply the *pseudoinverse*. It generalizes the concept of the inverse to arbitrary (possibly non-square or rank-deficient) matrices. The pseudoinverse is an involution that commutes with the transpose operator. Furthermore, $\mathbf{A}\mathbf{A}^+$ is the orthogonal projector onto the column space $\text{Span}_{\mathbb{R}}(\mathbf{A})$, while $\mathbf{A}^+\mathbf{A}$ is the orthogonal projector onto the row space $\text{Span}_{\mathbb{R}}(\mathbf{A}^T)$. As a result, the image of $\mathbf{A}^+$ is the same as the row space of $\mathbf{A}$, and its kernel coincides with that of $\mathbf{A}^T$, i.e., $\text{Span}_{\mathbb{R}}(\mathbf{A}^+) = \text{Span}_{\mathbb{R}}(\mathbf{A}^T)$, and $\ker(\mathbf{A}^+) = \ker(\mathbf{A}^T)$.

In the special case where $\mathbf{A}$ is square and invertible, we have $\mathbf{A}^+ = \mathbf{A}^{-1}$. More generally, if $\mathbf{A}$ has full column rank (i.e., its columns are linearly independent), we have $\mathbf{A}^+\mathbf{A} = \mathbf{I}_m$. The pseudoinverse can also be computed from the singular value decomposition (SVD) of $\mathbf{A}$. If $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$ is an SVD of $\mathbf{A}$, then its pseudoinverse is $\mathbf{A}^+ = \mathbf{V}\mathbf{D}^+\mathbf{U}^T$, where $\mathbf{D}^+$ is obtained by inverting the nonzero singular values of $\mathbf{D}$. For unitary matrices $\mathbf{U}, \mathbf{V}$, we also have the identity $(\mathbf{U}\mathbf{A}\mathbf{V}^T)^+ = \mathbf{V}\mathbf{A}^+\mathbf{U}^T$. Note that, in general, $(\mathbf{A}\mathbf{B})^+ \neq \mathbf{B}^+\mathbf{A}^+$. Equality holds only in certain cases.

We later need the following technical lemma on pseudoinverses of positive semi-definite matrices.

Lemma 2.1. *Let d, m be in $\mathbb{N}^\times$. For any $\mathbf{B} \in \mathcal{S}_m^{++}(\mathbb{R})$ and $\mathbf{A} \in \mathbb{R}^{d \times m}$ that has full column rank, it holds that $(\mathbf{A}\mathbf{B}\mathbf{A}^T)^+ = \mathbf{A}^{+T}\mathbf{B}^{-1}\mathbf{A}^+$ and $\text{Span}_{\mathbb{R}}(\mathbf{A}\mathbf{B}\mathbf{A}^T) = \text{Span}_{\mathbb{R}}((\mathbf{A}\mathbf{B}\mathbf{A}^T)^+) = \text{Span}_{\mathbb{R}}(\mathbf{A})$.*

Proof. We denote $\mathbf{A}\mathbf{B}\mathbf{A}^T$ by $\mathbf{C}$. First, we naturally have $\text{Span}_{\mathbb{R}}(\mathbf{C}) \subseteq \text{Span}_{\mathbb{R}}(\mathbf{A})$. Then, let $\mathbf{y} = \mathbf{A}\mathbf{x} \in \text{Span}_{\mathbb{R}}(\mathbf{A})$ with $\mathbf{x} \in \mathbb{R}^m$. As $\mathbf{B} > 0$, it is invertible. We then deduce that $\mathbf{y} = \mathbf{A}\mathbf{B}(\mathbf{B}^{-1}\mathbf{x}) = \mathbf{A}\mathbf{B}\mathbf{A}^T(\mathbf{A}^{+T}\mathbf{B}^{-1}\mathbf{x}) \in \text{Span}_{\mathbb{R}}(\mathbf{C})$ where we used the fact that $\mathbf{A}^T\mathbf{A}^{+T} = (\mathbf{A}^+\mathbf{A})^T = \mathbf{I}_m$. So $\text{Span}_{\mathbb{R}}(\mathbf{C}) = \text{Span}_{\mathbb{R}}(\mathbf{A})$. As $\mathbf{C}$ is symmetric, we then get $\text{Span}_{\mathbb{R}}(\mathbf{C}^+) = \text{Span}_{\mathbb{R}}(\mathbf{C}^T) = \text{Span}_{\mathbb{R}}(\mathbf{C}) = \text{Span}_{\mathbb{R}}(\mathbf{A})$.

Now, let us define $\mathbf{D} = \mathbf{A}^{+T}\mathbf{B}^{-1}\mathbf{A}^+$. We check $\mathbf{D}$ verifies the four Moore–Penrose conditions. We use extensively the fact that $\mathbf{A}^+\mathbf{A} = \mathbf{I}_m$, and also define $\mathbf{P} = \mathbf{A}\mathbf{A}^+$ which is the orthogonal projector onto $\text{Span}_{\mathbb{R}}(\mathbf{A})$. In particular, we have $\mathbf{P}^T = \mathbf{P}$.

$$(1) \quad \mathbf{C}\mathbf{D}\mathbf{C} = \mathbf{A}\mathbf{B}\mathbf{A}^T\mathbf{A}^{+T}\mathbf{B}^{-1}\mathbf{A}^+\mathbf{A}\mathbf{B}\mathbf{A}^T = \mathbf{A}\mathbf{B}\mathbf{A}^T = \mathbf{C}$$

- (2) $\mathbf{DCD} = \mathbf{A}^{+T} \mathbf{B}^{-1} \mathbf{A}^+ \mathbf{A} \mathbf{B} \mathbf{A}^T \mathbf{A}^{+T} \mathbf{B}^{-1} \mathbf{A}^+ = \mathbf{A}^{+T} \mathbf{B}^{-1} \mathbf{A}^+ = \mathbf{D}$
- (3) $\mathbf{CD} = \mathbf{A} \mathbf{B} \mathbf{A}^T \mathbf{A}^{+T} \mathbf{B}^{-1} \mathbf{A}^+ = \mathbf{A} \mathbf{A}^+ = \mathbf{P} \in \mathcal{S}_d(\mathbb{R})$
- (4) $\mathbf{DC} = \mathbf{A}^{+T} \mathbf{B}^{-1} \mathbf{A}^+ \mathbf{A} \mathbf{B} \mathbf{A}^T = \mathbf{A}^{+T} \mathbf{A}^T = \mathbf{P}^T = \mathbf{P} \in \mathcal{S}_d(\mathbb{R})$

It proves that $\mathbf{D} = \mathbf{C}^+$ as desired. $\square$

2.2 Lattices

A real d -dimensional lattice $\mathcal{L}$ is a discrete subgroup of $(\mathbb{R}^d, +)$, and its rank is defined as $r = \dim \text{Span}_{\mathbb{R}}(\mathcal{L})$. It admits an infinite number of basis $\mathbf{B} \in \mathbb{R}^{d \times r}$, and we write the lattice generated by $\mathbf{B}$ as $\mathcal{L}(\mathbf{B})$ when $\mathbf{B}$ has linearly independent columns. The dual lattice of $\mathcal{L}$ is defined by $\mathcal{L}^* = \{\mathbf{x} \in \text{Span}_{\mathbb{R}}(\mathcal{L}) : \forall \mathbf{y} \in \mathcal{L}, \mathbf{x}^T \mathbf{y} \in \mathbb{Z}\}$. If $\mathcal{L} = \mathcal{L}(\mathbf{B})$, it then holds that $\mathcal{L}^* = \mathcal{L}(\mathbf{B}^{+T})$. In other words, the dual basis is $\mathbf{B}^{+T} = \mathbf{B}(\mathbf{B}^T \mathbf{B})^{-1}$.

2.3 Algebraic Number Theory

Although our sampler works for arbitrary number fields, we decide for the sake of clarity to present only power-of-two cyclotomic fields, through their polynomial quotient field interpretation, as they are the prominent case in lattice cryptography. We refer to the extensive literature for a more complete background. For n a power of two, we define the power-of-two cyclotomic field of degree n as $\mathcal{K} = \mathbb{Q}[x]/\langle x^n + 1 \rangle$. We also define $\mathcal{K}_{\mathbb{R}} = \mathbb{R}[x]/\langle x^n + 1 \rangle$ and its ring of integers $\mathcal{R} = \mathbb{Z}[x]/\langle x^n + 1 \rangle$. For any integer $q \geq 2$, we define the quotient ring $\mathcal{R}_q = \mathcal{R}/q\mathcal{R} = \mathbb{Z}_q[x]/\langle x^n + 1 \rangle$.

We present the following notions for $\mathcal{R}$ but the same concepts also apply analogously to $\mathcal{K}, \mathcal{K}_{\mathbb{R}}$ by changing the underlying polynomial ring. Any element in $\mathcal{R}$ can be naturally represented as a vector in $\mathbb{Z}^n$ via its coefficients in the basis $\{1, x, \dots, x^{n-1}\}$. We denote this coefficient embedding by τ : for $r = \sum_{i=0}^{n-1} r_i x^i$, we define $\tau(r) = [r_0 | \dots | r_{n-1}]^T$. We also define the conjugate of an element $r \in \mathcal{R}$ as $r^* = r(x^{-1})$. In power-of-two cyclotomics, this operation corresponds to reversing and negating all coefficients except the first, so that $\tau(r^*) = [r_0 | -r_{n-1} | \dots | -r_1]^T$. We equip $\mathcal{R}$ with the standard ℓ^p norms under the embedding τ , namely, $\|r\|_p := \|\tau(r)\|_p$. We also define the subset $T_1 = \tau^{-1}(\{0, 1\}^n)$. Finally, we define the multiplication matrix map M_τ defined by the relation $\tau(rs) = M_\tau(r)\tau(s)$ for all pairs of elements r, s . In the power-of-two cyclotomic case, $M_\tau(r)$ corresponds to a nega-circulant matrix whose first column is $\tau(r)$. Note that M_τ is a ring isomorphism.

These definitions naturally extend to vectors and matrices over $\mathcal{R}$ by applying the operations component-wise, except that the conjugate of a matrix is defined as its conjugate transpose. In particular, $M_\tau(\mathbf{A})$ denotes the block-wise application of M_τ to each entry of $\mathbf{A}$, and it holds that $M_\tau(\mathbf{A}^*) = M_\tau(\mathbf{A})^T$. Finally, for matrices over $\mathcal{R}_q$, we define the q -ary (structured) lattice notation by defining $\mathcal{L}_q^{\mathbf{u}}(\mathbf{A}) = \{\mathbf{x} \in \mathcal{R}^m : \mathbf{A}\mathbf{x} = \mathbf{u} \bmod q\mathcal{R}\}$, for $\mathbf{A} \in \mathcal{R}_q^{d \times m}$ and $\mathbf{u} \in \mathcal{R}_q^d$, and $\mathcal{L}_q^\perp(\mathbf{A}) = \mathcal{L}_q^{\mathbf{0}}(\mathbf{A})$. To be thorough, it corresponds to a lattice (coset) through

its embedding via τ , i.e., $\tau(\mathcal{L}_q^{\mathbf{u}}(\mathbf{A})) = \mathcal{L}_q^{\tau(\mathbf{u})}(M_\tau(\mathbf{A})) = \{\mathbf{x} \in \mathbb{Z}^{nm} : M_\tau(\mathbf{A})\mathbf{x} = \tau(\mathbf{u}) \bmod q\mathbb{Z}\}$.

We later need the following lemma from [LS18] on the modular invertibility of small norm elements.

Lemma 2.2 ([LS18, Cor. 1.2]). *Let $\mathcal{R}$ be the power-of-two cyclotomic ring of degree n . Let $\kappa \in [n]$ be a power of two, and let b be a prime such that $b \bmod 4\kappa = 2\kappa + 1$. Then, for all $r \in \mathcal{R}$ such that $\|r\|_\infty < b^{1/\kappa}/\sqrt{\kappa}$, it holds that $r \bmod q\mathcal{R} \in \mathcal{R}_q^\times$.*

2.4 Probabilities

For a finite set S , we call $U(S)$ the uniform distribution over S . We define the centered binomial distribution $\mathcal{B}_1$ over $\mathcal{R}$ obtained by sampling each coefficient according to ψ_1 , with $\psi_1(-1) = \psi_1(1) = 1/4$ and $\psi_1(0) = 1/2$. For $\delta \geq 1$ and two distributions $\mathcal{P}_1, \mathcal{P}_2$ of same support S , we write $\mathcal{P}_1 \approx_\delta \mathcal{P}_2$ if for all $x \in S$, $\mathcal{P}_1(x) \in [1/\delta, \delta]\mathcal{P}_2(x)$. When $\delta = 1 + o(1)$, we later say that $\mathcal{P}_1$ and $\mathcal{P}_2$ are marginally close.

Relative Error Lemma. Our trapdoor switching argument later relies on the relative error lemma by [Pre17]. The latter gives a bound on the Rényi divergence of two distributions when they are marginally close. Combined with the probability preservation property of the Rényi divergence, we get the following lemma.

Lemma 2.3 ([Pre17, Lem. 3] adapted). *Let $\mathcal{P}_1, \mathcal{P}_2$ be two distributions over the same support S . Let $\delta \geq 1$ and assume $\mathcal{P}_1 \approx_\delta \mathcal{P}_2$. It then holds that for any $a > 1$, and any event $E \subseteq S$, we have*

$$\mathcal{P}_1(E) \leq \left(1 + \frac{a(a-1)(\delta-1)^2}{2(2-\delta)^{a+1}}\right)^{1/a} \mathcal{P}_2(E)^{1-1/a}.$$

Discrete Gaussians. We now define discrete Gaussian distributions and related results. For $\mathbf{S} \in \mathcal{S}_d^+(\mathbb{R})$, and $\mathbf{c} \in \mathbb{R}^d$, we first define the Gaussian function of center $\mathbf{c}$ and covariance $\mathbf{S}$, denoted $\rho_{\sqrt{\mathbf{S}}, \mathbf{c}}$, by

$$\forall \mathbf{x} \in \mathbb{R}^d, \rho_{\sqrt{\mathbf{S}}, \mathbf{c}}(\mathbf{x}) = \exp(-\pi(\mathbf{x} - \mathbf{c})^T \mathbf{S}^+(\mathbf{x} - \mathbf{c})) \mathbf{1}_{\text{Span}_{\mathbb{R}}(\mathbf{S})}(\mathbf{x} - \mathbf{c}).$$

where $\mathbf{1}_A$ is the indicator function of set A . For a countable set $A \subset \mathbb{R}^d$ such that $(A - \mathbf{c}) \cap \text{Span}_{\mathbb{R}}(\mathbf{S}) \neq \emptyset$, we define the discrete Gaussian distribution over A by its probability mass function $\mathcal{D}_{A, \sqrt{\mathbf{S}}, \mathbf{c}} : \mathbf{x} \in \mathcal{L} \mapsto \rho_{\sqrt{\mathbf{S}}, \mathbf{c}}(\mathbf{x}) / \rho_{\sqrt{\mathbf{S}}, \mathbf{c}}(A)$. This definition encompasses discrete Gaussians over lattices but also lattice cosets. When $\mathbf{c} = \mathbf{0}$, we omit the subscript, and when $\mathbf{S} = s^2 \mathbf{I}_d$ for $s > 0$, we replace the subscript $\sqrt{\mathbf{S}}$ by s and characterize the distribution as *spherical*. We extend the definition to *module Gaussians*, which are defined via their coefficient embedding, i.e., $\mathcal{D}_{\mathcal{M}, \sqrt{\mathbf{S}}} = \tau^{-1}(\mathcal{D}_{\tau(\mathcal{M}), \sqrt{\mathbf{S}}})$ where $\tau(\mathcal{M})$ is a module lattice (coset).

Discrete Gaussians benefit from similar properties as continuous Gaussians as long as the width s is sufficiently large to smooth out their discrete peculiarities.

In particular, Micciancio and Regev [MR07] defined the *smoothing parameter* of a lattice as the threshold above which the discrete Gaussian yields smooth regularity properties. The smoothing parameter of $\mathcal{L}$ is parameterized by some smoothing loss $\varepsilon > 0$ and defined as $\eta_\varepsilon(\mathcal{L}) = \min\{s > 0 : \rho_{1/s}(\mathcal{L}^*) = 1 + \varepsilon\}$. A recent work by Espitau, Wallet and Yu [EWY23] gives an exact expression of $\eta_\varepsilon(\mathcal{L})$ with tight approximations for remarkable lattices.

Lemma 2.4 ([EWY23, Lem. 5]). *For any $d \in \mathbb{N}^\times$ and $\varepsilon > 0$, it holds that $\eta_\varepsilon(\mathbb{Z}^d) \approx \sqrt{\ln(2d/\varepsilon)/\pi}$, where the estimate gets more precise as ε goes to 0.*

For non-spherical Gaussian distributions, if $\mathbf{S} \in \mathcal{S}_d^+(\mathbb{R})$ and $\mathcal{L} \subset \text{Span}_{\mathbb{R}}(\mathbf{S})$, we say that $\sqrt{\mathbf{S}} \geq \eta_\varepsilon(\mathcal{L})$ if $1 \geq \eta_\varepsilon(\sqrt{\mathbf{S}}^+ \mathcal{L})$. We give an equivalent characterization here. Additionally, if $\mathbf{S} \in \mathcal{S}_d^{++}(\mathbb{R})$ and $\mathcal{L}$ is full rank, $\sqrt{\mathbf{S}} \geq \eta_\varepsilon(\mathcal{L})$ is equivalent to $\mathbf{S} \geq \eta_\varepsilon(\mathcal{L})^2 \mathbf{I}_d$ for the Loewner order. This does not hold true if $\mathbf{S}$ is singular.

Lemma 2.5. *Let $d \in \mathbb{N}^\times$, $\varepsilon > 0$, $\mathbf{S} \in \mathcal{S}_d^+(\mathbb{R})$ and $\mathcal{L} \subset \text{Span}_{\mathbb{R}}(\mathbf{S})$ a lattice. It holds that $\sqrt{\mathbf{S}} \geq \eta_\varepsilon(\mathcal{L})$ if and only if $\rho_{\sqrt{\mathbf{S}}^+}(\mathcal{L}^*) \leq 1 + \varepsilon$.*

Exceeding the smoothing parameters yields important smoothing properties. The first one we need is the fact that the Gaussian mass of a lattice coset is close to (and maximal at) that of the lattice itself. This result, originally from [MR07], can be extended to degenerate non-spherical Gaussian using the Poisson summation formula $\rho_{\sqrt{\mathbf{S}}}(\mathcal{L} + \mathbf{c}) = \sqrt{\det^+ \mathbf{S}} \text{Vol}(\mathcal{L})^{-1} \sum_{\omega \in \mathcal{L}^*} e^{i2\pi \omega^T \mathbf{c}} \rho_{\sqrt{\mathbf{S}}^+}(\omega)$, where $\det^+ \mathbf{S}$ is the pseudo-determinant (the product of all non-zero eigenvalues), and $\text{Vol}(\mathcal{L})$ the volume of $\mathcal{L}$.

Lemma 2.6 ([MR07] adapted). *Let $d \in \mathbb{N}^\times$, $\mathcal{L}$ a lattice, $\varepsilon > 0$, $\mathbf{S} \in \mathcal{S}_d^+(\mathbb{R})$ such that $\sqrt{\mathbf{S}} \geq \eta_\varepsilon(\mathcal{L})$. Then for any $\mathbf{c} \in \text{Span}_{\mathbb{R}}(\mathcal{L})$, it holds that $\rho_{\sqrt{\mathbf{S}}}(\mathcal{L} + \mathbf{c}) \in [(1 - \varepsilon)/(1 + \varepsilon), 1] \rho_{\sqrt{\mathbf{S}}}(\mathcal{L})$.*

We then recall the regularity lemma from [GPV08] which states that if $\mathbf{x}$ is drawn above the smoothing parameter of $\mathcal{L}_q^\perp(\mathbf{A})$, then $\mathbf{A}\mathbf{x} \bmod q$ is close to uniform.

Lemma 2.7 ([GPV08, Lem. 5.2] adapted). *Let n, d, q be in $\mathbb{N}^\times$, and $\mathbf{A} \in \mathcal{R}_q^{d \times m}$ be primitive, i.e., $\mathbf{A}\mathcal{R}_q^m = \mathcal{R}_q^d$. Let $\varepsilon \in (0, 1)$ and $\mathbf{S} \in \mathcal{S}_{nm}^+(\mathbb{R})$ such that $\sqrt{\mathbf{S}} \geq \eta_\varepsilon(\mathcal{L}_q^\perp(\mathbf{A}))$. Then, if we define $\mathcal{P} = \mathbf{A}\mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{S}}} \bmod q\mathcal{R}$, we have that for all $\mathbf{u} \in \mathcal{R}_q^d$, $\mathcal{P}(\mathbf{u}) \in [\frac{1-\varepsilon}{1+\varepsilon}, 1 + \varepsilon] q^{-nd}$.*

Concentration Bounds. We first recall the concentration bounds for the (integer) spectral norm. The spectral norm of a real matrix $\mathbf{R}$ is defined as the largest singular value of $\mathbf{R}$. It can also be defined by $\|\mathbf{R}\|_2 = \max_{\mathbf{x} \in \mathbb{R}^m \setminus \{\mathbf{0}\}} \|\mathbf{R}\mathbf{x}\|_2 / \|\mathbf{x}\|_2$. When $\mathbf{R}$ has integer coefficients, we define the integer spectral norm as $\|\mathbf{R}\|_{2, \mathbb{Z}} = \max_{\mathbf{x} \in \mathbb{Z}^m \setminus \{\mathbf{0}\}} \|\mathbf{R}\mathbf{x}\|_2 / \|\mathbf{x}\|_2$. The latter is implicitly defined in [AGJ⁺24] and formalized in [JS25a]. We extend these notations to matrices over $\mathcal{R}$ with their embedding $M_\tau(\mathbf{R})$.

We can then study concentration bounds of these quantities for random matrices $\mathbf{R}$. Bounds on the spectral norm have been well studied in random matrix theory [Ver12], and empirically verified for structured matrices such as $M_\tau(\mathbf{R})$ in many works from lattice cryptography, e.g., [MP12,GMPW20,AGJ+24]. Concentration bounds on the integer spectral norm correspond to Johnson-Lindenstrauss bounds which were proven for example in [JRS23,AGJ+24] in the unstructured case, and sometimes tightened heuristically [GHL22,LNP22]. These were also empirically verified for structured matrices and used in several constructions. Here, we recall the empirically verified bound from [AGJ+24].

Lemma 2.8 ([AGJ+24, Lem. 2.2 & 2.4]). *Let $\mathcal{R}$ be a power-of-two cyclotomic ring of degree n . Let $d, m \in \mathbb{N}^\times$. It empirically holds that $\mathbb{P}_{\mathbf{R} \sim \mathcal{B}_1^{d \times m}}[\|\mathbf{R}\|_2 \leq \frac{3}{4}(\sqrt{nd} + \sqrt{nm} + 6)] = 1/O(1)$ (non-negligible), and $\mathbb{P}_{\mathbf{R} \sim \mathcal{B}_1^{d \times m}}[\|\mathbf{R}\|_{2,\mathbb{Z}} \leq \sqrt{nd/2}] = 1/C$ with $C \approx 2$.*

We also later use the following concentration bound for centered discrete Gaussian distributions.

Lemma 2.9 ([Ban93, Lem. 1.5]). *Let $d \in \mathbb{N}^\times$ and $\mathcal{L} \subset \mathbb{R}^d$ be a lattice of rank d . For any $s > 0$ and $c > 1/\sqrt{2\pi}$, $\mathbb{P}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{L},s}}[\|\mathbf{x}\|_2 > cs\sqrt{d}] < (c\sqrt{2\pi}de^{-\pi c^2})^d$.*

Rejection Sampling. The security proof of our signature scheme of Section 4.1, as well as our anonymous credentials and blind signatures of Appendix A and B, relies on the following rejection sampling result from [LNS21].

Lemma 2.10 ([LNS21, Lem. 3.2]). *Let $d \in \mathbb{N}^\times$, and $S \subset \mathcal{R}^d$ such that $\max_{\mathbf{x} \in S} \|\mathbf{x}\|_2 \leq B$. Let $\mathcal{D}_S$ be a distribution over S , $M > 1$, $\alpha = \sqrt{\pi/\ln M}$ and $s = \alpha B$. We then define the following distributions.*

$\mathcal{P}_1$	<i>Sample $\mathbf{s} \leftarrow \mathcal{D}_S$, $\mathbf{y} \leftarrow \mathcal{D}_{\mathcal{R}^d,s}$ and set $\mathbf{z} = \mathbf{y} + \mathbf{v}$. Then, sample $u \leftarrow U([0, 1])$. If $\langle \tau(\mathbf{z}), \tau(\mathbf{s}) \rangle < 0$ or if $u > \frac{1}{M} \exp(\pi(\ \tau(\mathbf{s})\ _2^2 - 2\langle \tau(\mathbf{z}), \tau(\mathbf{s}) \rangle)/s^2)$, output $\perp$. Otherwise, output $(\mathbf{s}, \mathbf{z})$.</i>
$\mathcal{P}_2$	<i>Sample $\mathbf{s} \leftarrow \mathcal{D}_S$, and $\mathbf{z} \leftarrow \mathcal{D}_{\mathcal{R}^d,s}$. Then, sample $u \leftarrow U([0, 1])$. If $\langle \tau(\mathbf{z}), \tau(\mathbf{s}) \rangle < 0$ or if $u > \frac{1}{M}$, output $\perp$. Otherwise, output $(\mathbf{s}, \mathbf{z})$.</i>

Conditioned on not aborting, it holds that $\mathcal{P}_1$ and $\mathcal{P}_2$ are identical.

2.5 Hardness Assumptions

The security of our constructions is based on the well-known lattice assumptions *Module Short Integer Solution* (M-SIS) and *Module Learning With Errors* (M-LWE) problems [LS15]. We recall their formal definition here. When the parameters are clear from the context, we define the hardness bound of problem $P \in \{\text{M-LWE}, \text{M-SIS}\}$ as $\varepsilon_P = \sup_{\mathcal{A}} \text{ppT} \text{Adv}_P[\mathcal{A}]$.

Definition 2.1 (M-SIS). *Let $\mathcal{R}$ be the power-of-two cyclotomic ring of degree n . Let $d, m, q \in \mathbb{N}^\times$ and $\beta > 0$ with $m > d$. The Module Short Integer Solution*

problem $\text{M-SIS}_{n,d,m,q,\beta}$ asks to find $\mathbf{x} \in \mathcal{L}_q^\perp([\mathbf{I}_d|\mathbf{A}]) \setminus \{\mathbf{0}\}$ such that $\|\mathbf{x}\|_2 \leq \beta$, given $\mathbf{A} \leftarrow U(\mathcal{R}_q^{d \times m-d})$. The advantage of an adversary $\mathcal{A}$ is $\text{Adv}_{\text{M-SIS}}[\mathcal{A}] = \mathbb{P}_{\mathbf{x} \leftarrow \mathcal{A}(\mathbf{A})} [\mathbf{x} \in \mathcal{L}_q^\perp([\mathbf{I}_d|\mathbf{A}]) \wedge 0 < \|\mathbf{x}\|_2 \leq \beta]$.

Definition 2.2 (M-LWE). Let $\mathcal{R}$ be the power-of-two cyclotomic ring of degree n . Let $d, m, q \in \mathbb{N}^\times$ and $\mathcal{D}$ be a distribution over $\mathcal{R}$. The Module Learning With Errors problem $\text{M-LWE}_{n,d,m,q,\mathcal{D}}$ asks to distinguish distributions $\mathcal{P}_1$ and $\mathcal{P}_2$ which are defined by

$\mathcal{P}_1$ Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{s}_1, \mathbf{s}_2 \leftarrow \mathcal{D}^m \times \mathcal{D}^d$, and define $\mathbf{b} = \mathbf{s}_1 + \mathbf{A}\mathbf{s}_2 \bmod q\mathcal{R}$.
Output $(\mathbf{A}, \mathbf{b})$.

$\mathcal{P}_2$ Sample $\mathbf{A} \leftarrow U(\mathcal{R}_q^{m \times d})$, $\mathbf{b} \leftarrow U(\mathcal{R}_q^m)$. Output $(\mathbf{A}, \mathbf{b})$.

We define $\text{Adv}_{\text{M-LWE}}[\mathcal{A}] = |\mathbb{P}_{(\mathbf{A}, \mathbf{b}) \sim \mathcal{P}_1} [\mathcal{A}(\mathbf{A}, \mathbf{b}) = 1] - \mathbb{P}_{(\mathbf{A}, \mathbf{b}) \sim \mathcal{P}_2} [\mathcal{A}(\mathbf{A}, \mathbf{b}) = 1]|$ as the advantage of an adversary $\mathcal{A}$.

3 New Tag-Friendly Gadget Sampler

If we recall our discussion in the introduction, sampling some preimage $\mathbf{v}$ of a vector $\mathbf{u}$ for the matrix $\mathbf{A}_T = [\mathbf{I}_d|\mathbf{A}|\mathbf{T}\mathbf{G} - \mathbf{B}]$ is not difficult with the knowledge of the trapdoor $(\mathbf{R}_1, \mathbf{R}_2)$ such that $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2$. Indeed, the very structure of $\mathbf{G}$ enables sampling $\mathbf{z}$ such that $\mathbf{T}\mathbf{G}\mathbf{z} = \mathbf{u}$, and then set $\mathbf{v} = \mathbf{M}\mathbf{z}$, where $\mathbf{M} = [\mathbf{R}_1^T|\mathbf{R}_2^T|\mathbf{I}_{dk}]^T$. However, this solution is insecure, leading [MP12] to introduce some well-crafted Gaussian perturbation $\mathbf{p}$ to $\mathbf{v}$ and adapt the previous process. Concretely, one now uses the nice features of the tagged gadget $\mathbf{T}\mathbf{G}$ to find a Gaussian $\mathbf{z}$ satisfying $\mathbf{T}\mathbf{G}\mathbf{z} = \mathbf{u} - \mathbf{A}_T\mathbf{p}$. One can then note that $\mathbf{v} = \mathbf{M}\mathbf{z} + \mathbf{p}$ is still a valid preimage of $\mathbf{u}$ by $\mathbf{A}_T$. Moreover, if $\mathbf{p}$ and $\mathbf{z}$ are properly distributed, then one can prove that $\mathbf{v}$ does not leak any information on the trapdoor and hence the security of the sampler.

This approach was generalized by [JS25b] to the case of truncated gadget, where $\mathbf{A}_T = [\mathbf{I}_d|\mathbf{A}|\mathbf{T}\mathbf{G}_H - \mathbf{B}]$. Here, one still samples $\mathbf{z}$ such that $\mathbf{T}\mathbf{G}\mathbf{z} = \mathbf{u} - \mathbf{A}_T\mathbf{p}$ for some perturbation $\mathbf{p}$, but now splits $\mathbf{z}$ into $\mathbf{z}_L$ and $\mathbf{z}_H$ to reflect the components $\mathbf{G}_L$ and $\mathbf{G}_H$ of the gadget $\mathbf{G} = [\mathbf{G}_L|\mathbf{G}_H]$ (see Section 1 for the definition of $\mathbf{G}_L$, $\mathbf{G}_H$ and related parameters). One can then set $\mathbf{v} = \mathbf{M}\mathbf{z} + \mathbf{p}$, but for a different $\mathbf{M}$:

$$\mathbf{M} = \begin{bmatrix} \mathbf{T}\mathbf{G}_L & \mathbf{R}_1 \\ \mathbf{0} & \mathbf{R}_2 \\ \mathbf{0} & \mathbf{I}_{dk_H} \end{bmatrix}$$

Identifying the right perturbation in this case is tricky, in particular because the distribution of $\mathbf{v}$ is now influenced by both secret (e.g., $\mathbf{R}_1$ and $\mathbf{R}_2$) and public (e.g., $\mathbf{G}_L$) matrices. Proceeding as in [MP12] would have led to generate a perturbation hiding the influence of $\mathbf{G}_L$, which would have been both useless and very inefficient. Additionally, as $\mathbf{M}$ is not full column-rank because of $\mathbf{G}_L$, the security analysis turns out to be very complex, which was one of the main obstacle in the attempt of [CGM19]. This led the authors of [JS25b] to factor $\mathbf{M}$ as $\mathbf{K}\mathbf{L}$, where $\mathbf{K}$ is meant to accumulate all the public elements, and $\mathbf{L}$ the

secret ones while being full column rank. Examples of such a factorization are provided in Section 3.3. The point of this approach was not only being able to prove security in the worst-case (using the full rank condition on $\mathbf{L}$), but also allowing the perturbation parameters to only depend on $\mathbf{L}$, then yielding much better performance.

In our paper, we retain this factorization approach but have to revisit the security analysis to account for our new tagged gadget. Indeed, as we explain in the introduction, we depart from $\mathbf{T}\mathbf{G}$ used by all previous works to instead rely on $[\mathbf{G}_L | \mathbf{G}_{H,\mathbf{t}}]$, where $\mathbf{G}_{H,\mathbf{t}}$ is defined in Equation (1) below. As a consequence, we now work with

$$\mathbf{M} = \begin{bmatrix} \mathbf{G}_L & \mathbf{R}_1 \\ \mathbf{0} & \mathbf{R}_2 \\ \mathbf{0} & \mathbf{I}_{dk_H} \end{bmatrix},$$

which removes any dependency in $\mathbf{T}$. Of course, we have to redefine the gadget sampling procedure to accomodate its unusual structure and, in the process, identify some requirement on q to make it work. Concretely we require a composite modulus of the form $q = q_L b_H^{k_H}$. This new sampling procedure in turn requires a new security analysis and identification of parameters.

3.1 Description

Let us now give a high-level technical description of how our sampler works. Throughout the rest of the paper, we let $\mathcal{R}$ be the power-of-two cyclotomic ring of degree n (although this generalizes to arbitrary algebraic integer rings). Then, let $d, q_L, b_L, b_H, k_L, k_H$ be positive integers such that $k_L = \lceil \log_{b_L} q_L \rceil$, and finally define $q_H = b_H^{k_H}$ and $q = q_L q_H$. Then, we define $\mathbf{G}_L = [1|b_L| \dots |b_L^{k_L-1}] \otimes \mathbf{I}_d$, and for any $\mathbf{t} = (t_0, \dots, t_{dk_H-1}) \in (\mathcal{R}_{b_H}^\times)^{dk_H}$, we define

$$\mathbf{G}_{H,\mathbf{t}} = \left[t_i \cdot b_H^{\lfloor i/d \rfloor} \cdot \mathbf{e}_{(i \bmod d)} \right]_{i \in [0, dk_H-1]}. \quad (1)$$

Our sampler aims at securely producing short vectors $\mathbf{v}$ such that $\mathbf{A}_{\mathbf{t}}\mathbf{v} = \mathbf{u} \bmod q\mathcal{R}$ for any $\mathbf{u} \in \mathcal{R}_q^d$, with

$$\mathbf{A}_{\mathbf{t}} = [\mathbf{I}_d | \mathbf{A}| q_L \mathbf{G}_{H,\mathbf{t}} - (\mathbf{R}_1 + \mathbf{A}\mathbf{R}_2)] \bmod q\mathcal{R}$$

and $\mathbf{R}_1, \mathbf{R}_2$ short matrices being the secret trapdoor.

Step 1: Sampling with Tag-Independent Quality. Let us first assume $\mathbf{t} = t \cdot \mathbf{1}$, in which case $\mathbf{G}_{H,\mathbf{t}} = t\mathbf{G}_H$ where we define $\mathbf{G}_H = [1|b_H| \dots |b_H^{k_H-1}] \otimes \mathbf{I}_d$. As explained above, the approach in [JS25b] would lead to sampling a full preimage $\mathbf{z}$ for the tagged gadget $t\mathbf{G}$ that would then be truncated, resulting in a tag-dependent error. We can remove this downside by instead working with the tagged gadget $[\mathbf{G}_L | q_L t\mathbf{G}_H]$ which requires adapting the sampling of $\mathbf{z}$.

To sample $\mathbf{z}$ such that $[\mathbf{G}_L | q_L t\mathbf{G}_H]\mathbf{z} = \mathbf{w} \bmod q\mathcal{R}$, we proceed by successively sampling $\mathbf{z}_L$ and $\mathbf{z}_H$, while leveraging the fact that $q = q_L q_H$. We first compute $\mathbf{w} = \mathbf{u} - \mathbf{A}_{\mathbf{t}}\mathbf{p} \bmod q\mathcal{R}$ as usual, and then $\ell = \mathbf{w} \bmod q_L$ (as q_L divides

q). We then sample $\mathbf{z}_L$ such that $\mathbf{G}_L \mathbf{z}_L = \ell \bmod q_L \mathcal{R}$. When lifting this equation modulo q , we then have $\mathbf{w} - \mathbf{G}_L \mathbf{z}_L = q_L \mathbf{h} \bmod q\mathcal{R}$. We can then sample $\mathbf{z}_H$ such that $t\mathbf{G}_H \mathbf{z}_H = \mathbf{h} \bmod q_H \mathcal{R}$, which would yield $\mathbf{w} - \mathbf{G}_L \mathbf{z}_L = q_L t\mathbf{G}_H \mathbf{z}_H \bmod q\mathcal{R}$. One important feature is that although the tag does not symmetrically affect all the tagged gadget, we are still able to sample $\mathbf{z}_L$ and $\mathbf{z}_H$ with a quality independent of t , as opposed to [JS25b].

The benefits of this split process are twofold. First, it removes the tag dependency so as to reduce the size of the truncation error term from $t\mathbf{G}_L \mathbf{z}_L$ to $\mathbf{G}_L \mathbf{z}_L$. Reducing this error essentially allows us to truncate more entries for the same security as [JS25b], which gives better compactness as we have fewer dimensions in $\mathbf{G}_H$. Second, it enables to choose the parameters of $\mathbf{G}_L$ and $\mathbf{G}_H$ *independently*, yielding further performance improvements.

Step 2: Enhanced Tag Management for Trapdoor Switching. However, one problem still remains unscathed, that of the partial trapdoor slot needed for the subsequent trapdoor switching argument discussed in Section 1.2. Despite optimizing the dimension of the main matrix $\mathbf{A}_t = [\mathbf{I}_d | \mathbf{A} | q_L t\mathbf{G}_H - \mathbf{B}]$, we still need to extend it with a block $\mathbf{A}_3$ of k_H columns to prove security of all advanced constructions, where inverting ROM-produced syndromes is not an option. Note that this extension does not significantly affect the sampling process we described as finding a preimage $\mathbf{x}$ of $\mathbf{u}$ for $[\mathbf{A}_t | \mathbf{A}_3]$ can be done by sampling $\mathbf{v}_3$ for a distribution $\mathcal{D}$ and then sampling a preimage of $\mathbf{u} - \mathbf{A}_3 \mathbf{v}_3$ for $\mathbf{A}_t$. However, each column of $\mathbf{A}_3$ adds a dimension to the preimage, which impacts its size. With the advances of [AGJ⁺24], this extra block is independent of the module rank d (which drives security), but still depends on the gadget dimension k_H (which drives efficiency).

This brings us to our second step. So far, we have only looked at the case where $\mathbf{t} = t(1, \dots, 1)$ and treated the tagged part $q_L t\mathbf{G}_H$ monolithically. Tackling the general case of $\mathbf{G}_{H,\mathbf{t}}$ would give a much more fine-grained tag management as different tags t_i are applied to each column. We then get the ability to change the effective tag column by column and thus only need one extra column in the trapdoor switching process. More precisely, to compensate the loss of the trapdoors hidden in the i -th column of $\mathbf{B}$, one can use a single column $\mathbf{A}_3 = \mathbf{a}_3$ defined as follows.

$$\mathbf{a}_3 = t'_i b_H^{\lfloor i/d \rfloor} \cdot \mathbf{e}_{(i \bmod d)} - (\mathbf{r}'_{1,i} + \mathbf{A} \mathbf{r}'_{2,i}) \bmod q\mathcal{R}$$

Concretely, this means that we now sample with a different effective tag $\mathbf{t}' = (t_0, \dots, t_{i-1}, t'_i, t_{i+1}, \dots, t_{dk_H-1})$, and change the effective trapdoor $\mathbf{R}_1, \mathbf{R}_2$ by incorporating $\mathbf{r}'_{1,i}$ and $\mathbf{r}'_{2,i}$ accordingly. This way, we can iteratively change all columns of $\mathbf{B}$, while retaining the ability to generate well-distributed preimages, and therefore prove security under standard assumptions.

It then remains to explain how we can effectively sample preimages with this tagged gadget. We proceed just like in the first step up to the sampling of $\mathbf{z}_H$, where we now need to sample $\mathbf{z}_H$ such that $\mathbf{G}_{H,\mathbf{t}} \mathbf{z}_H = \mathbf{h} = (\mathbf{w} - \mathbf{G}_L \mathbf{z}_L) / q_L \bmod q_H \mathcal{R}$. When $\mathbf{t} \neq t(1, \dots, 1)$, we cannot rely on prior approaches consisting of

reducing the problem to sampling over $\mathcal{L}_{q_H}^{t^{-1}\mathbf{h}}(\mathbf{G}_H)$. The structure of our new gadget requires to work with each component in an iterative way.

Concretely, for each entry index $i \in [0, d-1]$, we must find a way to sample $\mathbf{z}_i = [[\mathbf{z}_H]_{i+0d}, \dots, [\mathbf{z}_H]_{i+(k_H-1)d}]^T \in \mathcal{R}^{k_H}$ such that $\mathbf{g}_{H,\mathbf{t}_i}^T \mathbf{z}_i = [\mathbf{h}]_i \bmod q_H \mathcal{R}$, with $\mathbf{g}_{H,\mathbf{t}_i}^T = [t_{i+jd} \cdot b_H^j]_{j \in [0, k_H-1]}$ and $\mathbf{t}_i = [t_{i+jd}]_{j \in [0, k_H-1]}$. For that, we first set $h_{i,0} = [\mathbf{h}]_i$. We then compute the first corrected syndrome $c_{i,0} = t_{i+0d}^{-1} h_{i,0} \bmod b_H \mathcal{R}$ and sample z_0 from $\mathcal{D}_{b_H \mathcal{R} + c_{i,0}, s_H}$ (which can be done efficiently with uncentered spherical sampling over $\mathbb{Z}^n$). We then update h for the following step by defining $h_{i,1} = (h_{i,0} - t_{i+0d} z_0) / b_H$ which is in $\mathcal{R}$ based on how z_0 is sampled. Iterating this process, at step j , we compute the corrected syndrome $c_{i,j} = t_{i+jd}^{-1} h_{i,j} \bmod b_H \mathcal{R}$, sample z_j from $\mathcal{D}_{b_H \mathcal{R} + c_{i,j}, s_H}$, and compute $h_{i,j+1} = (h_{i,j} - t_{i+jd} z_j) / b_H$. In the end, we sample all the z_j components of $\mathbf{z}_H$ while correcting the carries induced by all previous steps. By definition of the $h_{i,j}$, we can easily check that

$$[\mathbf{h}]_i = h_{i,0} = \sum_{j=0}^{k_H-1} t_{i+jd} b_H^j \cdot z_j + b_H^{k_H} h_{i,k_H},$$

which gives $\mathbf{g}_{H,\mathbf{t}_i}^T \mathbf{z}_i = [\mathbf{h}]_i \bmod q_H \mathcal{R}$, as long as $q_H = b_H^{k_H}$. We now give the formal description of our sampler in Algorithm 3.1

Algorithm 3.1: TSampler($\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u}, \mathbf{t}, \mathbf{K}, \mathbf{L}, \mathbf{S}, \mathbf{S}_L, \mathbf{S}_H$)

Input: trapdoor matrices $\mathbf{R}_1, \mathbf{R}_2 \in \mathcal{R}^{d \times dk_H}$, matrix $\mathbf{A} \in \mathcal{R}_q^{d \times d}$, syndrome $\mathbf{u} \in \mathcal{R}_q^d$, tag $\mathbf{t} \in (\mathcal{R}_{b_H}^\times)^{dk_H}$, matrices $\mathbf{K} \in \mathcal{R}^{d(2+k_H) \times r}$ and $\mathbf{L} \in \mathcal{R}^{r \times d(k_L+k_H)}$ such that $\mathbf{KL} = \begin{bmatrix} \mathbf{G}_L & \mathbf{R}_1 \\ \mathbf{0} & \mathbf{R}_2 \\ \mathbf{0} & \mathbf{I}_{dk_H} \end{bmatrix}$, covariance matrices $\mathbf{S} \in \mathcal{S}_{nr}^{++}(\mathbb{R})$, $\mathbf{S}_L \in \mathcal{S}_{ndk_L}^{++}(\mathbb{R})$, $\mathbf{S}_H \in \mathcal{S}_n^{++}(\mathbb{R})$.

1. $\mathbf{p} \leftarrow \mathcal{D}_{\mathcal{R}^r, \sqrt{\mathbf{S}_p}}$ with $\mathbf{S}_p = \mathbf{S} - M_\tau(\mathbf{L}) \text{diag}(\mathbf{S}_L, \mathbf{I}_{dk_H} \otimes \mathbf{S}_H) M_\tau(\mathbf{L})^T$.
2. $\mathbf{w} \leftarrow \mathbf{u} - [\mathbf{I}_d | \mathbf{A} | q_L \mathbf{G}_H, \mathbf{t} - (\mathbf{R}_1 + \mathbf{A} \mathbf{R}_2)] \mathbf{K} \mathbf{p} \bmod q \mathcal{R}$.
3. $\ell \leftarrow \mathbf{w} \bmod q_L \mathcal{R}$.
4. $\mathbf{z}_L \leftarrow \mathcal{D}_{\mathcal{L}_{q_L}^\ell(\mathbf{G}_L), \sqrt{\mathbf{S}_L}}$.
5. $\mathbf{h} \leftarrow (\mathbf{w} - \mathbf{G}_L \mathbf{z}_L) / q_L \bmod q_H \mathcal{R}$.
6. **For** $i = 0, \dots, d-1$ **do**
7. $h_{i,0} \leftarrow [\mathbf{h}]_i$.
8. **For** $j = 0, \dots, k_H-1$ **do**
9. $c_{i,j} \leftarrow t_{i+jd}^{-1} \cdot h_{i,j} \bmod b_H \mathcal{R}$.
10. $[\mathbf{z}_H]_{i+jd} \leftarrow \mathcal{D}_{b_H \mathcal{R} + c_{i,j}, \sqrt{\mathbf{S}_H}}$.
11. $h_{i,j+1} \leftarrow (h_{i,j} - t_{i+jd} [\mathbf{z}_H]_{i+jd}) / b_H \in \mathcal{R}$.
12. $\mathbf{z} \leftarrow [\mathbf{z}_L^T | \mathbf{z}_H^T]^T$.
13. $\mathbf{v}' \leftarrow \mathbf{p} + \mathbf{L} \mathbf{z}$.
14. $\mathbf{v} \leftarrow \mathbf{K} \mathbf{v}'$.

Output: $\mathbf{v} \in \mathcal{L}_q^u(\mathbf{A}_T)$

Observe that step 10 of Algorithm 3.1 can be carried with an integer sampler. Indeed, sampling from $\mathcal{D}_{b_H \mathcal{R} + c_{i,j}, \sqrt{\mathbf{S}_H}}$ can be done exactly by sampling $[\mathbf{z}_H]_{i+jd}$ from $c_{i,j} + b_H \mathcal{D}_{\mathcal{R}, \sqrt{\mathbf{S}_H}/b_H, -c_{i,j}/b_H}$. As we shall see in the security analysis, we

need $\sqrt{\mathbf{S}_H}/b_H \geq \eta_{\varepsilon_H}(\mathbb{Z}^n)$ so we can smooth out the coset shifts $c_{i,j}$ and have $\rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R} + c_{i,j}) \approx \rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R})$.

3.2 Security Analysis of TSampler

We now prove security of our sampler by relying on the blueprint of [JS25b], which provides a comprehensive and thorough analysis methodology for gadget-based samplers, that we adapt to our case. In particular, the analysis goes through using very similar arguments if the Gaussian parameters of $\mathbf{z}_L$ and the components of $\mathbf{z}_H$ are above the smoothing parameter of their respective lattices, in order to smooth out the dependency between $\mathbf{z}_L$ and $\mathbf{z}_H$, but also the dependency with the perturbation as usual. As a result, we identify similar conditions as [JS25b].

To facilitate future implementation works, we also generalize the security statement, and in particular the distance to the ideal distribution when plugging inexact samplers for step 1 and 4. We denote by $\mathcal{P}_p, \mathcal{P}_L$ the respective distributions of these samplers and, although the distribution may not exactly match that of the ideal case, we assume their support does.

The main theorem below reflects the high complexity of sampler security analysis. In practice, it essentially states that the sampler outputs correct preimages of $\mathbf{A}_t$ while statistically hiding $\mathbf{L}$ (and hence the trapdoor), provided that some conditions related to the gadget and other parameters are satisfied. We provide in Sections 3.3 and 3.5 some concrete instantiations that do satisfy all these conditions. Examples of applications and associated performance are provided in Section 4.

Theorem 3.1. *Let $d, r, q_L, b_L, b_H, k_L, k_H$ be in $\mathbb{N}^\times$ such that $k_L = \lceil \log_{b_L} q_L \rceil$, and $\varepsilon_L, \varepsilon_H, \varepsilon$ in $(0, 1)$. Then, define $q_H = b_H^{k_H}$ and $q = q_L q_H$, as well as $\mathbf{G}_L = [1|b_L| \dots |b_L^{k_L-1}|] \otimes \mathbf{I}_d$. Let $\mathbf{S} \in \mathcal{S}_{nr}^{++}(\mathbb{R})$, $\mathbf{S}_L \in \mathcal{S}_{ndk_L}^{++}(\mathbb{R})$, $\mathbf{S}_H \in \mathcal{S}_n^{++}(\mathbb{R})$, and define $\mathbf{S}_G = \text{diag}(\mathbf{S}_L, \mathbf{I}_{dk_H} \otimes \mathbf{S}_H)$. Let $\mathbf{A} \in \mathcal{R}_q^{d \times d}$, $\mathbf{R}_1, \mathbf{R}_2 \in \mathcal{R}^{d \times dk_H}$, and $\mathbf{t} = [t_i]_{i \in [0, dk_H-1]} \in (\mathcal{R}_{b_H}^\times)^{dk_H}$. Then define $\mathbf{G}_{H,\mathbf{t}} = [t_i \cdot b_H^{\lfloor i/d \rfloor} \cdot \mathbf{e}_{(i \bmod d)}]_{i \in [0, dk_H-1]}$. We then let $\mathbf{K}$ be in $\mathcal{R}^{d(2+k_H) \times r}$ and $\mathbf{L}$ in $\mathcal{R}^{r \times d(k_L+k_H)}$.*

We denote by $\mathcal{P}_p, \mathcal{P}_L$ the distribution outputted by a given choice of samplers in steps 1, and 4 of Algorithm 3.1. We assume that $\mathcal{P}_x \approx_{\alpha_x} \mathcal{P}_x^$ for $x \in \{p, L\}$, with $\alpha_x \geq 1$ and where the $\mathcal{P}_x^*$ represent the ideal distributions identified in Algorithm 3.1. We finally denote by $\mathcal{P}$ the output distribution of $\text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u}, \mathbf{t}, \mathbf{K}, \mathbf{L}, \mathbf{S}, \mathbf{S}_L, \mathbf{S}_H)$ and assume the following conditions.*

- ① $\mathbf{KL} = \begin{bmatrix} \mathbf{G}_L & \mathbf{R}_1 \\ \mathbf{0} & \mathbf{R}_2 \\ \mathbf{0} & \mathbf{I}_{dk_H} \end{bmatrix} \bmod q\mathcal{R}$
- ② $\text{rank}(\tilde{\mathbf{L}}) = nd(k_L + k_H)$
- ③ $\mathbf{S}_L \geq \eta_{\varepsilon_L}(\mathcal{L}_{q_L}^\perp(\mathbf{G}_L))^2 \mathbf{I}_{ndk_L}$ and $\mathbf{S}_L > \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)})^2 \mathbf{I}_{ndk_L}$
- ④ $\mathbf{S}_H \geq (b_H \eta_{\varepsilon_H}(\mathbb{Z}^n))^2 \mathbf{I}_n$ and $\mathbf{S}_H > \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)})^2 \mathbf{I}_n$

$$\textcircled{5} \mathbf{S} > \tilde{\mathbf{L}} \mathbf{S}_{\mathbf{G}} \tilde{\mathbf{L}}^T$$

$$\textcircled{6} \mathbf{S} \geq \tilde{\mathbf{L}} \left(\mathbf{S}_{\mathbf{G}} + \left(\frac{1}{\eta_{\varepsilon}(\mathbb{Z}^{nd(k_L+k_H)})^2} \mathbf{I}_{nd(k_L+k_H)} - \mathbf{S}_{\mathbf{G}}^{-1} \right)^{-1} \right) \tilde{\mathbf{L}}^T$$

where $\tilde{\mathbf{L}} = M_{\tau}(\mathbf{L})$. It then holds that $\text{Supp}(\mathcal{P}) = \mathbf{K} \mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_{\mathbf{t}} \mathbf{K}) \subseteq \mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_{\mathbf{t}})$, and that $\mathcal{P} \approx_{\delta} \mathbf{K} \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_{\mathbf{t}} \mathbf{K}), \sqrt{\mathbf{S}}}$ for

$$\delta = \alpha_p^2 \cdot \alpha_L^2 \cdot \frac{1+\varepsilon}{1-\varepsilon} \cdot \frac{1+\varepsilon_L}{1-\varepsilon_L} \cdot \left(\frac{1+\varepsilon_H}{1-\varepsilon_H} \right)^{dk_H},$$

where $\mathbf{A}_{\mathbf{t}} = [\mathbf{I}_d | \mathbf{A} | q_L \mathbf{G}_{H,\mathbf{t}} - (\mathbf{R}_1 + \mathbf{A} \mathbf{R}_2)] \bmod q\mathcal{R}$.

Proof. For clarity and conciseness, we use $\tilde{\mathbf{M}}$ to denote multiplication matrix embedding of $\mathbf{M} \in \mathcal{R}^{x \times y}$, namely $\tilde{\mathbf{M}} = M_{\tau}(\mathbf{M}) \in \mathbb{R}^{nx \times ny}$. Similarly, for vectors $\mathbf{x}$ over $\mathcal{R}$, we may use the notation $\tilde{\mathbf{x}}$ for their coefficient vector, i.e., $\tilde{\mathbf{x}} = \tau(\mathbf{x})$. We also define $k = k_L + k_H$ for clarity.

Step 1: Correctness. We first show in Lemma 3.1 that the support of $\mathcal{P}$ is that intended, thus showing the sampler produces correct preimages.

Lemma 3.1. *It holds that $\text{Supp}(\mathcal{P}) = \mathbf{K} \mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_{\mathbf{t}} \mathbf{K}) \subseteq \mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_{\mathbf{t}})$*

Proof (Lemma 3.1). Let $\mathbf{v} = \mathbf{K} \mathbf{v}'$ be in $\text{Supp}(\mathcal{P})$ outputted by the sampler with $\mathbf{v}' = \mathbf{p} + \mathbf{L} \mathbf{z}$. By condition $\textcircled{1}$, it first holds that

$$\begin{aligned} \mathbf{A}_{\mathbf{t}} \mathbf{K} \mathbf{L} &= \mathbf{A}_{\mathbf{t}} \begin{bmatrix} \mathbf{G}_L & \mathbf{R}_1 \\ \mathbf{0} & \mathbf{R}_2 \\ \mathbf{0} & \mathbf{I}_{dk_H} \end{bmatrix} = [\mathbf{G}_L | \mathbf{R}_1 + \mathbf{A} \mathbf{R}_2 + q_L \mathbf{G}_{H,\mathbf{t}} - (\mathbf{R}_1 + \mathbf{A} \mathbf{R}_2)] \\ &= [\mathbf{G}_L | q_L \mathbf{G}_{H,\mathbf{t}}] \bmod q\mathcal{R} \end{aligned}$$

We then have that $\mathbf{A}_{\mathbf{t}} \mathbf{K} \mathbf{v}' = \mathbf{A}_{\mathbf{t}} \mathbf{K} \mathbf{p} + [\mathbf{G}_L | q_L \mathbf{G}_{H,\mathbf{t}}] \mathbf{z} \bmod q\mathcal{R}$. First, notice that $\mathbf{h}$ is indeed well defined with entries in $\mathcal{R}_{q_H}$. This is because by construction $\mathbf{G}_L \mathbf{z}_L = \mathbf{w} \bmod q_L \mathcal{R}$. Hence, when lifting in $\mathcal{R}_q$, there exists $\mathbf{h} \in \mathcal{R}^d$ such that $\mathbf{G}_L \mathbf{z}_L = \mathbf{w} + q_L \mathbf{h} \bmod q\mathcal{R}$ and therefore $(\mathbf{w} - \mathbf{G}_L \mathbf{z}_L)/q_L = \mathbf{h} \bmod q_H \mathcal{R}$.

We now focus on $\mathbf{G}_{H,\mathbf{t}} \mathbf{z}_H \bmod q_H \mathcal{R}$. Let $i \in [0, d-1]$. We first verify that all the $h_{i,0}, \dots, h_{i,k_H}$ are indeed in $\mathcal{R}$. First, $h_{i,0} \in \mathcal{R}$ as we have seen above. Then, assume $h_{i,j} \in \mathcal{R}$. Based on how $[\mathbf{z}_H]_{i+jd}$ is sampled, we have $t_{i+jd}[\mathbf{z}_H]_{i+jd} = h_{i,j} + b_H a_{i,j}$ with $a_{i,j} \in \mathcal{R}$. So $h_{i,j+1} = -a_{i,j} \in \mathcal{R}$. By induction, all the $h_{i,j}$ are in $\mathcal{R}$. Then, we can easily see by induction again that

$$\begin{aligned} [\mathbf{h}]_i &= h_{i,0} = t_{i+0d}[\mathbf{z}_H]_{i+0d} + b_H h_{i,1} \\ &= t_{i+0d}[\mathbf{z}_H]_{i+0d} + b_H t_{i+1d}[\mathbf{z}_H]_{i+1d} + b_H^2 h_{i,2} \\ &= \dots \\ &= \sum_{j=0}^{k_H-1} t_{i+jd} b_H^j [\mathbf{z}_H]_{i+jd} + b_H^{k_H} h_{i,k_H}. \end{aligned}$$

Hence $\sum_{j=0}^{k_H-1} t_{i+jd} b_H^j [\mathbf{z}_H]_{i+jd} = [\mathbf{h}]_i \bmod q_H \mathcal{R}$. As this is true for all i , we end up with $\mathbf{G}_{H,t} \mathbf{z}_H = \mathbf{h} \bmod q_H \mathcal{R}$. So there exists $\mathbf{a}' \in \mathcal{R}^d$ such that $\mathbf{G}_{H,t} \mathbf{z}_H = (\mathbf{w} - \mathbf{G}_L \mathbf{z}_L)/q_L + q_H \mathbf{a}'$. Hence, we have $q_L \mathbf{G}_{H,t} \mathbf{z}_H = \mathbf{w} - \mathbf{G}_L \mathbf{z}_L + q_L q_H \mathbf{a}'$ which means $[\mathbf{G}_L | q_L \mathbf{G}_{H,t}] \mathbf{z} = \mathbf{w} = \mathbf{u} - \mathbf{A}_t \mathbf{K} \mathbf{p} \bmod q \mathcal{R}$. We then conclude $\mathbf{v}' \in \mathcal{L}_q^u(\mathbf{A}_t \mathbf{K})$ and therefore $\mathbf{v} \in \mathbf{K} \mathcal{L}_q^u(\mathbf{A}_t \mathbf{K}) \subseteq \mathcal{L}_q^u(\mathbf{A}_t)$.

Reciprocally, let $\mathbf{v} = \mathbf{K} \mathbf{v}' \in \mathbf{K} \mathcal{L}_q^u(\mathbf{A}_t \mathbf{K})$. Let $\mathbf{z} \in \mathcal{R}^{d(k_L+k_H)}$ and define $\mathbf{p} = \mathbf{v}' - \mathbf{L} \mathbf{z}$. Then $\mathbf{p} \in \mathcal{R}^r$ because $\mathbf{L}$ has coefficients in $\mathcal{R}$ and $\mathbf{v}'$ as well (because $\mathcal{L}_q^u(\mathbf{A}_t \mathbf{K}) \subseteq \mathcal{R}^r$). Then, $\mathbf{A}_t \mathbf{K} \mathbf{L} \mathbf{z} = \mathbf{A}_t \mathbf{K} \mathbf{v}' - \mathbf{A}_t \mathbf{K} \mathbf{p} = \mathbf{u} - \mathbf{A}_t \mathbf{K} \mathbf{p} \bmod q \mathcal{R}$. As $\mathbf{A}_t \mathbf{K} \mathbf{L} = [\mathbf{G}_L | q_L \mathbf{G}_{H,t}] \bmod q \mathcal{R}$, we deduce

$$\mathbf{G}_L \mathbf{z}_L + q_L \mathbf{G}_{H,t} \mathbf{z}_H = \mathbf{u} - \mathbf{A}_t \mathbf{K} \mathbf{p} \bmod q \mathcal{R}$$

Because $q_L | q$, we then have $\mathbf{G}_L \mathbf{z}_L = \mathbf{u} - \mathbf{A}_t \mathbf{K} \mathbf{p} \bmod q_L \mathcal{R} = \ell$. So $\mathbf{z}_L \in \mathcal{L}_{q_L}^\ell(\mathbf{G}_L)$. Then, we directly have $\mathbf{G}_{H,t} \mathbf{z}_H = (\mathbf{w} - \mathbf{G}_L \mathbf{z}_L)/q_L \bmod q_H \mathcal{R} = \mathbf{h}$. Taking it componentwise gives $\sum_{j=0}^{k_H-1} t_{i+jd} b_H^j [\mathbf{z}_H]_{i+jd} = [\mathbf{h}]_i \bmod q_H \mathcal{R}$. Taking progressively modulo b_H following the update rule of the algorithm, we get that each $[\mathbf{z}_H]_{i+jd}$ is in the right coset of $b_H \mathcal{R}$ as in the sampler. Because $\mathbf{p}, \mathbf{z}_L, \mathbf{z}_H$ are in the support of their respective distributions, it holds $\mathbf{v}$ is in the support of $\mathcal{P}$. $\square$

Step 2: Linear Algebra Computations. We now look at the elements we will need to analyze $\mathcal{P}$. First, we define $\mathbf{S}_y = \tilde{\mathbf{L}} \mathbf{S}_G \tilde{\mathbf{L}}^T$ and $\mathbf{S}_p = \mathbf{S} - \mathbf{S}_y$. We let $V = \text{Span}_{\mathbb{R}}(\tilde{\mathbf{L}}) \subset \mathbb{R}^{nr}$, and define $\mathbf{P} = \tilde{\mathbf{L}} \tilde{\mathbf{L}}^+$ to be the orthogonal projector onto V . In particular, we have $\mathbf{P}^+ = \mathbf{P}$ and $\mathbf{P}^T = \mathbf{P}$. By condition (2), $\tilde{\mathbf{L}}$ has linearly independent columns which means $\tilde{\mathbf{L}}^+ \tilde{\mathbf{L}} = \mathbf{I}_{ndk}$. We will use this identity extensively. This linear independence of the columns of $\tilde{\mathbf{L}}$ also allows us to define a lattice $\mathcal{L}' = \mathcal{L}(\tilde{\mathbf{L}}) = \tilde{\mathbf{L}} \mathbb{Z}^{ndk}$. We also define the lattice $\mathcal{L} = \mathbb{Z}^{nr} \cap V$. We can easily check that $\mathcal{L}' \subseteq \mathcal{L}$ by observing $\tilde{\mathbf{L}} \mathbb{Z}^{ndk} \subset V$ and $\tilde{\mathbf{L}} \mathbb{Z}^{ndk} \subseteq \mathbb{Z}^{nr}$ as $\tilde{\mathbf{L}}$ has integer coefficients. We can also check that $\text{Span}_{\mathbb{R}}(\mathcal{L}) = \text{Span}_{\mathbb{R}}(\mathcal{L}') = V$ which entails that both lattices have rank ndk . Although we do not need it, our concrete choice of $\mathbf{L}$ in Equation (3) later ensures $\mathcal{L} = \mathcal{L}'$.

We now define $\mathbf{S}_3 = (\mathbf{P}(\mathbf{S}_p^+ + \mathbf{S}_y^+) \mathbf{P})^+$. Condition (5) yields $\mathbf{S}_p > 0$ and thus $\mathbf{S}_p \in GL_{nr}(\mathbb{R})$. We then have $\mathbf{S}_p^+ = \mathbf{S}_p^{-1}$ in the expression of $\mathbf{S}_3$.

Lemma 2.1 directly leads to $\mathbf{S}_y^+ = \tilde{\mathbf{L}}^{+T} \mathbf{S}_G^{-1} \tilde{\mathbf{L}}^+$ and $\text{Span}_{\mathbb{R}}(\mathbf{S}_y) = \text{Span}_{\mathbb{R}}(\mathbf{S}_y^+) = V$ by taking $\mathbf{A} = \tilde{\mathbf{L}}$ and $\mathbf{B} = \mathbf{S}_G$. This is because (2) ensures $\tilde{\mathbf{L}}$ has full column rank, and (3) and (4) ensure $\mathbf{S}_G$ is positive definite. As a consequence, we observe that $\mathbf{P} \mathbf{S}_y^+ \mathbf{P} = \mathbf{S}_y^+$. By using $\mathbf{P} = \tilde{\mathbf{L}} \tilde{\mathbf{L}}^+$, we can thus re-express $\mathbf{S}_3$ as

$$\mathbf{S}_3 = (\tilde{\mathbf{L}}^{+T} (\tilde{\mathbf{L}}^T \mathbf{S}_p^{-1} \tilde{\mathbf{L}} + \mathbf{S}_G^{-1}) \tilde{\mathbf{L}}^+)^+.$$

We define $\mathbf{S}' = \tilde{\mathbf{L}}^T \mathbf{S}_p^{-1} \tilde{\mathbf{L}} + \mathbf{S}_G^{-1}$. As $\tilde{\mathbf{L}}^T \mathbf{S}_p^{-1} \tilde{\mathbf{L}} \geq 0$, we directly have that $\mathbf{S}' \geq \mathbf{S}_G > 0$, which proves $\mathbf{S}' > 0$. Using Lemma 2.1 again with $\mathbf{A} = \tilde{\mathbf{L}}^{+T}$, which also has full column rank, and $\mathbf{B} = \mathbf{S}'$, we get

$$\mathbf{S}_3 = \tilde{\mathbf{L}} \mathbf{S}'^{-1} \tilde{\mathbf{L}}^T, \quad \text{and} \quad \text{Span}_{\mathbb{R}}(\mathbf{S}_3) = \text{Span}_{\mathbb{R}}(\tilde{\mathbf{L}}^{+T}) = \text{Span}_{\mathbb{R}}(\tilde{\mathbf{L}}) = V.$$

Step 3: Distribution. Let us now look at the distribution of the sampler. For that, we study the distribution $\mathcal{P}'$ of $\mathbf{v}'$ defined in step 13 of Algorithm 3.1. We then want to show that $\mathcal{P}'$ is close to $\mathcal{D}_{\mathcal{L}_q^u(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}$ to conclude on the theorem statement by multiplying by $\mathbf{K}$. As we have shown in Lemma 3.1, we already have $\text{Supp}(\mathcal{P}') = \mathcal{L}_q^u(\mathbf{A}_T \mathbf{K})$. Let $\mathbf{y}$ be a sample from $\mathcal{P}'$. There exists $\mathbf{p} \in \mathcal{R}^r$, $\mathbf{z}_L \in \mathcal{L}_{q_L}^\ell(\mathbf{G}_L)$, and $\mathbf{z}_H$ whose entries are in the right cosets of $b_H \mathcal{R}$ such that $\mathbf{y} = \mathbf{p} + \mathbf{L}\mathbf{z}$ with $\mathbf{z} = [\mathbf{z}_L^T | \mathbf{z}_H^T]^T$. Observe that ℓ depends on $\mathbf{p}$ and the $c_{i,j}$ depends on $\mathbf{p}$ and previously sampled terms. We then have $\mathbf{p} = \mathbf{y} - \mathbf{L}\mathbf{z}$ showing that $\tilde{\mathbf{p}} \in \tilde{\mathbf{y}} + \mathcal{L}' \subseteq \tilde{\mathbf{y}} + \mathcal{L}$. Also, once $\mathbf{p}$ and $\mathbf{y}$ are fixed, the value $\mathbf{z}$ can take is unique. Indeed, as $\tilde{\mathbf{L}}$ has linearly independent columns, if there exists $\tilde{\mathbf{z}}'$ such that $\tilde{\mathbf{L}}\tilde{\mathbf{z}}' = \tilde{\mathbf{y}} - \tilde{\mathbf{p}} = \tilde{\mathbf{L}}\tilde{\mathbf{z}}$, then $\tilde{\mathbf{z}}' = \tilde{\mathbf{z}}$.

For clarity, we denote $[\mathbf{z}_H]_i$ by z_i below. By the chain rule of conditional probabilities, it holds that the conditional distribution of $\mathbf{z}_H$ given $\mathbf{p}, \mathbf{z}_L$ is given by

$$\begin{aligned} \mathbb{P}[\mathbf{z}_H | \mathbf{p}, \mathbf{z}_L] &= \prod_{i=0}^{d-1} \prod_{j=0}^{k_H-1} \mathbb{P}[z_{i+jd} | \mathbf{p}, \mathbf{z}_L, z_{i+0d}, \dots, z_{i+(j-1)d}] \\ &= \prod_{i=0}^{d-1} \prod_{j=0}^{k_H-1} \frac{\rho_{\sqrt{\mathbf{S}_H}}(z_{i+jd})}{\rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R} + c_{i,j})} \\ &= \rho_{\sqrt{\mathbf{I}_{dk_H} \otimes \mathbf{S}_H}}(\mathbf{z}_H) \cdot \prod_{i=0}^{d-1} \prod_{j=0}^{k_H-1} \frac{1}{\rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R} + c_{i,j})} \end{aligned}$$

The numerator then features $\rho_{\sqrt{\mathbf{I}_{dk_H} \otimes \mathbf{S}_H}}(\mathbf{z}_H)$. For clarity, we denote the denominator by $D(\tilde{\mathbf{p}})$ as it depends on $\tilde{\mathbf{p}}$, i.e., $D(\tilde{\mathbf{p}}) = \prod_{i=0}^{d-1} \prod_{j=0}^{k_H-1} \rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R} + c_{i,j})$. Looking at the numerator, and the relation between $\mathbf{p}$ and $\mathbf{z}$, we have the following identity.

Lemma 3.2. *We have $\rho_{\sqrt{\mathbf{S}_L}}(\tilde{\mathbf{z}}_L) \rho_{\sqrt{\mathbf{I}_{dk_H} \otimes \mathbf{S}_H}}(\tilde{\mathbf{z}}_H) = \rho_{\sqrt{\mathbf{S}_y}}(\tilde{\mathbf{y}} - \tilde{\mathbf{p}})$*

Proof (Lemma 3.2). As we have $\mathbf{S}_y^+ = \tilde{\mathbf{L}}^+ \mathbf{S}_G^{-1} \tilde{\mathbf{L}}^+$, we directly obtain $\tilde{\mathbf{L}}^T \mathbf{S}_y^+ \tilde{\mathbf{L}} = \mathbf{S}_G^{-1}$ using the fact that $\tilde{\mathbf{L}}^+ \tilde{\mathbf{L}} = \mathbf{I}_{ndk}$. As a result, we have the following equalities.

$$\begin{aligned} \rho_{\sqrt{\mathbf{S}_y}}(\tilde{\mathbf{y}} - \tilde{\mathbf{p}}) &= \exp(-\pi(\tilde{\mathbf{y}} - \tilde{\mathbf{p}})^T \mathbf{S}_y^+ (\tilde{\mathbf{y}} - \tilde{\mathbf{p}})) \cdot \mathbf{1}_{\text{Span}_{\mathbb{R}}(\mathbf{S}_y)}(\tilde{\mathbf{y}} - \tilde{\mathbf{p}}) \\ &= \exp(-\pi \tilde{\mathbf{z}}^T \tilde{\mathbf{L}}^T \mathbf{S}_y^+ \tilde{\mathbf{L}} \tilde{\mathbf{z}}) \cdot \mathbf{1}_{\text{Span}_{\mathbb{R}}(\mathbf{S}_y)}(\tilde{\mathbf{L}} \tilde{\mathbf{z}}) \\ &= \exp(-\pi \tilde{\mathbf{z}}^T \mathbf{S}_G^{-1} \tilde{\mathbf{z}}) \cdot \mathbf{1}_V(\tilde{\mathbf{L}} \tilde{\mathbf{z}}) \\ &= \exp(-\pi \tilde{\mathbf{z}}^T \mathbf{S}_G^{-1} \tilde{\mathbf{z}}) \\ &= \exp(-\pi(\tilde{\mathbf{z}}_L^T \mathbf{S}_L^{-1} \tilde{\mathbf{z}}_L + \tilde{\mathbf{z}}_H^T (\mathbf{I}_{dk_H} \otimes \mathbf{S}_H)^{-1} \tilde{\mathbf{z}}_H)) \\ &= \rho_{\sqrt{\mathbf{S}_L}}(\tilde{\mathbf{z}}_L) \rho_{\sqrt{\mathbf{I}_{dk_H} \otimes \mathbf{S}_H}}(\tilde{\mathbf{z}}_H), \end{aligned}$$

where we also used that $\text{Span}_{\mathbb{R}}(\mathbf{S}_y) = V$ and thus $\tilde{\mathbf{L}} \tilde{\mathbf{z}} \in V$ by definition of V . $\square$

Noting $\mathbf{P}[\mathbf{z}_L|\mathbf{p}] = \mathcal{P}_L(\mathbf{z}_L)$ (where the dependency in $\mathbf{p}$ is implicit in the notation $\mathcal{P}_L$), we can then express $\mathcal{P}'$ as follows.

$$\begin{aligned}
\mathcal{P}'(\mathbf{y}) &= \sum_{\tilde{\mathbf{p}} \in \tilde{\mathbf{y}} + \mathcal{L}} \mathcal{P}_p(\mathbf{p}) \cdot \mathcal{P}_L(\mathbf{z}_L) \cdot \mathbb{P}[\mathbf{z}_H|\mathbf{p}, \mathbf{z}_L] \\
&\in [\alpha^{-1}, \alpha] \sum_{\tilde{\mathbf{p}} \in \tilde{\mathbf{y}} + \mathcal{L}} \mathcal{D}_{\mathbb{Z}^{nr}, \sqrt{\mathbf{S}_p}}(\tilde{\mathbf{p}}) \cdot \mathcal{D}_{\tau(\mathcal{L}_{q_L}^{\ell}(\mathbf{G}_L)), \sqrt{\mathbf{S}_L}}(\tilde{\mathbf{z}}_L) \cdot \mathbb{P}[\mathbf{z}_H|\mathbf{p}, \mathbf{z}_L] \\
&\in [\alpha^{-1}, \alpha] \sum_{\tilde{\mathbf{p}} \in \tilde{\mathbf{y}} + \mathcal{L}} \frac{\rho_{\sqrt{\mathbf{S}_p}}(\tilde{\mathbf{p}}) \rho_{\sqrt{\mathbf{S}_L}}(\tilde{\mathbf{z}}_L) \rho_{\sqrt{\mathbf{I}_{dk_H} \otimes \mathbf{S}_H}}(\tilde{\mathbf{z}}_H)}{\rho_{\sqrt{\mathbf{S}_p}}(\mathbb{Z}^{nr}) \rho_{\sqrt{\mathbf{S}_L}}(\tau(\mathcal{L}_{q_L}^{\ell}(\mathbf{G}_L))) D(\tilde{\mathbf{p}})} \\
&\in [\alpha^{-1}, \alpha] \sum_{\tilde{\mathbf{p}} \in \tilde{\mathbf{y}} + \mathcal{L}} \frac{\rho_{\sqrt{\mathbf{S}_p}}(\tilde{\mathbf{p}}) \rho_{\sqrt{\mathbf{S}_y}}(\tilde{\mathbf{p}} - \tilde{\mathbf{y}})}{\rho_{\sqrt{\mathbf{S}_p}}(\mathbb{Z}^{nr}) \rho_{\sqrt{\mathbf{S}_L}}(\mathcal{L}_{q_L}^{\ell}(\tilde{\mathbf{G}}_L)) D(\tilde{\mathbf{p}})}
\end{aligned}$$

where $\alpha = \alpha_p \alpha_L$. The last equation uses Lemma 3.2 and that $\tau(\mathcal{L}_q^{\mathbf{x}}(\mathbf{M})) = \mathcal{L}_q^{\tau(\mathbf{x})}(M_{\tau}(\mathbf{M}))$. Note that $M_{\tau}(\mathbf{G}_L)$ is simply the extended gadget as it is composed of only integer entries, i.e., $\tilde{\mathbf{G}}_L = \mathbf{G}_L \otimes \mathbf{I}_n$. We now use the identity proven in [JS24, App. A] (the full version of [JS25b]), which fixes [MP12, Fact 5.6], to obtain that

$$\rho_{\sqrt{\mathbf{S}_p}}(\tilde{\mathbf{p}}) \rho_{\sqrt{\mathbf{S}_y}}(\tilde{\mathbf{p}} - \tilde{\mathbf{y}}) = \rho_{\sqrt{\mathbf{S}}}(\tilde{\mathbf{y}}) \rho_{\sqrt{\mathbf{S}_3}}(\tilde{\mathbf{p}} - \mathbf{c}).$$

where $\mathbf{c} = \mathbf{x} + \mathbf{S}_3 \mathbf{P}(-\mathbf{S}_p^{-1} \mathbf{x} + \mathbf{S}_y^+ \tilde{\mathbf{y}})$ where $\mathbf{x} = \tilde{\mathbf{y}} - \mathbf{P} \tilde{\mathbf{y}}$ the unique element of $(\tilde{\mathbf{y}} + V) \cap V^{\perp}$. Notice it clearly holds that $\mathbf{c} \in \mathbf{x} + V$. Going back to our computation of $\mathcal{P}'(\mathbf{y})$, we have

$$\begin{aligned}
\mathcal{P}'(\mathbf{y}) &\in [\alpha^{-1}, \alpha] \sum_{\tilde{\mathbf{p}} \in \tilde{\mathbf{y}} + \mathcal{L}} \frac{\rho_{\sqrt{\mathbf{S}}}(\tilde{\mathbf{y}}) \rho_{\sqrt{\mathbf{S}_3}}(\tilde{\mathbf{p}} - \mathbf{c})}{\rho_{\sqrt{\mathbf{S}_p}}(\mathbb{Z}^{nr}) \rho_{\sqrt{\mathbf{S}_L}}(\mathcal{L}_{q_L}^{\ell}(\tilde{\mathbf{G}}_L)) D(\tilde{\mathbf{p}})} \\
&\in [\alpha^{-1}, \alpha] \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}(\mathbf{y}) \frac{\rho_{\sqrt{\mathbf{S}}}(\tau(\mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_T \mathbf{K})))}{\rho_{\sqrt{\mathbf{S}_p}}(\mathbb{Z}^{nr})} \sum_{\tilde{\mathbf{p}} \in \tilde{\mathbf{y}} + \mathcal{L}} \frac{\rho_{\sqrt{\mathbf{S}_3}}(\tilde{\mathbf{p}} - \mathbf{c})}{\rho_{\sqrt{\mathbf{S}_L}}(\mathcal{L}_{q_L}^{\ell}(\tilde{\mathbf{G}}_L)) D(\tilde{\mathbf{p}})}
\end{aligned}$$

We now smooth out the shifts from the denominator using the appropriate smoothing conditions. First, using condition ③, we get $\sqrt{\mathbf{S}_L} \geq \eta_{\varepsilon_L}(\mathcal{L}_{q_L}^{\perp}(\mathbf{G}_L))$. Lemma 2.6 then yields $\rho_{\sqrt{\mathbf{S}_L}}(\mathcal{L}_{q_L}^{\ell}(\tilde{\mathbf{G}}_L)) \in [\frac{1-\varepsilon_L}{1+\varepsilon_L}, 1] \rho_{\sqrt{\mathbf{S}_L}}(\mathcal{L}_{q_L}^{\perp}(\tilde{\mathbf{G}}_L))$. Similarly, condition ④ in combination with Lemma 2.6 gives that $\rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R} + c_{i,j}) \in [(1-\varepsilon_H)/(1+\varepsilon_H), 1] \rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R})$ as $\sqrt{\mathbf{S}_H} \geq b_H \eta_{\varepsilon_H}(\mathbb{Z}^n) = \eta_{\varepsilon_H}(b_H \mathcal{R})$. As a result, we have $D(\tilde{\mathbf{p}}) \in [((1-\varepsilon_H)/(1+\varepsilon_H))^{dk_H}, 1] \rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R})^{dk_H}$. It then yields

$$\mathcal{P}'(\mathbf{y}) \in [\alpha^{-1}, \alpha'] \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}(\mathbf{y}) \frac{\rho_{\sqrt{\mathbf{S}}}(\tau(\mathcal{L}_q^{\mathbf{u}}(\mathbf{A}_T \mathbf{K}))) \rho_{\sqrt{\mathbf{S}_3}}(\mathcal{L} + \tilde{\mathbf{y}} - \mathbf{c})}{\rho_{\sqrt{\mathbf{S}_p}}(\mathbb{Z}^{nr}) \rho_{\sqrt{\mathbf{S}_L}}(\mathcal{L}_{q_L}^{\perp}(\tilde{\mathbf{G}}_L)) \rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R})^{dk_H}}$$

with $\alpha' = \alpha \frac{1+\varepsilon_L}{1-\varepsilon_L} (\frac{1+\varepsilon_H}{1-\varepsilon_H})^{dk_H}$. We now prove the following lemma to smooth out the coset $\mathcal{L} + \tilde{\mathbf{y}} - \mathbf{c}$ and conclude.

Lemma 3.3. *It holds that $\sqrt{\mathbf{S}_3} \geq \eta_{\varepsilon}(\mathcal{L})$. In particular, we have that for all $\mathbf{a} \in V$, $\rho_{\sqrt{\mathbf{S}_3}}(\mathcal{L} + \mathbf{a}) \in [(1-\varepsilon)/(1+\varepsilon), 1] \rho_{\sqrt{\mathbf{S}_3}}(\mathcal{L})$*

Proof (Lemma 3.3). For clarity, we set $N = nd(k_L + k_H)$ and $\eta = \eta_\varepsilon(\mathbb{Z}^N)$. We here follow the same proof methodology as [JS25b]. First, by condition (3), we have that $\mathbf{S}_L > \eta^2 \mathbf{I}_{ndk_L}$, while (4) gives $\mathbf{I}_{dk_H} \otimes \mathbf{S}_H > \eta^2 \mathbf{I}_{ndk_H}$. As a result, we obtain $\mathbf{S}_G > \eta^2 \mathbf{I}_N$ and thus that $\mathbf{M} := \eta^{-2} \mathbf{I}_N - \mathbf{S}_G^{-1}$ is positive definite. As such, the matrix appearing in condition (6), i.e., $\mathbf{M}^{-1}$, exists.

The latter condition can then be written as $\mathbf{S}_p \geq \tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T$. We now let $\delta > 0$ and $\mathbf{P}_\perp = \mathbf{I}_{nr} - \mathbf{P}$ be the orthogonal projector onto $V^\perp = \ker(\tilde{\mathbf{L}}^T)$. Because $\mathbf{P}_\perp \geq 0$, we then deduce

$$\mathbf{S}_{p,\delta} := \mathbf{S}_p + \delta \mathbf{P}_\perp \geq \tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T + \delta \mathbf{P}_\perp. \quad (2)$$

Also, we have $\mathbf{S}_{p,\delta} \geq \mathbf{S}_p > 0$ by $\delta \mathbf{P}_\perp \geq 0$ and condition (5). We now show $\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T + \delta \mathbf{P}_\perp$ is invertible by studying its kernel. Let $\mathbf{x} \in \ker(\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T + \delta \mathbf{P}_\perp)$ and decompose it uniquely onto the direct sum $V \oplus V^\perp$ as $\mathbf{x} = \mathbf{x}_V + \mathbf{x}_{V^\perp}$ with $\mathbf{x}_V \in V$ and $\mathbf{x}_{V^\perp} \in V^\perp$. We then have

$$\mathbf{0} = (\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T + \delta \mathbf{P}_\perp) \mathbf{x} = \tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T \mathbf{x}_V + \delta \mathbf{x}_{V^\perp}.$$

Indeed, $V^\perp = \ker(\tilde{\mathbf{L}}^T)$ which yields $\tilde{\mathbf{L}}^T \mathbf{x}_{V^\perp} = \mathbf{0}$, and we naturally have $\mathbf{P}_\perp \mathbf{x}_V = \mathbf{0}$ as well. Then, because $\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T \mathbf{x}_V \in V$ and $\delta \mathbf{x}_{V^\perp} \in V^\perp$, and that $V \oplus V^\perp = \mathbb{R}^{nr}$, we get $\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T \mathbf{x}_V = \mathbf{0}$ and $\mathbf{x}_{V^\perp} = \mathbf{0}$. We deduce that $\tilde{\mathbf{L}}^T \mathbf{x}_V = \mathbf{M} \tilde{\mathbf{L}}^+ (\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T \mathbf{x}_V) = \mathbf{0}$ and thus $\mathbf{x}_V \in \ker(\tilde{\mathbf{L}}^T) = V^\perp$. So $\mathbf{x}_V \in V \cap V^\perp = \{\mathbf{0}\}$, meaning $\mathbf{x}_V = \mathbf{0}$. As we directly have $\mathbf{x}_{V^\perp} = \mathbf{0}$ as well, it proves $\mathbf{x} = \mathbf{0}$. We hence get that $\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T + \delta \mathbf{P}_\perp$ is in $GL_{nr}(\mathbb{R})$.

As a result, we get $\mathbf{S}_{p,\delta}^{-1} \leq (\tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T + \delta \mathbf{P}_\perp)^{-1} =: \mathbf{N}$, which implies $\tilde{\mathbf{L}}^T \mathbf{S}_{p,\delta}^{-1} \tilde{\mathbf{L}} \leq \tilde{\mathbf{L}}^T \mathbf{N} \tilde{\mathbf{L}}$. Let us now prove $\tilde{\mathbf{L}}^T \mathbf{N} \tilde{\mathbf{L}} = \mathbf{M}$. For that, we first show that for all $\mathbf{v} \in V$, $\mathbf{N} \mathbf{v} = \tilde{\mathbf{L}}^{+T} \mathbf{M} \tilde{\mathbf{L}}^+ \mathbf{v}$ or equivalently that $\mathbf{N}^{-1} \tilde{\mathbf{L}}^{+T} \mathbf{M} \tilde{\mathbf{L}}^+ \mathbf{v} = \mathbf{v}$. Let $\mathbf{v} \in V$.

$$\begin{aligned} \text{(a)} \quad & \tilde{\mathbf{L}} \mathbf{M}^{-1} \tilde{\mathbf{L}}^T \tilde{\mathbf{L}}^{+T} \mathbf{M} \tilde{\mathbf{L}}^+ \mathbf{v} = \tilde{\mathbf{L}} \tilde{\mathbf{L}}^+ \mathbf{v} = \mathbf{P} \mathbf{v} = \mathbf{v} \\ \text{(b)} \quad & \delta \mathbf{P}_\perp \tilde{\mathbf{L}}^{+T} \mathbf{M} \tilde{\mathbf{L}}^+ \mathbf{v} = \mathbf{0} \quad \text{as } \tilde{\mathbf{L}}^{+T} \mathbf{M} \tilde{\mathbf{L}}^+ \mathbf{v} \in \text{Span}_{\mathbb{R}}(\tilde{\mathbf{L}}^{+T}) = V \end{aligned}$$

So $\tilde{\mathbf{L}}^{+T} \mathbf{M} \tilde{\mathbf{L}}^+ \mathbf{x} = \mathbf{N} \mathbf{x}$ for all $\mathbf{x}$ in V . Hence, for all $\mathbf{x} \in \mathbb{R}^N$, we get $\mathbf{N} \tilde{\mathbf{L}} \mathbf{x} = \tilde{\mathbf{L}}^{+T} \mathbf{M} \tilde{\mathbf{L}}^+ \tilde{\mathbf{L}} \mathbf{x} = \tilde{\mathbf{L}}^{+T} \mathbf{M} \mathbf{x}$, thus proving $\mathbf{N} \tilde{\mathbf{L}} = \tilde{\mathbf{L}}^{+T} \mathbf{M}$, and therefore $\tilde{\mathbf{L}}^T \mathbf{N} \tilde{\mathbf{L}} = \mathbf{M}$.

It then entails $\tilde{\mathbf{L}}^T \mathbf{S}_{p,\delta}^{-1} \tilde{\mathbf{L}} \leq \mathbf{M}$ which is equivalent to $\tilde{\mathbf{L}}^T \mathbf{S}_{p,\delta}^{-1} \tilde{\mathbf{L}} + \mathbf{S}_G^{-1} \leq \eta^{-2} \mathbf{I}_N$. Taking the limit when δ goes to 0 using the continuity of the inverse over $GL_{nr}(\mathbb{R})$, we get $\mathbf{S}' \leq \eta^{-2} \mathbf{I}_N$. It then implies $\mathbf{S}'^{-1} \geq \eta^2 \mathbf{I}_N$, and thus $\mathbf{S}_3 \geq \eta^2 \tilde{\mathbf{L}} \tilde{\mathbf{L}}^T$. This Loewner inequality is then sufficient to argue that $\rho_{\sqrt{\mathbf{S}_3}}(\mathcal{L}^*) \leq 1 + \varepsilon$. Indeed, it holds that

$$\begin{aligned} 1 + \varepsilon &= \sum_{\mathbf{y} \in \mathbb{Z}^N} \exp(-\pi \eta^2 \mathbf{y}^T \mathbf{y}) \\ &= \sum_{\mathbf{y} \in \mathbb{Z}^N} \exp(-\pi \eta^2 \mathbf{y}^T \tilde{\mathbf{L}}^+ \tilde{\mathbf{L}} \tilde{\mathbf{L}}^+ \mathbf{y}) \quad (\text{as } \tilde{\mathbf{L}}^+ \tilde{\mathbf{L}} = \mathbf{I}_N) \end{aligned}$$

$$\begin{aligned}
&= \sum_{\mathbf{y} \in \mathcal{L}'^*} \exp(-\pi \eta^2 \mathbf{y}^T \tilde{\mathbf{L}} \tilde{\mathbf{L}}^T \mathbf{y}) && (\text{as } \mathcal{L}'^* = \tilde{\mathbf{L}}^{+T} \mathbb{Z}^{ndk}) \\
&\geq \sum_{\mathbf{y} \in \mathcal{L}^*} \exp(-\pi \eta^2 \mathbf{y}^T \tilde{\mathbf{L}} \tilde{\mathbf{L}}^T \mathbf{y}) && (\text{as } \mathcal{L}^* \subseteq \mathcal{L}'^*) \\
&\geq \sum_{\mathbf{y} \in \mathcal{L}^*} \exp(-\pi \mathbf{y}^T \mathbf{S}_3 \mathbf{y})
\end{aligned}$$

Because $\mathcal{L}^* \subset V = \text{Span}_{\mathbb{R}}(\mathbf{S}_3) = \text{Span}_{\mathbb{R}}(\mathbf{S}_3^+)$, the last sum is equal to $\rho_{\sqrt{\mathbf{S}_3^+}}(\mathcal{L}^*)$. Lemma 2.5 then yields $\sqrt{\mathbf{S}_3} \geq \eta_\varepsilon(\mathcal{L})$. Using Lemma 2.6 and the fact that $V = \text{Span}_{\mathbb{R}}(\mathbf{S}_3)$ gives the second part of the statement. $\square$

Noting $\mathbf{c} \in \mathbf{x} + V = \tilde{\mathbf{y}} + V$, it holds that $\mathbf{a} = \tilde{\mathbf{y}} - \mathbf{c} \in V$. Applying Lemma 3.3 with $\mathbf{a}$ yields

$$\mathcal{P}'(\mathbf{y}) \in [a, b] \mathcal{D}_{\mathcal{L}_q^u(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}(\mathbf{y}) \frac{\rho_{\sqrt{\mathbf{S}}}(\tau(\mathcal{L}_q^u(\mathbf{A}_T \mathbf{K}))) \rho_{\sqrt{\mathbf{S}_3}}(\mathcal{L})}{\rho_{\sqrt{\mathbf{S}_p}}(\mathbb{Z}^{nr}) \rho_{\sqrt{\mathbf{S}_L}}(\mathcal{L}_{q_L}^\perp(\tilde{\mathbf{G}}_L)) \rho_{\sqrt{\mathbf{S}_H}}(b_H \mathcal{R})^{dk_H}},$$

with $a = \alpha^{-1}(1 + \varepsilon)/(1 - \varepsilon)$ and $b = \alpha'$. We then get that $\mathcal{P}'(\mathbf{y}) \in [a, b] \cdot c \cdot \mathcal{D}_{\mathcal{L}_q^u(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}(\mathbf{y})$. Summing over the entire support of $\mathcal{P}'$, which matches that of $\mathcal{D}_{\mathcal{L}_q^u(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}$, we get $ac \leq 1 \leq bc$ and thus $c \in [b^{-1}, a^{-1}]$. Hence, $\mathcal{P}' \approx_{b/a} \mathcal{D}_{\mathcal{L}_q^u(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}$, which then shows that $\mathcal{P} \approx_{b/a} \mathbf{K} \mathcal{D}_{\mathcal{L}_q^u(\mathbf{A}_T \mathbf{K}), \sqrt{\mathbf{S}}}$. Because b/a can be expressed as

$$\delta := \frac{b}{a} = \alpha_p^2 \cdot \alpha_L^2 \cdot \frac{1 + \varepsilon}{1 - \varepsilon} \cdot \frac{1 + \varepsilon_L}{1 - \varepsilon_L} \cdot \left(\frac{1 + \varepsilon_H}{1 - \varepsilon_H} \right)^{dk_H},$$

we get the desired claim. $\square$

3.3 KL Factorization and Spectral Analysis

We now instantiate **TSampler** with a concrete **KL** factorization and analyze conditions ① through ⑥ in this case. As the sampler is only proven secure with respect to $\mathbf{L}$, we should provide a factorization such that $\mathbf{K}$ is only composed of public elements, and $\mathbf{L}$ contains every element that needs to remain independent and hidden from the preimages. We can do so with a similar factorization as that of [JS25b] but which no longer features the tag thanks to our sampling method. This tag-independence allows for choosing tighter parameters, thus improving efficiency, and gives more freedom as we no longer require tags to be small. We factor it with

$$\mathbf{K} = \begin{bmatrix} \mathbf{G}_L & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_d & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{dk_H} \end{bmatrix}, \quad \text{and} \quad \mathbf{L} = \begin{bmatrix} \mathbf{I}_d & \mathbf{0} & \mathbf{R}_1 \\ \mathbf{0} & \mathbf{I}_{d(k_L-1)} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R}_2 \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{dk_H} \end{bmatrix}. \quad (3)$$

We can easily see by a block product that $\mathbf{KL}$ satisfies the equation of (1), and that $M_\tau(\mathbf{L})$ is full column rank, hence verifying (2). Next, we select $\mathbf{S}_L = s_L^2 \mathbf{I}_{ndk_L}$ and $\mathbf{S}_H = s_H^2 \mathbf{I}_n$, with $s_L = \sqrt{b_L^2 + 1} \eta_{\varepsilon_L}(\mathbb{Z}^{ndk_L})$ and $s_H = b_H \eta_{\varepsilon_H}(\mathbb{Z}^n)$, and adjust $\varepsilon_L, \varepsilon_H, \varepsilon$ so that we verify $s_L, s_H > \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)})$ thus satisfying conditions (3) and (4). We discuss this adjustment later on but the conclusion is that any $\varepsilon \in (0, 1)$ fits due to the fact that $s_L, s_H \gg \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)})$ for typical parameters n, d, b_x, k_x .

We now look at (5) and (6). We have $\mathbf{S}_G = \text{diag}(s_L^2 \mathbf{I}_{ndk_L}, s_H^2 \mathbf{I}_{ndk_H})$ with possibly different s_L, s_H . Noting $\eta = \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)})$, we can express

$$\mathbf{S}_G + \left(\frac{1}{\eta^2} \mathbf{I} - \mathbf{S}_G^{-1} \right)^{-1} = \text{diag} \left(\frac{s_L^4}{s_L^2 - \eta^2} \mathbf{I}_{ndk_L}, \frac{s_H^4}{s_H^2 - \eta^2} \mathbf{I}_{ndk_H} \right)$$

We can therefore study both conditions at once as in [JS25b] by studying the spectral properties of $\mathbf{S}' = \mathbf{S} - \tilde{\mathbf{L}} \text{diag}(a_L \mathbf{I}_{ndk_L}, a_H \mathbf{I}_{ndk_H}) \tilde{\mathbf{L}}^T$. In particular, $\mathbf{S}' > 0$ with $a_x = s_x^2$ is then equivalent to (5), while $\mathbf{S}' \geq 0$ with $a_x = s_x^4/(s_x^2 - \eta^2)$ is equivalent to (6). We thus prove the following result.

Theorem 3.2. *Let N, N_L, N_H be in $\mathbb{N}^\times$ with $N_H \geq N$, and $a_L, a_H > 0$. Let $\mathbf{F}_1, \mathbf{F}_2 \in \mathbb{R}^{N \times N_H}$. We define $\mathbf{L} \in \mathbb{R}^{(N+N_L+N_H) \times (N_L+N_H)}$ as*

$$\mathbf{L} = \begin{bmatrix} \mathbf{I}_N & \mathbf{0} & \mathbf{F}_1 \\ \mathbf{0} & \mathbf{I}_{N_L-N} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{F}_2 \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{N_H} \end{bmatrix},$$

and $\mathbf{S} = \text{diag}(s_1^2 \mathbf{I}_N, s_2^2 \mathbf{I}_{N_L-N}, s_3^2 \mathbf{I}_N, s_4^2 \mathbf{I}_{N_H})$. If $\mathbf{F}_2 = \mathbf{U} \mathbf{D}'_2 \mathbf{V}^T$ is a singular value decomposition of $\mathbf{F}_2$, it holds that $\mathbf{S} > \mathbf{L} \text{diag}(a_L \mathbf{I}_{N_L}, a_H \mathbf{I}_{N_H}) \mathbf{L}^T$ if and only if

$$\begin{aligned} s_1^2 &> a_L + a_H \cdot \|\mathbf{F}_1 \mathbf{V} \mathbf{D} \mathbf{V}^T \mathbf{F}_1\|_2 & s_2^2 &> a_L \\ s_3^2 &> a_H s_4^2 / (s_4^2 - a_H) \cdot \|\mathbf{F}_2\|_2^2 & s_4^2 &> a_H, \end{aligned}$$

where

$$\mathbf{D} = \text{diag} \left(\text{diag} \left(\frac{s_3^2 s_4^2}{s_4^2 (s_3^2 - a_H s_i(\mathbf{F}_2)^2) - s_3^2 a_H} \right), \frac{s_4^2}{s_4^2 - a_H} \mathbf{I}_{N_H-N} \right).$$

The condition on s_1 is implied by

$$s_1^2 > a_L + a_H \frac{s_3^2 s_4^2}{s_4^2 (s_3^2 - a_H \|\mathbf{F}_2\|_2^2) - s_3^2 a_H} \|\mathbf{F}_1\|_2^2.$$

We have the same result for $\mathbf{S} \geq \mathbf{L} \text{diag}(a_L \mathbf{I}_{N_L}, a_H \mathbf{I}_{N_H}) \mathbf{L}^T$ but with large inequalities for s_1 and s_2 .

Proof. We follow the proof methodology from [JS25b, Lem. 4.1]. As it contains many similarities with our structure, we may refer to their proof directly in order to avoid redundancy and improve readability of our proof. We start by defining $\mathbf{S}_G = \text{diag}(a_L \mathbf{I}_{N_L}, a_H \mathbf{I}_{N_H})$ and $N'_L = N_L - N$ for clarity. Computing $\mathbf{S}' = \mathbf{S} - \mathbf{L} \mathbf{S}_G \mathbf{L}^T$, we get $\mathbf{S}'$ equals

$$\begin{array}{c}
\mathbf{A} \qquad \qquad \qquad \mathbf{B} \\
\left[\begin{array}{cc|cc}
(s_1^2 - a_L)\mathbf{I}_N - a_H\mathbf{F}_1\mathbf{F}_1^T & \mathbf{0} & -a_H\mathbf{F}_1\mathbf{F}_2^T & -a_H\mathbf{F}_1 \\
\mathbf{0} & (s_2^2 - a_L)\mathbf{I}_{N'_L} & \mathbf{0} & \mathbf{0} \\
\hline
-a_H\mathbf{F}_2\mathbf{F}_1^T & \mathbf{0} & s_3^2\mathbf{I}_N - a_H\mathbf{F}_2\mathbf{F}_2^T & -a_H\mathbf{F}_2 \\
-a_H\mathbf{F}_1^T & \mathbf{0} & -a_H\mathbf{F}_2^T & (s_4^2 - a_H)\mathbf{I}_{N_H}
\end{array} \right] \\
\mathbf{B}^T \qquad \qquad \qquad \mathbf{C}
\end{array}$$

The Schur complement characterization then gives that $\mathbf{S}' > 0$ (resp. ≥ 0) if and only if $\mathbf{C} > 0$ and $\mathbf{S}'/\mathbf{C} = \mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^T > 0$ (resp. ≥ 0). Because $N_H \geq N$, there exists $\mathbf{U} \in O_N(\mathbb{R})$, $\mathbf{V} \in O_{N_H}(\mathbb{R})$ and $\mathbf{D}'_2 = [\mathbf{D}_2|\mathbf{0}]$ with $\mathbf{D}_2 = \text{diag}(s_1(\mathbf{F}_2), \dots, s_N(\mathbf{F}_2))$ that define a singular value decomposition of $\mathbf{F}_2$ with $s_1(\mathbf{F}_2) \geq \dots \geq s_N(\mathbf{F}_2) \geq 0$ and $\mathbf{F}_2 = \mathbf{U}\mathbf{D}'_2\mathbf{V}^T$. The condition for $\mathbf{C} > 0$ and the subsequent expression of $\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^T$ rigorously follows from the proof of [JS25b, Lem. 4.1]. In particular, we have

$$\mathbf{C} > 0 \iff s_4^2 > a_H \quad \text{and} \quad s_3^2 > \frac{s_4^2 a_H}{s_4^2 - a_H} \|\mathbf{F}_2\|_2^2,$$

and $\mathbf{B}\mathbf{C}^{-1}\mathbf{B} = \text{diag}(\mathbf{Y}, \mathbf{0})$, with

$$\begin{aligned}
\mathbf{Y} &= a_H\mathbf{F}_1\mathbf{V}\text{diag}\left(\mathbf{D}', \frac{a_H}{s_4^2 - a_H}\mathbf{I}_{N_H-N}\right)\mathbf{V}^T\mathbf{F}_1^T, \\
\text{with } \mathbf{D}' &= \text{diag}\left(\frac{s_3^2 a_H + s_4^2 a_H s_i(\mathbf{F}_2)^2}{s_4^2(s_3^2 - a_H s_i(\mathbf{F}_2)^2) - s_3^2 a_H}\right)
\end{aligned}$$

We then have $\mathbf{S}'/\mathbf{C} = \text{diag}((s_1^2 - a_L)\mathbf{I}_N - a_H\mathbf{F}_1\mathbf{F}_1^T - \mathbf{Y}, (s_2^2 - a_L)\mathbf{I}_{N'_L})$. We can express the first diagonal block as $(s_1^2 - a_L)\mathbf{I}_N - a_H\mathbf{F}_1\mathbf{V}\mathbf{D}\mathbf{V}^T\mathbf{F}_1^T$ where $\mathbf{D} = \mathbf{I}_{N_H} + \text{diag}(\mathbf{D}', a_H/(s_4^2 - a_H) \cdot \mathbf{I}_{N_H-N})$. The expression of $\mathbf{D}$ then simplifies to

$$\mathbf{D} = \text{diag}\left(\text{diag}\left(\frac{s_3^2 s_4^2}{s_4^2(s_3^2 - a_H s_i(\mathbf{F}_2)^2) - s_3^2 a_H}\right), \frac{s_4^2}{s_4^2 - a_H}\mathbf{I}_{N_H-N}\right)$$

We then have $\mathbf{S}'/\mathbf{C} > 0$ if and only if $s_2^2 > a_L$ and $s_1^2 - a_L > a_H \|\mathbf{F}_1\mathbf{V}\mathbf{D}\mathbf{V}^T\mathbf{F}_1^T\|_2$ as desired. The same holds for the condition $\mathbf{S}'/\mathbf{C} \geq 0$ with large inequalities on s_1^2, s_2^2 . The sufficient condition on s_1 then directly follows from the submultiplicativity of $\|\cdot\|_2$, the fact that $\mathbf{V}$ is orthogonal, and the fact that the maximal entry of $\mathbf{D}$ is the top left entry (i.e., $i = 1$). $\square$

The conditions on s_1, s_2, s_3, s_4 are interdependent which makes it unclear how to select them while satisfying the constraints. We now give a sufficient methodology to select them with respect to s_L, s_H and η . For that we give the following corollary, which consists in applying Theorem 3.2 with $a_x = s_x^2$ and $a_x = s_x^4/(s_x^2 - \eta^2)$.

Corollary 3.1. Let d, k_L, k_H be in $\mathbb{N}^\times$, $s_L, s_H > 0$ and $\varepsilon \in (0, 1)$. We assume $s_L, s_H > \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)}) =: \eta$. Let $\mathbf{R}_1, \mathbf{R}_2 \in \mathcal{R}^{d \times dk_H}$. Let

$$\begin{aligned} s_1 &= \sqrt{\frac{s_L^4}{s_L^2 - \eta^2} + 3 \frac{s_H^4}{s_H^2 - \eta^2} \|\mathbf{R}_1\|_2^2} & s_2 &= \frac{s_L^2}{\sqrt{s_L^2 - \eta^2}} \\ s_3 &= \sqrt{3} \frac{s_H^2}{\sqrt{s_H^2 - \eta^2}} \|\mathbf{R}_2\|_2^2 & s_4 &= \sqrt{3} \frac{s_H^2}{\sqrt{s_H^2 - \eta^2}} \end{aligned}$$

Then, for $\mathbf{S} = \text{diag}(s_1^2 \mathbf{I}_{nd}, s_2^2 \mathbf{I}_{nd(k_L-1)}, s_3^2 \mathbf{I}_{nd}, s_4^2 \mathbf{I}_{ndk_H})$ and $\mathbf{L}$ as in Equation (3), it holds that $\mathbf{S} > M_\tau(\mathbf{L}) \text{diag}(s_L^2 \mathbf{I}_{ndk_L}, s_H^2 \mathbf{I}_{ndk_H}) M_\tau(\mathbf{L})^T$, and also that $\mathbf{S} \geq M_\tau(\mathbf{L}) \text{diag}(\frac{s_L^4}{s_L^2 - \eta^2} \mathbf{I}_{ndk_L}, \frac{s_H^4}{s_H^2 - \eta^2} \mathbf{I}_{ndk_H}) M_\tau(\mathbf{L})^T$.

Proof. We first apply Theorem 3.2 with strict inequalities and $a_x = s_x^2$. We apply it using the embedding of $\mathbf{L}$, i.e., with $\mathbf{F}_i = M_\tau(\mathbf{R}_i)$, $N = nd$, $N_L = ndk_L$ and $N_H = ndk_H$. We get the (sufficient) conditions below

$$\begin{aligned} s_1^2 &> s_L^2 + s_H^2 \frac{s_3^2 s_4^2}{s_4^2 (s_3^2 - s_H^2 \|\mathbf{R}_2\|_2^2) - s_3^2 s_H^2} \|\mathbf{R}_1\|_2^2 & s_2^2 &> s_L^2 \\ s_3^2 &> s_H^2 \frac{s_4^2}{s_4^2 - s_H^2} \|\mathbf{R}_2\|_2^2 & s_4^2 &> s_H^2, \end{aligned}$$

We then apply the theorem with large inequalities and $a_x = s_x^4 / (s_x^2 - \eta^2)$, which gives the conditions

$$\begin{aligned} s_1^2 &\geq a_L + a_H \frac{s_3^2 s_4^2}{s_4^2 (s_3^2 - a_H \|\mathbf{R}_2\|_2^2) - s_3^2 a_H} \|\mathbf{R}_1\|_2^2 & s_2^2 &\geq a_L \\ s_3^2 &> a_H \frac{s_4^2}{s_4^2 - a_H} \|\mathbf{R}_2\|_2^2 & s_4^2 &> a_H, \end{aligned}$$

Because $a_x > s_x^2$, the latter conditions imply the former. We then let $c_3, c_4 > 1$ be free variables and select

$$\begin{aligned} s_1^2 &= a_L + a_H \frac{s_3^2 s_4^2}{s_4^2 (s_3^2 - a_H \|\mathbf{R}_2\|_2^2) - s_3^2 a_H} \|\mathbf{R}_1\|_2^2 & s_2^2 &= a_L \\ s_3^2 &= c_3^2 a_H \frac{s_4^2}{s_4^2 - a_H} \|\mathbf{R}_2\|_2^2 & s_4^2 &= c_4^2 a_H, \end{aligned}$$

They verify the conditions and yield the result for any such pair (c_3, c_4) . Let us now simplify these expressions. We have

$$\begin{aligned} \frac{s_4^2}{s_4^2 - a_H} &= \frac{c_4^2}{c_4^2 - 1}, \\ \text{and } \frac{s_3^2 s_4^2}{s_4^2 (s_3^2 - a_H \|\mathbf{R}_2\|_2^2) - s_3^2 a_H} &= \frac{c_3^2 c_4^2}{(c_3^2 - 1)(c_4^2 - 1)} \end{aligned}$$

Hence $s_3^2 = c_3^2 c_4^2 / (c_4^2 - 1) \cdot a_H \|\mathbf{R}_2\|_2^2$, and $s_1^2 = a_L + c_3^2 c_4^2 / ((c_3^2 - 1)(c_4^2 - 1)) \cdot a_H \|\mathbf{R}_1\|_2^2$. To balance the different slacks, we decide to take c_3, c_4 so that $c_4^2 = c_3^2 c_4^2 / (c_4^2 - 1) = c_3^2 c_4^2 / ((c_3^2 - 1)(c_4^2 - 1))$, which gives $c_3 = \sqrt{2}$ and $c_4 = \sqrt{3}$ which are both indeed larger than 1, and giving the formulae from the corollary statement. $\square$

After projecting with $\mathbf{K}$, the output of **TSampler** is close² to $\mathcal{D}_{\mathcal{L}_q^u(\mathbf{A}_T), \sqrt{S'}}$ where $\mathbf{S}' = M_\tau(\mathbf{K})\mathbf{S}M_\tau(\mathbf{K})^T = \text{diag}(s_{1,1}^2 \mathbf{I}_{nd}, s_3^2 \mathbf{I}_{nd}, s_4^2 \mathbf{I}_{ndk_H})$ with

$$s_{1,1} = \sqrt{\frac{s_L^4}{s_L^2 - \eta^2} \cdot \frac{b_L^{2k_L} - 1}{b_L^2 - 1} + 3 \frac{s_H^4}{s_H^2 - \eta^2} \|\mathbf{R}_1\|_2^2}.$$

From now on, we use the latter factorization $\mathbf{K}, \mathbf{L}$, and select parameters according to Corollary 3.1. In that case, we specify the parameters of **TSampler** differently, and use the notation $\text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u}, \mathbf{t}, s_1, s_2, s_3, s_4, s_L, s_H)$.

3.4 Compact Trapdoor Switching for **TSampler**

We now formalize the trapdoor switching result that procures a certain compactness advantage within the security proofs of advanced authentication mechanisms. It follows the idea introduced in [AGJ⁺24] but in a much more compact way. We indeed tailored our sampler to enable a more efficient trapdoor switching strategy, which allows for changing the public key $\mathbf{R}_1 + \mathbf{A}\mathbf{R}_2$ into $\mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 + q_L t^+ \mathbf{G}_H$ column by column, then requiring only a single column for the partial trapdoor slot. We can abstract this argument and essentially show that we can sample preimages for the matrix $\overline{\mathbf{A}}$ defined by

$$[\mathbf{I}_d | \mathbf{A} | q_L \mathbf{G}_{H,\mathbf{t}} - (\mathbf{R}_1 + \mathbf{A}\mathbf{R}_2) | q_L t'_j b_H^{[j/d]} \mathbf{e}_{(j \bmod d)} - (\mathbf{r}'_{1,j} + \mathbf{A}\mathbf{r}'_{2,j})] \bmod q\mathcal{R}$$

in two different ways: (1) using **TSampler** with trapdoor $\mathbf{R}_1, \mathbf{R}_2$ and tag $\mathbf{t}$, or (2) using **TSampler** with trapdoor $\mathbf{R}'_1, \mathbf{R}'_2$ and tag $\mathbf{t}'$ where $\mathbf{R}'_i$ is obtained from $\mathbf{R}_i$ by changing the j -th column with $\mathbf{r}'_{i,j}$, and $\mathbf{t}'$ from $\mathbf{t}$ by changing the j -th entry with t'_j . This is sufficient to perform this public key change in the security proof unbeknownst to the adversary. Let us now give the formal trapdoor switching lemma.

Lemma 3.4. *Let d, q_L, b_L, k_L, k_H be in $\mathbb{N}^\times$ and such that $k_L = \lceil \log_{b_L} q_L \rceil$, and $\varepsilon_L, \varepsilon_H, \varepsilon$ in $(0, 1)$. Then, define $q_H = b_H^{k_H}$ and $q = q_L q_H$, as well as $\mathbf{G}_L = [1 | b_L | \dots | b_L^{k_L-1}] \otimes \mathbf{I}_d$. Let $\mathbf{A} \in \mathcal{R}_q^{d \times d}$, $\mathbf{u} \in \mathcal{R}_q^d$, $\mathbf{R}_1, \mathbf{R}_2 \in \mathcal{R}^{d \times dk_H}$, and $\mathbf{t} = [t_i]_{i \in [0, dk_H-1]} \in (\mathcal{R}_{b_H}^\times)^{dk_H}$ and define $\mathbf{G}_{H,\mathbf{t}} = [t_i b_H^{[i/d]} \mathbf{e}_{(i \bmod d)}]_{i \in [0, dk_H-1]}$. Let $j \in [0, dk_H-1]$ and also $\mathbf{r}'_{1,j}, \mathbf{r}'_{2,j} \in \mathcal{R}^d$, and $t'_j \in \mathcal{R}_{b_H}^\times$. We define*

$$\begin{aligned} \mathbf{R}'_i &= [\mathbf{r}_{i,0} | \dots | \mathbf{r}_{i,j-1} | \mathbf{r}'_{i,j} | \mathbf{r}_{i,j+1} | \dots | \mathbf{r}_{i,dk_H-1}] \\ \mathbf{t}' &= (t_0, \dots, t_{j-1}, t'_j, t_{j+1}, \dots, t_{dk_H-1}), \end{aligned}$$

² As noted in [JS25b], there is no need here for “cryptographic closeness” as $\mathbf{K}$ is public. If one desires stronger guarantees, one could use [GMPW20] for this argument.

for $i \in \{1, 2\}$ and where the $\mathbf{r}_{i,\ell}$ are the columns of $\mathbf{R}_i$. Let $s_L > 0$ be such that $s_L \geq \sqrt{b_L^2 + 1}\eta_{\varepsilon_L}(\mathbb{Z}^{ndk_L})$ and $s_L > \eta_{\varepsilon}(\mathbb{Z}^{nd(k_L+k_H)}) =: \eta$. Let $s_H > 0$ be such that $s_H \geq b_H\eta_{\varepsilon_H}(\mathbb{Z}^n)$ and $s_H > \eta$. Then, let

$$\begin{aligned} s_1 &= \sqrt{\frac{s_L^4}{s_L^2 - \eta^2} + 3\frac{s_H^4}{s_H^2 - \eta^2}B_{R,1}^2} & s_2 &= \frac{s_L^2}{\sqrt{s_L^2 - \eta^2}} \\ s_3 &= \sqrt{3}\frac{s_H^2}{\sqrt{s_H^2 - \eta^2}}B_{R,2} & s_4 &= \sqrt{3}\frac{s_H^2}{\sqrt{s_H^2 - \eta^2}} \end{aligned}$$

where $B_{R,i} \geq \max(\|\mathbf{R}_i\|_2, \|\mathbf{R}'_i\|_2)$. We call $\mathbf{g}_{H,j} = b_H^{\lfloor j/d \rfloor} \mathbf{e}_{(j \bmod d)}$ where $\mathbf{e}_i$ is the zero vector with a single 1 at the i -th entry. We then define $\bar{\mathbf{A}} = [\mathbf{I}_d | \mathbf{A}]_{qL} \mathbf{G}_{H,\mathbf{t}} - (\mathbf{R}_1 + \mathbf{A}\mathbf{R}_2)_{qL} \mathbf{t}'_j \mathbf{g}_{H,j} - (\mathbf{r}'_{1,j} + \mathbf{A}\mathbf{r}'_{2,j}) \bmod q\mathcal{R}$, as well as the following distributions.

$$\begin{aligned} \mathcal{P}_1 & \quad \text{Sample } v_3 \leftarrow \mathcal{D}_{\mathcal{R},s_4}, \text{ compute } \mathbf{u}' = \mathbf{u} - (q_L \mathbf{t}'_j \mathbf{g}_{H,j} - (\mathbf{r}'_{1,j} + \mathbf{A}\mathbf{r}'_{2,j}))v_3 \bmod q\mathcal{R}, \\ & \quad \text{sample } \mathbf{v} \leftarrow \text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u}', \mathbf{t}, s_1, s_2, s_3, s_4, s_L, s_H), \text{ and output } (\mathbf{v}, v_3). \\ \mathcal{P}_2 & \quad \text{Sample } v_{2,j} \leftarrow \mathcal{D}_{\mathcal{R},s_4}, \text{ compute } \mathbf{u}' = \mathbf{u} - (q_L \mathbf{t}_j \mathbf{g}_{H,j} - (\mathbf{r}_{1,j} + \mathbf{A}\mathbf{r}_{2,j}))v_{2,j} \bmod \\ & \quad q\mathcal{R}, \text{ sample } \mathbf{w} \leftarrow \text{TSampler}(\mathbf{R}'_1, \mathbf{R}'_2, \mathbf{A}, \mathbf{u}', \mathbf{t}', s_1, s_2, s_3, s_4, s_L, s_H). \text{ Parse } \mathbf{w} \\ & \quad \text{as } (\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{w}_2) \text{ with } \mathbf{v}_{1,i} \in \mathcal{R}^d \text{ and } \mathbf{w}_2 \in \mathcal{R}^{dk_H}. \text{ Parse } \mathbf{w}_2 \text{ as} \\ & \quad (v_{2,0}, \dots, v_{2,j-1}, v_3, v_{2,j+1}, \dots, v_{2,dk_H-1}) \text{ and compute } \mathbf{v}_2 = [v_{2,i}]_{i \in [0, dk_H-1]}. \\ & \quad \text{Output } (\mathbf{v} = (\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2), v_3). \end{aligned}$$

It holds that $\mathcal{P}_1$ and $\mathcal{P}_2$ have the same support and that $\mathcal{P}_1 \approx_{\delta^2} \mathcal{P}_2$ and $\mathcal{P}_2 \approx_{\delta^2} \mathcal{P}_1$ where δ is defined in Theorem 3.1.

Proof. We denote by $\mathcal{P}'_1$ the distribution of $(\mathbf{v}', v_3)$ where $\mathbf{v}'$ is the random vector in step 13 of Algorithm 3.1. We then have $\mathbf{v} = \mathbf{K}\mathbf{v}'$ with $\mathbf{K} = \text{diag}(\mathbf{G}_L, \mathbf{I}_d, \mathbf{I}_{dk_H})$. For clarity, we denote by $\mathcal{P}'^{\mathbf{u}'}$ the distribution of $\mathbf{v}'$ where the sampler is called for a specific syndrome $\mathbf{u}'$. For any given $\mathbf{u}'$, Theorem 3.1 gives that $\mathcal{P}'^{\mathbf{u}'}$ is δ -close to $\mathcal{D}_{\mathcal{L}_q^{\mathbf{u}'}(\mathbf{A}_t\mathbf{K})}(\mathbf{u}', \sqrt{\mathbf{S}})$ where $\mathbf{S} = \text{diag}(s_1^2 \mathbf{I}_{nd}, s_2^2 \mathbf{I}_{nd(k_L-1)}, s_3^2 \mathbf{I}_{nd}, s_4^2 \mathbf{I}_{ndk_H})$. The latter is exactly equal to the conditional probability distribution “ $\mathcal{D}_{\mathcal{R}^{d(1+k_L+k_H)}, \sqrt{\mathbf{S}}}$ conditioned on $\mathbf{A}_t \mathbf{K} \mathbf{v}' = \mathbf{u}' \bmod q\mathcal{R}$ ”. Hence, conditioned on $v_3 = \bar{v}_3$, it holds that $\mathbf{v}'$ is δ -close to “ $\mathcal{D}_{\mathcal{R}^{d(1+k_L+k_H)}, \sqrt{\mathbf{S}'}}$ conditioned on $\bar{\mathbf{A}}\mathbf{K}'[\mathbf{v}'^T | \bar{v}_3]^T = \mathbf{u} \bmod q\mathcal{R}$ ”, where we simply reorganized based on the definition of $\mathbf{u}'$ and defined $\mathbf{K}' = \text{diag}(\mathbf{K}, 1)$. In other words, we have

$$\begin{aligned} \mathcal{P}'_1(\bar{\mathbf{v}}', \bar{v}_3) &= \mathcal{D}_{\mathcal{R},s_4}(\bar{v}_3) \cdot \mathcal{P}'_2^{\mathbf{u}'(\bar{v}_3)}(\bar{\mathbf{v}}') \\ &\in [\tfrac{1}{\delta}, \delta] \mathcal{D}_{\mathcal{R},s_4}(\bar{v}_3) \left(\mathcal{D}_{\mathcal{R}^{d(1+k_L+k_H)}, \sqrt{\mathbf{S}}} \left| \bar{\mathbf{A}}\mathbf{K}' \begin{bmatrix} \mathbf{v}' \\ \bar{v}_3 \end{bmatrix} = \mathbf{u} \bmod q\mathcal{R} \right) (\bar{\mathbf{v}}') \right. \\ &\in [\tfrac{1}{\delta}, \delta] \left(\mathcal{D}_{\mathcal{R}^{d(1+k_L+k_H)+1}, \sqrt{\mathbf{S}'}} \left| \bar{\mathbf{A}}\mathbf{K}' \begin{bmatrix} \mathbf{v}' \\ v_3 \end{bmatrix} = \mathbf{u} \bmod q\mathcal{R} \right) (\bar{\mathbf{v}}', \bar{v}_3) \right. \\ &\in [\tfrac{1}{\delta}, \delta] \mathcal{D}_{\mathcal{L}_q^{\mathbf{u}}(\bar{\mathbf{A}}\mathbf{K}'), \sqrt{\mathbf{S}'}}(\bar{\mathbf{v}}', \bar{v}_3) \end{aligned}$$

where $\mathbf{S}' = \text{diag}(\mathbf{S}, s_4^2 \mathbf{I}_n)$. We then have $\mathcal{P}_1 = \mathbf{K}'\mathcal{P}'_1$ and therefore $\mathcal{P}_1 \approx_{\delta} \mathbf{K}'\mathcal{D}_{\mathcal{L}_q^{\mathbf{u}}(\bar{\mathbf{A}}\mathbf{K}'), \sqrt{\mathbf{S}'}}$. In particular, the support of $\mathcal{P}_1$ is $\mathbf{K}'\mathcal{L}_q^{\mathbf{u}}(\bar{\mathbf{A}}\mathbf{K}') \subseteq \mathcal{L}_q^{\mathbf{u}}(\mathbf{A})$.

Using the same reasoning on $\mathcal{P}_2$ but by conditioning on $v_{2,j} = \overline{v_{2,j}}$ and re-arranging the equation $\mathbf{A}_{\mathbf{t}'} \mathbf{w} = \mathbf{u}' \bmod q\mathcal{R}$ gives

$$\mathcal{P}_2 \approx_\delta \mathbf{K}' \mathcal{D}_{\mathcal{L}_q^u(\overline{\mathbf{A}}\mathbf{K}'), \sqrt{S'}} \quad \text{and} \quad \text{Supp}(\mathcal{P}_2) = \mathbf{K}' \mathcal{L}_q^u(\overline{\mathbf{A}}\mathbf{K}') \subseteq \mathcal{L}_q^u(\overline{\mathbf{A}}).$$

As they are both δ -close to the same ideal distribution, it then directly yields that $\mathcal{P}_1 \approx_{\delta^2} \mathcal{P}_2$ and $\mathcal{P}_2 \approx_{\delta^2} \mathcal{P}_1$ as desired. $\square$

3.5 Implementation Considerations and Concrete Subsamplers

We now discuss certain implementation details related to our sampler, in particular on the instantiation of distributions $\mathcal{P}_p, \mathcal{P}_L$ with concrete subsamplers, as well as precomputations. For the rest of this section, we assume we have access to a perfect integer sampler, i.e., we can sample $\mathcal{D}_{\mathbb{Z}, s, c}$ for any $s > 0, c \in \mathbb{R}$ exactly.

3.5.1 Gadget Sampling. One of the perks of our sampler is that it decouples the sampling of $\mathbf{z}_L$ and $\mathbf{z}_H$, allowing us to choose independent gadget parameters b_x, k_x, q_x . In particular, as there is no constraint on b_L and q_L , we can choose the latter to be smooth as well by selecting $q_L = b_L^{k_L}$. With this specific form, we can actually update the sampling of $\mathbf{z}_L$ and mimic what we presented for $\mathbf{z}_H$, except that all the tags are equal to 1 for the $\mathbf{G}_L$ part.

More precisely, we would replace steps 3 to 5 from Algorithm 3.1 with the following steps. For each i in $[0, d-1]$, we define $\ell_{i,0} = [\mathbf{w}]_i$. Then, we compute $c_{i,0} = \ell_{i,0} \bmod b_L \mathcal{R}$, sample $[\mathbf{z}_L]_{i+0d}$ from $\mathcal{D}_{b_L \mathcal{R} + c_{i,0}, s_L}$ and update ℓ for the next step as $\ell_{i,1} = (\ell_{i,0} - [\mathbf{z}_L]_{i+0d})/b_L \in \mathcal{R}$. We then keep going so that at step j we compute $c_{i,j} = \ell_{i,j} \bmod b_L \mathcal{R}$, we sample $[\mathbf{z}_L]_{i+jd}$ from $\mathcal{D}_{b_L \mathcal{R} + c_{i,j}, s_L}$ and update with $\ell_{i,j+1} = (\ell_{i,j} - [\mathbf{z}_L]_{i+jd})/b_L \in \mathcal{R}$. Just like for $\mathbf{z}_H$, in the end, we have

$$[\mathbf{w}_i] = \sum_{j=0}^{k_L-1} b_L^j [\mathbf{z}_L]_{i+jd} + b_L^{k_L} \ell_{i,k_L}$$

We can then change step 7 of Algorithm 3.1 so that $h_{i,0} = \ell_{i,k_L}$, or merge the i loops directly. Overall, we only rely on an integer sampler for all the entries of $\mathbf{z}_L$, which yields an exact distribution for $\mathcal{P}_L$ if we assume an exact integer sampler, hence $\alpha_L = 1$. The small subtlety is the smoothing argument for $\mathbf{z}_L$ from the proof of Theorem 3.1 which is slightly different. It can be handled similarly as the one for $\mathbf{z}_H$, but then slightly changes condition (3), as well as the overall loss δ . More precisely, condition (3) becomes

$$s_L \geq b_L \eta_{\varepsilon_L}(\mathbb{Z}^n), \quad \text{and} \quad s_L > \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)})$$

and the final marginal distance becomes

$$\delta = \alpha_p^2 \cdot \frac{1+\varepsilon}{1-\varepsilon} \cdot \left(\frac{1+\varepsilon_L}{1-\varepsilon_L} \right)^{dk_L} \cdot \left(\frac{1+\varepsilon_H}{1-\varepsilon_H} \right)^{dk_H}. \quad (4)$$

The sampler would then be modified as follows. Here we use the parameters $s_L, s_H, s_1, \dots, s_4$ as set in Section 3.3, and in particular the **KL** factorization of Equation (3), but it could be extended to the general factorization and covariances as in Section 3.1.

Algorithm 3.2: TSampler($\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u}, \mathbf{t}, s_1, s_2, s_3, s_4, s_L, s_H$)

1. $\mathbf{p} \leftarrow \mathcal{D}_{\mathcal{R}^{d(k_L+k_H)}, \sqrt{s_p}}$ with $\mathbf{S}_p = \mathbf{S} - M_\tau(\mathbf{L})\text{diag}(s_L^2 \mathbf{I}_{ndk_L}, s_H^2 \mathbf{I}_{ndk_H})M_\tau(\mathbf{L})^T$.
2. $\mathbf{w} \leftarrow \mathbf{u} - [\mathbf{I}_d | \mathbf{A}] q_L \mathbf{G}_{H,t} - (\mathbf{R}_1 + \mathbf{A}\mathbf{R}_2) \mathbf{K}\mathbf{p} \bmod q\mathcal{R}$.
3. **For** $i = 0, \dots, d-1$ **do**
4. $\ell_{i,0} \leftarrow [\mathbf{w}]_i$.
5. **For** $j = 0, \dots, k_L - 1$ **do**
6. $c_{i,j}^\ell \leftarrow \ell_{i,j} \bmod b_L \mathcal{R}$.
7. $[\mathbf{z}_L]_{i+jd} \leftarrow \mathcal{D}_{b_L \mathcal{R} + c_{i,j}^\ell, s_L}$.
8. $\ell_{i,j+1} \leftarrow (\ell_{i,j} - [\mathbf{z}_L]_{i+jd})/b_L \in \mathcal{R}$.
9. $h_{i,0} \leftarrow \ell_{i,k_L}$.
10. **For** $j = 0, \dots, k_H - 1$ **do**
11. $c_{i,j}^h \leftarrow t_{i+jd}^{-1} \cdot h_{i,j} \bmod b_H \mathcal{R}$.
12. $[\mathbf{z}_H]_{i+jd} \leftarrow \mathcal{D}_{b_H \mathcal{R} + c_{i,j}^h, s_H}$.
13. $h_{i,j+1} \leftarrow (h_{i,j} - t_{i+jd}[\mathbf{z}_H]_{i+jd})/b_H \in \mathcal{R}$.
14. $\mathbf{z} \leftarrow [\mathbf{z}_L^T | \mathbf{z}_H^T]^T$.
15. $\mathbf{v}' \leftarrow \mathbf{p} + \mathbf{L}\mathbf{z}$.
16. $\mathbf{v} \leftarrow \mathbf{K}\mathbf{v}'$.

Output: $\mathbf{v} \in \mathcal{L}_q^u(\mathbf{A}_T)$

While the choice of $q_L = b_L^{k_L}$ is not a problem in all the applications we consider in Section 4, one might need to use a different q_L . In such a case, it is still possible to rely on Klein's sampler [Kle00] for the gadget sampling of $\mathbf{z}_L$, resulting in an inexact distribution $\mathcal{P}_L$ that would affect the overall loss δ . Fortunately, Klein's sampler has been thoroughly analyzed by Prest [Pre17] for general lattices. In particular, for the lattice $\mathcal{L}_q^\perp(\mathbf{G}_L)$, choosing $s_L \geq \sqrt{b_L^2 + 1}\eta_{\varepsilon_L}(\mathbb{Z}^{ndk_L})$ gives a marginal loss $\alpha_L = ((1 + \varepsilon_L/ndk_L)/(1 - \varepsilon_L/ndk_L))^{ndk_L} \approx 1 + 2\varepsilon_L$. This loss could be refined as the gadget lattice has a very specific structure and is not arbitrary (see, e.g., [Jeu24, Sec. 8.2.3]).

3.5.2 Perturbation Sampling. Let us now discuss the perturbation sampling (step 1 of Algorithms 3.1 and 3.2) producing the distribution $\mathcal{P}_p$. The latter has been covered in [AGJ⁺24] for their elliptic perturbation sampler, and at length in [JS25b] for their tag-dependent perturbation sampler. Our sampler follows the one from [JS25b], which is closer in essence, but with tweaks to match our new tag-independent covariance matrix

$$\mathbf{S}_p = \begin{bmatrix} (s_1^2 - s_L^2)\mathbf{I}_d - s_H^2 \mathbf{R}_1 \mathbf{R}_1^* & \mathbf{0} & -s_H^2 \mathbf{R}_1 \mathbf{R}_2^* & -s_H^2 \mathbf{R}_1 \\ \mathbf{0} & (s_2^2 - s_L^2)\mathbf{I}_{d(k_L-1)} & \mathbf{0} & \mathbf{0} \\ -s_H^2 \mathbf{R}_2 \mathbf{R}_1^* & \mathbf{0} & s_3^2 \mathbf{I}_d - s_H^2 \mathbf{R}_2 \mathbf{R}_2^* & -s_H^2 \mathbf{R}_2 \\ -s_H^2 \mathbf{R}_1^* & \mathbf{0} & -s_H^2 \mathbf{R}_2^* & (s_4^2 - s_H^2)\mathbf{I}_{dk_H} \end{bmatrix}.$$

Note that here, we work directly over $\mathcal{K}_{\mathbb{R}}^{d(1+k_L+k_H) \times d(1+k_L+k_H)}$ as we later leverage the tower of power-of-two cyclotomic fields for better computational efficiency. When parsing $\mathbf{p}$ as $[\mathbf{p}_1^T | \mathbf{p}_2^T | \mathbf{p}_3^T | \mathbf{p}_4^T]^T$ to match the blocks' dimensions of $\mathbf{S}_p$, we observe that $\mathbf{p}_2$ can be sampled independently from the others. We

thus start by sampling it from $\mathcal{D}_{\mathcal{R}^{d(k_L-1)}, \sqrt{s_2^2 - s_L^2}}$, which is essentially a spherical sampler over $\mathbb{Z}^{nd(k_L-1)}$ and thus efficiently sampleable. We must then sample $(\mathbf{p}_1, \mathbf{p}_3, \mathbf{p}_4)$ with covariance $\mathbf{S}'_p$ defined from $\mathbf{S}_p$ by removing the second block row and second block column. To do so, we use the successive Schur complement convolution sampler based on [GM18], which was already used in prior implemented works [AGJ+24, JS25b]. We first sample $\mathbf{p}_4$ from $\mathcal{D}_{\mathcal{R}^{dk_H}, \sqrt{s_4^2 - s_H^2}}$ (which is again a spherical integer sampler) and then update the sampling of $(\mathbf{p}_1, \mathbf{p}_3)$ to account for the dependency. We indeed sample them with covariance $\mathbf{S}''_p = \mathbf{S}'_p / (s_4^2 - s_H^2) \mathbf{I}_{dk_H}$ and center $\mathbf{c}$ where

$$\mathbf{S}''_p = \begin{bmatrix} (s_1^2 - s_L^2) \mathbf{I}_d - \frac{s_4^2 s_H^2}{s_4^2 - s_H^2} \mathbf{R}_1 \mathbf{R}_1^* & -\frac{s_4^2 s_H^2}{s_4^2 - s_H^2} \mathbf{R}_1 \mathbf{R}_2^* \\ -\frac{s_4^2 s_H^2}{s_4^2 - s_H^2} \mathbf{R}_2 \mathbf{R}_1^* & s_3^2 \mathbf{I}_d - \frac{s_4^2 s_H^2}{s_4^2 - s_H^2} \mathbf{R}_2 \mathbf{R}_2^* \end{bmatrix}$$

$$\mathbf{c} = -\frac{s_H^2}{s_4^2 - s_H^2} \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \end{bmatrix} \mathbf{p}_4.$$

We then use the iterative version from [AGJ+24, JS25b] to sample $\mathbf{p}_1, \mathbf{p}_3$. It involves computing successive Schur complements of $\mathbf{S}''_p$ and updating the centers to sample the entries of $\mathbf{p}_1, \mathbf{p}_3$ successively. We describe the process in Algorithm 3.3, but we note that as opposed to [JS25b], we can precompute all the Schur complements ahead of time as follows. We let $\mathbf{S}_{2d} = \mathbf{S}''_p$ and define $\mathbf{S}_i$ to be the top left $i \times i$ submatrix. We update all the $\mathbf{S}_i$ with the following update rule:

1. **For** $i = 2d, \dots, 1$ **do**
2. Parse $\mathbf{S}_i$ as $\mathbf{S}_i = \begin{bmatrix} \mathbf{S}'_i & \mathbf{s}_i \\ \mathbf{s}_i^* & f_i \end{bmatrix}$
3. $\mathbf{S}_{i-1} \leftarrow \mathbf{S}'_i - f_i^{-1} \mathbf{s}_i \mathbf{s}_i^*$
4. $\mathbf{f}_i \leftarrow f_i^{-1} \mathbf{s}_i$ ($\mathbf{f}_i$ can be stored in place of $\mathbf{s}_i$)

In the end, $\mathbf{S}_{2d} \in \mathcal{K}_{\mathbb{R}}^{2d \times 2d}$ contains all the f_i on the diagonal and all the $\mathbf{f}_i = f_i^{-1} \mathbf{s}_i$ in the columns above the diagonal. The $\mathbf{s}_i^*$ below the diagonal can be discarded after the key generation. From these precomputations, we can design the following perturbation sampler.

Algorithm 3.3: PerturbationSampler($\mathbf{R}_1, \mathbf{R}_2$)

Precomputed: $(f_i)_{i \in [2d]}$ and $(\mathbf{f}_i = f_i^{-1} \mathbf{s}_i)_{i \in [2d]}$.

1. $\mathbf{p}_2 \leftarrow \mathcal{D}_{\mathcal{R}^{d(k_L-1)}, \sqrt{s_2^2 - s_L^2}}$ $\triangleright$ sampling over $\mathbb{Z}^{nd(k_L-1)}$
2. $\mathbf{p}_4 \leftarrow \mathcal{D}_{\mathcal{R}^{dk_H}, \sqrt{s_4^2 - s_H^2}}$ $\triangleright$ sampling over $\mathbb{Z}^{ndk_H}$
3. $\mathbf{c} \leftarrow -\frac{s_H^2}{s_4^2 - s_H^2} \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \end{bmatrix} \mathbf{p}_4$
4. **For** $i = 2d, \dots, 1$ **do**
5. Parse $\mathbf{c}$ as $\mathbf{c} = \begin{bmatrix} \mathbf{c}' \\ d_i \end{bmatrix}$
6. $\bar{p}_i \leftarrow \text{SampleFz}(f_i, d_i)$ $\triangleright \sim \mathcal{D}_{\mathcal{R}, \sqrt{M_{\tau}(f_i)}, d_i}$ [GM18]
7. $\mathbf{c} \leftarrow \mathbf{c}' + (\bar{p}_i - d_i) \mathbf{f}_i$

8. $\mathbf{p}_1 \leftarrow [\bar{p}_1 | \dots | \bar{p}_d]^T$
9. $\mathbf{p}_3 \leftarrow [\bar{p}_{d+1} | \dots | \bar{p}_{2d}]^T$

Output: $\mathbf{p} = [\mathbf{p}_1 | \mathbf{p}_2 | \mathbf{p}_3 | \mathbf{p}_4]^T$ (close to $\mathcal{D}_{\mathcal{R}^{d(1+k_L+k_H)}, \sqrt{M_\tau(\mathbf{S}_p)}}$)

The security statement for this sampler immediately follows from [GM18, Thm. 4.1], which was also used for the perturbation samplers of [AGJ⁺24, JS25b].

Lemma 3.5 (Adapted from [GM18, Thm. 4.1]). *The distribution $\mathcal{P}_p$ of $\text{PerturbationSampler}(\mathbf{R}_1, \mathbf{R}_2)$ verifies $\mathcal{P}_p \approx_{\alpha_p} \mathcal{D}_{\mathcal{R}^{d(1+k_L+k_H)}, \sqrt{M_\tau(\mathbf{S}_p)}}$ with $\alpha_p = ((1 + \varepsilon)/(1 - \varepsilon))^{3d(n-1)+1} \approx 1 + (6d(n-1) + 2)\varepsilon$.*

3.5.3 Combining Everything. We can now plug the expression of α_p from Lemma 3.5 into the main result of Theorem 3.1 with our modified sampling of $\mathbf{z}_L$, that is in Equation (4). It gives an overall marginal distance of

$$\begin{aligned} \delta &= \left(\frac{1 + \varepsilon_L}{1 - \varepsilon_L} \right)^{dk_L} \left(\frac{1 + \varepsilon_H}{1 - \varepsilon_H} \right)^{dk_H} \left(\frac{1 + \varepsilon}{1 - \varepsilon} \right)^{6d(n-1)+3} \\ &\approx 1 + 2 \cdot (dk_L \varepsilon_L + dk_H \varepsilon_H + (6d(n-1) + 3)\varepsilon) \end{aligned} \quad (5)$$

When choosing $\varepsilon_L = \varepsilon/dk_L$ and $\varepsilon_H = \varepsilon/dk_H$, we get $\delta \approx 1 + (12d(n-1) + 10)\varepsilon$. We now go back on the choice of ε to satisfy conditions (3) and (4). As a simple example, we select $s_x = b_x \eta_{\varepsilon/dk_x}(\mathbb{Z}^n)$ so that the first part of these conditions is satisfied. We now look at the condition $s_x > \eta_{\varepsilon}(\mathbb{Z}^{nd(k_L+k_H)})$. If we approximate the smoothing of $\mathbb{Z}^N$ with the tight estimate of Lemma 2.4, the condition is equivalent to $\varepsilon < \min_x (2ndk_x \cdot (k_x/(k_L + k_H))^{1/(b_x^2-1)})$. Because this right-hand side is very likely to be much larger than 1 for typical parameters, any choice of $\varepsilon \in (0, 1)$ fits. Hence, choosing s_L, s_H as above is sufficient to satisfy conditions (3) and (4) as desired.

4 Applications

We highlight the benefits of our new sampler by plugging it into some privacy-enhanced authentication mechanisms and comparing the resulting performances with those obtained when using the state-of-the-art sampler in [JS25b]. As we discussed, the enhanced tag management allows us to choose more compact parameters (reducing k_H), and also solve the double trapdoor problem in a more compact way, both of which reduce the final preimage dimension. Moreover, our sampler opens for a wider range of parameters, resulting in further gains. As these advanced authentication mechanisms rely on the standard model signature framework of [AGJ⁺24], we first describe a more compact standard model signature based on TSampler, which might be of independent interest.

4.1 Standard Model Signature

We start from the scheme presented in [JS25b], itself based on that of [AGJ⁺24], which yielded, so far, the shortest signatures in the standard model. The very

features of our sampler means that we can use it without affecting the overall structure of the signature, nor the ones of the tags that are still defined as $t\mathbf{1} = t(1, \dots, 1)$

At a high-level, the public key is $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 \bmod q\mathcal{R}$, and $(\mathbf{R}_1, \mathbf{R}_2)$ is the secret key composed of short matrices. To generate a signature on a message m , which is a binary polynomial, the signer starts by choosing a sparse binary polynomial t . Then, it samples v_3 over $\mathcal{R}$ with parameter s_4 , and uses `TSampler` to sample a Gaussian $\mathbf{v} = (\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2)$ such that

$$\mathbf{v}_{1,1} + \mathbf{A}\mathbf{v}_{1,2} + (q_L \mathbf{G}_{H,t\mathbf{1}} - (\mathbf{R}_1 + \mathbf{A}\mathbf{R}_2))\mathbf{v}_2 = \mathbf{u} + m\mathbf{d} - v_3\mathbf{a}_3 \bmod q\mathcal{R}.$$

The message appears in an Ajtai commitment $m\mathbf{d}$, which will then interface with other primitives like anonymous credentials, blind signatures, etc. Depending on the use case, one can require higher-dimensional messages $\mathbf{m}$ to encode longer messages or attributes, in which case this term is replaced by $\mathbf{D}\mathbf{m}$. But we note that $\mathbf{d}$ (or $\mathbf{D}$), as well as $\mathbf{A}, \mathbf{u}, \mathbf{a}_3$ can be common to many signers and be part of the public parameters. In the standalone signature use case, we keep the message to be a single polynomial m , which can be generated by hashing the actual signed message into a binary polynomial. But we insist that this hash would not need to be modelled as a random oracle in the security proof, thus making the signature secure in the standard model. We would only need collision resistance of the hash function.

4.1.1 Description and Security Analysis. We now give a full description and security proof of the scheme. The parameter Q represents the maximal number of emitted signature per key pair.

Algorithm 4.1: Setup(λ)

1. Choose $n \geq \kappa$ to be powers of two.
2. Choose $d \in \mathbb{N}^\times$.
3. Choose $b_H \in \mathbb{N}^\times$ a small prime s.t. $b_H = 2\kappa + 1 \bmod 4\kappa$ and $b_H > \sqrt{\kappa}^\kappa$.
4. Choose $k_H, b_L, k_L \in \mathbb{N}^\times$
5. $q_L \leftarrow b_L^{k_L}$
6. $q_H \leftarrow b_H^{k_H}$
7. $q \leftarrow q_L q_H$
8. Choose $w \in \mathbb{N}^\times$ st $\binom{n}{w} \geq Q$.
9. $\mathcal{T}_w \leftarrow \{t \in T_1 : \|t\|_1 = w\}$.
10. $B_R \leftarrow \frac{3}{4}(\sqrt{nd} + \sqrt{ndk_H} + 6)$
11. For $x \in \{L, H\}$, define $\mathbf{G}_x = [1 \dots |b_x^{k_x-1}|] \otimes \mathbf{I}_d \in \mathcal{R}^{d \times dk_x}$
12. Choose $\varepsilon \in (0, 1)$ $\triangleright$ typically $\varepsilon \approx 2^{-40}$
13. $s_L \leftarrow b_L \cdot \sqrt{\ln(2ndk_L/\varepsilon)/\pi}$ $\triangleright s_L \approx b_L \eta_\varepsilon(\mathbb{Z}^{ndk_L}) \geq b_L \eta_{\varepsilon/dk_L}(\mathbb{Z}^n)$ by Lemma 2.4
14. $s_H \leftarrow b_H \cdot \sqrt{\ln(2ndk_H/\varepsilon)/\pi}$ $\triangleright s_H \approx b_H \eta_\varepsilon(\mathbb{Z}^{ndk_H}) \geq b_H \eta_{\varepsilon/dk_H}(\mathbb{Z}^n)$ by Lemma 2.4
15. $\eta \leftarrow \sqrt{\ln(2nd(k_L + k_H)/\varepsilon)/\pi}$ $\triangleright \eta \approx \eta_\varepsilon(\mathbb{Z}^{nd(k_L + k_H)})$ by Lemma 2.4
16. $s_1 \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$.
17. $s_2 \leftarrow s_L^2/\sqrt{s_L^2 - \eta^2}$.
18. $s_{1,1} \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) \cdot (b_L^{2k_L} - 1)/(b_L^2 - 1) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$.
19. $s_3 \leftarrow \sqrt{3s_H^2 B_R/\sqrt{s_H^2 - \eta^2}}$.
20. $s_4 \leftarrow \sqrt{3s_H^2/\sqrt{s_H^2 - \eta^2}}$.
21. $(\alpha_{1,1}, \alpha_{1,2}) \leftarrow (s_{1,1}/(n\sqrt{d/2}), s_3/(n\sqrt{d/2}))$.

22. $M_{1,i} \leftarrow \exp(\pi/\alpha_{1,i}^2)$.
23. $\mathbf{d} \leftarrow U(\mathcal{R}_q^d)$.
24. $\mathbf{A} \leftarrow U(\mathcal{R}_q^{d \times d})$.
25. $\mathbf{a}_3 \leftarrow U(\mathcal{R}_q^d)$.
26. $\mathbf{u} \leftarrow U(\mathcal{R}_q^d)$.
27. $\text{params} \leftarrow (n, d, b_H, k_H, q_H, b_L, k_L, q_L, q, w, B_R, \varepsilon, s_L, s_H, \eta, s_1, s_2, s_3, s_4, s_{1,1})$

Output: $\text{pp} = (\text{params}, \mathbf{A}, \mathbf{d}, \mathbf{a}_3, \mathbf{u})$

Algorithm 4.2: KeyGen(pp)

1. $\mathbf{R}_1, \mathbf{R}_2 \leftarrow \mathcal{B}_1^{d \times dk_H}$ conditioned on $\|\mathbf{R}_i\|_2 \leq B_R$.
2. $\mathbf{B} \leftarrow \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 \bmod q\mathcal{R} \in \mathcal{R}_q^{d \times dk_H}$.

Output: $\text{pk} = \mathbf{B}, \text{sk} = (\mathbf{R}_1, \mathbf{R}_2)$

Algorithm 4.3: Sign(sk, m, pk, pp, st)

1. $\mathbf{c} \leftarrow m\mathbf{d} \bmod q\mathcal{R}$.
2. $t \leftarrow F(\text{st}) \in \mathcal{T}_w$. $\triangleright t \in T_1, \|t\|_1 = w$
3. $\mathbf{v}_3 \leftarrow \mathcal{D}_{\mathcal{R}, s_4}$.
4. $(\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2) \leftarrow \text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u} + \mathbf{c} - \mathbf{v}_3\mathbf{a}_3 \bmod q\mathcal{R}, t\mathbf{1}, s_1, s_2, s_3, s_4, s_L, s_H)$
5. **if** $\|\mathbf{v}_{1,1}\|_2 > B_{1,1} \vee \|\mathbf{v}_{1,2}\|_2 > B_{1,2} \vee \|\mathbf{v}_2\|_2 > B_2 \vee \|\mathbf{v}_3\|_2 > B_3$ **goto** 3).
6. $\text{st} \leftarrow \text{st} + 1$.

Output: $\text{sig} = (t, \mathbf{v}_{1,2}, \mathbf{v}_2, \mathbf{v}_3)$

Algorithm 4.4: Verify(pk, m, sig, pp)

1. $\mathbf{v}_{1,1} \leftarrow \mathbf{u} + m\mathbf{d} - \mathbf{A}\mathbf{v}_{1,2} - (q_L t \mathbf{G}_H - \mathbf{B})\mathbf{v}_2 - \mathbf{v}_3\mathbf{a}_3 \bmod q\mathcal{R} \in \mathcal{R}^d$.
2. $b_1 \leftarrow \|\mathbf{v}_{1,1}\|_2 \leq B_{1,1}$. $\triangleright B_{1,1} = c_{nd}s_{1,1}\sqrt{nd}$
3. $b_2 \leftarrow \|\mathbf{v}_{1,2}\|_2 \leq B_{1,2}$. $\triangleright B_{1,2} = c_{nd}s_3\sqrt{nd}$
4. $b_3 \leftarrow \|\mathbf{v}_2\|_2 \leq B_2$. $\triangleright B_2 = c_{ndk_H}s_4\sqrt{ndk_H}$
5. $b_4 \leftarrow \|\mathbf{v}_3\|_2 \leq B_3$. $\triangleright B_3 = c_n s_4 \sqrt{n}$
6. $b_5 \leftarrow t \in \mathcal{T}_w$.

Output: $b_1 \wedge b_2 \wedge b_3 \wedge b_4 \wedge b_5$ (1 if valid, 0 otherwise)

The Gaussian tailcuts c_N can be adjusted to trade off smaller bounds (and thus better security) for computational efficiency (by avoiding too many rejections). We select them using Lemma 2.9 to achieve a rejection probability of at most $p = 2^{-20}$. That is, we choose c_N as the smallest $c > 1/\sqrt{2\pi}$ such that $(c\sqrt{2\pi}ee^{-\pi c^2})^N \leq p$. Note that by Lemma 2.2 and the way b_H is set, all the tags in $\mathcal{T}_w$ are units in $\mathcal{R}_{b_H}$.

Theorem 4.1. *The signature scheme of Algorithms 4.1, 4.2, 4.3 and 4.4 is unforgeable in the standard model based on the hardness of M-LWE, M-SIS. More precisely, the advantage of PPT adversary in breaking the unforgeability of the signature is upper-bounded by*

$$\begin{aligned} \text{Adv}[\mathcal{A}] \leq & 2 \left(h^{\circ dk_H} (C(\binom{n}{w} - Q)\varepsilon_{\text{M-SIS}}^{\bullet}) + C^2 \varepsilon_{\text{M-LWE}} \right. \\ & \left. + \frac{1+\varepsilon}{1-\varepsilon} \frac{4M_{1,1}M_{1,2}}{(1-p)^4} h^{\circ dk_H} (CQ\varepsilon_{\text{M-SIS}}^{\bullet}) \right) + \text{negl}(\lambda) \end{aligned}$$

where $\varepsilon_{\text{M-LWE}}$ the hardness bound of $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1}$, and $\varepsilon_{\text{M-SIS}}^{\bullet}$ and $\varepsilon_{\text{M-SIS}}^{\bullet\bullet}$ are the respective hardness bounds of $\text{M-SIS}_{n,d,2d+3,q,\beta_{\bullet}}$ and $\text{M-SIS}_{n,d,2d+1,q,\beta_{\bullet\bullet}}$. The constant $C \approx 2$ is the one from Lemma 2.8, and the probability p corresponds to the tailcut probability used to set the Gaussian verification bounds. The function

$h^{\circ dk_H}$ corresponds to the dk_H -th composition power of the function h defined by

$$h(x) = \varepsilon_{\text{M-LWE}} + \mu \left(2\varepsilon_{\text{M-LWE}} + \mu (\varepsilon_{\text{M-LWE}} + x)^{\frac{2\lambda-1}{2\lambda}} \right)^{\frac{2\lambda-1}{2\lambda}}.$$

The constant μ is defined by $\mu = \left(1 + \frac{\lambda(2\lambda-1)(\delta^2-1)^2}{(2-\delta^2)^{2\lambda+1}} \right)^{Q/2\lambda}$ where δ is defined in Theorem 3.1 and Equation (5). Finally, we have

$$\begin{aligned} \beta_{\bullet} &= \sqrt{\left(\sqrt{B_{1,1}^2 + B_{1,2}^2} + \sqrt{nd}B_2 \right)^2 + B_3^2 + n + 1} \\ \beta_{\bullet} &= \sqrt{\left(2\sqrt{B_{1,1}^2 + B_{1,2}^2} + \sqrt{nd}\sqrt{4B_2^2 + n} \right)^2 + 4B_3^2} \end{aligned}$$

Proof. Our unforgeability proof is very similar to that of the standard model signature from [AGJ⁺24] and [JS25b], with a few tweaks due to our sampler. The main difference occurs in the trapdoor switching argument. We carry the latter in a more progressive way, using M-LWE and Lemma 3.4, until we have changed the dk_H columns of $\mathbf{B}$, one by one. The final reduction to M-SIS is then concluded in the same way as [AGJ⁺24, JS25b].

Let us now give the fully detailed proof. We call G_0 the original unforgeability game. More precisely, the challenger generates $\text{pp} = \text{Setup}(\lambda)$ and $(\text{pk}, \text{sk}) = \text{KeyGen}(\text{pp})$ and provides the adversary $\mathcal{A}$ with pp, pk . Then, upon query by $\mathcal{A}$ on a message m , the challenger answers by giving $\text{sig} = \text{Sign}(\text{sk}, m, \text{pk}, \text{pp}, \text{st})$ to $\mathcal{A}$. After at most Q queries, the adversary returns a forgery m^*, sig^* such that m^* was never queried for a signature, and $\text{Verify}(\text{pk}, m^*, \text{sig}^*, \text{pp}) = 1$. We then progressively change G_0 so that the challenger can use said forgery to solve an M-SIS instance. Within the proof, we denote by Adv_i the advantage of $\mathcal{A}$ in producing a valid forgery in game G_i .

Game G_1 . We start by taking similar steps to [JS25b]. We first change the tag generation, by generating all the tags $\{t_i; i \in [Q]\} \subseteq \mathcal{T}_w$ at the beginning of the game. Concretely, the challenger computes $t_i = F(\text{st} + i)$ where st is the initial state. The state no longer has to be incremented at each signing query. We note here that F does not have any particular restriction besides being deterministic and such that all the generated t_i are different and belong to the tag space $\mathcal{T}_w$. Recall we chose w so that $|\mathcal{T}_w| \geq Q$, which allows for meeting these conditions. Because the view of $\mathcal{A}$ remains unchanged, we have

$$\text{Adv}_0 = \text{Adv}_1.$$

Game G_2 . We now introduce a branching in the challenger's behavior. This can be seen simply as combining the type 1 and type 2 unforgeability proofs of [AGJ⁺24], which has been done explicitly in [JS25a, JS25b]. The branching is done as follows. At the beginning of the game, the challenger samples all the tags as in G_1 , and samples $\rho \leftarrow U(\{1, 2\})$. Depending on the value of ρ , the challenger will act differently expecting a different type of forgery. If $\rho = 1$,

the challenger expects a type ❶ forgery: the tag t^* included in the forgery sig^* has never been obtain in a signature query, i.e., $t^* \notin \{t_i; i \in [Q]\}$. Conversely, if $\rho = 2$, the challenger expects a type ❷ forgery: the tag t^* from the forged signature is among the emitted ones, i.e., there exists $i \in [Q]$ such that $t^* = t_i$. When the adversary $\mathcal{A}$ returns a forgery at the end of the game, the challenger can efficiently check which type of forgery it is by checking whether or not t^* belongs to $\{t_i; i \in [Q]\}$. If its guess on the forgery type was wrong, the challenger aborts.

From now on, we may make changes that affect only a single branch. As a result, we now use $\text{Adv}_i^{\text{❶}}$ and $\text{Adv}_i^{\text{❷}}$ to denote the advantage of $\mathcal{A}$ in winning game G_i with a forgery of type ❶ or ❷ respectively. As opposed to what was claimed [JS25a, JS25b], i.e., $\text{Adv}_1 \leq 2 \max(\text{Adv}_2^{\text{❶}}, \text{Adv}_2^{\text{❷}})$, a direct probability computation shows $\text{Adv}_1 = 2(\text{Adv}_2^{\text{❶}} + \text{Adv}_2^{\text{❷}})$, as ρ is independent from the adversary's view. We note that this does not invalidate their claims, parameter selections, or security estimates, as they generally have $\text{Adv}_2^{\text{❷}} \gg \text{Adv}_2^{\text{❶}}$, in which case the sum and maximum are negligibly close. We then have

$$\text{Adv}_1 = 2 \left(\text{Adv}_2^{\text{❶}} + \text{Adv}_2^{\text{❷}} \right).$$

Game G_3 . The first modification we make that depends on the branch is the tag guess made by the challenger. If $\rho = 1$, the challenger samples $t^+ \leftarrow U(\mathcal{T}_w \setminus \{t_i; i \in [Q]\})$. On the other hand, if $\rho = 2$, the challenger samples $i^+ \leftarrow U([Q])$ and defines $t^+ = t_{i^+}$. So far, t^+ is never used and therefore independent of the adversary's view. Thence, we get

$$\text{Adv}_2^{\text{❶}} = \text{Adv}_3^{\text{❶}}, \text{ and } \text{Adv}_2^{\text{❷}} = \text{Adv}_3^{\text{❷}}.$$

Game G_4 ($\rho = 2$). In the branch $\rho = 2$, the challenger now changes the generation of $\mathbf{d}$ by hiding an M-LWE instance. It samples $\mathbf{s}_1, \mathbf{s}_2$ from $\mathcal{B}_1^d$ conditioned on $\|\mathbf{s}_i\|_{2, \mathbb{Z}} \leq \sqrt{nd}/2$ and defines $\mathbf{d} = \mathbf{s}_1 + \mathbf{A}\mathbf{s}_2 \bmod q\mathcal{R}$. The bound on $\|\mathbf{s}_i\|_{2, \mathbb{Z}}$ is empirically verified with probability $1/C$ for $C \approx 2$. Then, distinguishing view of the adversary between G_3 and G_4 corresponds to solving $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1}$ with a C^2 loss factor. Note that it only affects the branch for $\rho = 2$. We then get

$$\text{Adv}_3^{\text{❶}} = \text{Adv}_4^{\text{❶}}, \text{ and } \text{Adv}_3^{\text{❷}} \leq \text{Adv}_4^{\text{❷}} + C^2 \varepsilon_{\text{M-LWE}},$$

where $\varepsilon_{\text{M-LWE}}$ is the hardness bound of $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1}$.

Game G_5 ($\rho = 2$). The challenger slightly changes the way $\mathbf{u}$ is generated when $\rho = 2$. It samples $\mathbf{u}' \leftarrow U(\mathcal{R}_q^d)$ and $v_3 \leftarrow \mathcal{D}_{\mathcal{R}, s_4}$, before defining $\mathbf{u} = \mathbf{u}' + v_3 \mathbf{a}_3 \bmod q\mathcal{R}$. As $\mathbf{u}'$ is uniform and independent of $v_3 \mathbf{a}_3$, we directly have

$$\text{Adv}_4^{\text{❶}} = \text{Adv}_5^{\text{❶}}, \text{ and } \text{Adv}_4^{\text{❷}} = \text{Adv}_5^{\text{❷}}.$$

Game G_6 ($\rho = 2$). Yet again, the challenger now hides a preimage within the previously introduced $\mathbf{u}'$, when $\rho = 2$. Concretely, the challenger still samples v_3 from $\mathcal{D}_{\mathcal{R}, s_4}$, but it also samples $\mathbf{v}'$ from $\mathcal{D}_{\mathcal{R}^{d(1+k_L+k_H)}, \sqrt{\mathbf{S}}}$ where the covariance

matrix is that of the sampler, i.e., $\mathbf{S} = \text{diag}(s_1^2 \mathbf{I}_{nd}, s_2^2 \mathbf{I}_{nd(k_L-1)}, s_3^2 \mathbf{I}_{nd}, s_4^2 \mathbf{I}_{ndk_H})$. It then defines $\mathbf{u}' = \mathbf{A}_{t+1} \mathbf{K} \mathbf{v}' \bmod q\mathcal{R}$ and $\mathbf{u} = \mathbf{u}' + v_3 \mathbf{a}_3 \bmod q\mathcal{R}$, where $\mathbf{A}_{t+1} = [\mathbf{I}_d | \mathbf{A} | q_L \mathbf{G}_{H,t+1} - \mathbf{B}] \bmod q\mathcal{R}$. Because our parameter meet all of **TSampler**'s conditions, we have $\sqrt{\mathbf{S}} \geq \eta_\varepsilon(\mathcal{L}_q^\perp(\mathbf{A}_{t+1} \mathbf{K}))$. Therefore, if we note $\mathcal{P}_{\mathbf{u}'}$ the distribution of $\mathbf{u}'$ in G_6 , the regularity lemma of Lemma 2.7 yields $\mathcal{P}_{\mathbf{u}'} \in [(1-\varepsilon)/(1+\varepsilon), 1+\varepsilon]U(\mathcal{R}_q^d)$. It then gives

$$\text{Adv}_5^\bullet = \text{Adv}_6^\bullet, \text{ and } \text{Adv}_5^\bullet \leq \frac{1+\varepsilon}{1-\varepsilon} \text{Adv}_6^\bullet.$$

For later, we define $\mathbf{v} = \mathbf{K} \mathbf{v}'$ and parse it as $(\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2)$. It then holds that $\mathbf{v}_{1,1}$ follows $\mathcal{D}_{\mathcal{R}^d, s_{1,1}}$, $\mathbf{v}_{1,2}$ follows $\mathcal{D}_{\mathcal{R}^d, s_3}$ and $\mathbf{v}_2$ follows $\mathcal{D}_{\mathcal{R}^{dk_H}, s_4}$.

Game G_7 ($\rho = 2$). We now prepare for the next hybrid game by adding a rejection sampling step for the i^+ -th query when $\rho = 2$. In this branch, in the i^+ -th query, the challenger receives $m_{i^+} \in T_1$ which we denote by m^+ for clarity. It then proceeds legitimately, that is producing a preimage $(\mathbf{v}_{1,1}^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+)$ for the tag $t^+ \mathbf{1}$ with **TSampler**. Before returning the signature to $\mathcal{A}$ however, it rejects (and thus abort the game altogether) based on m^+ and the values of $\mathbf{v}_{1,1}, \mathbf{v}_{1,2}$ introduced in the previous game. That is, it samples reals $u_1, u_2 \leftarrow U([0, 1])$ and continues only if $u_1 \leq 1/M_{1,1}$, $u_2 \leq 1/M_{1,2}$, $\langle \tau(\mathbf{v}_{1,1}), \tau(\mathbf{s}_1 m^+) \rangle \geq 0$ and $\langle \tau(\mathbf{v}_{1,2}), \tau(\mathbf{s}_2 m^+) \rangle \geq 0$, and aborts otherwise.

If the test passes, the challenger outputs $\text{sig}_{i^+} = (t^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+)$. Notice it still returns the legitimately sampled preimage and not the one introduced in G_7 . Because the u_i are sampled uniformly independently from the adversary's view, and because $\mathbf{v}_{1,1}, \mathbf{v}_{1,2}$ follow a symmetric distribution that is independent of $\mathbf{s}_1 m^+, \mathbf{s}_2 m^+$, we directly get

$$\text{Adv}_6^\bullet = \text{Adv}_7^\bullet, \text{ and } \text{Adv}_6^\bullet = (2M_{1,1})(2M_{1,2})\text{Adv}_7^\bullet.$$

Game G_8 ($\rho = 2$). We now use the preimage hidden in $\mathbf{u}$ in the i^+ -th query when $\rho = 2$. As this preimage is essentially a signature for the zero-message, we however need to slightly change it so that it becomes a signature on m^+ . This is where the changes we have introduced thus far become important. Instead of sampling $(\mathbf{v}_{1,1}^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+)$ legitimately, we define them as $\mathbf{v}_{1,1}^+ = \mathbf{v}_{1,1} + \mathbf{s}_1 m^+$, $\mathbf{v}_{1,2}^+ = \mathbf{v}_{1,2} + \mathbf{s}_2 m^+$ and $(\mathbf{v}_2^+, v_3^+) = (\mathbf{v}_2, v_3)$. However, the distribution of the first two vectors is now a shifted Gaussian and does not match the expected distribution of signatures. We thus perform rejection sampling to ensure the signature is correctly distributed. Based on the changes introduced in G_7 , this can be done smoothly.

Before outputting $\text{sig}_{i^+} = (t^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+)$, the challenger samples $u_1, u_2 \leftarrow U([0, 1])$ and computes $\gamma_j = \langle \tau(\mathbf{v}_{1,j}^+), \tau(\mathbf{s}_j m^+) \rangle$ for $j \in \{1, 2\}$. It then continues only if

$$\gamma_1 \geq 0, \text{ and } u_1 \leq \frac{1}{M_{1,1}} \exp\left(\frac{\pi}{s_{1,1}^2} \left(\|\mathbf{s}_1 m^+\|_2^2 - 2\gamma_1\right)\right),$$

$$\text{and } \gamma_2 \geq 0, \text{ and } u_2 \leq \frac{1}{M_{1,2}} \exp \left(\frac{\pi}{s_3^2} \left(\|s_2 m^+\|_2^2 - 2\gamma_2 \right) \right).$$

and aborts otherwise. If it did not abort, sig_{i^+} is outputted and is a valid signature only if the bounds are verified. Because we have $\|s_j m^+\|_2 \leq \sqrt{nd/2} \|m^+\|_2$ based on G_4 , and the way the parameters $M_{1,j}$ are set, Lemma 2.10 shows that the distribution is the same. Hence, using the Gaussian tail bounds on $\mathbf{v}_{1,1}^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+$ after the rejection sampling passed, the challenger also aborts if the bounds are not verified. Because the bounds are verified each with probability $1 - p$, it yields

$$\text{Adv}_7^{\bullet} = \text{Adv}_8^{\bullet}, \text{ and } \text{Adv}_7^{\circ} = \frac{1}{(1-p)^4} \text{Adv}_8^{\circ}.$$

Trapdoor Switching Argument. We now present our new trapdoor switching argument. It proceeds in fairly the same way as the one described in [AGJ⁺24] but in a more fine-grained way. What we are aiming for is to change the public key to $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 + q_L \mathbf{G}_{H, \mathbf{t}_j}$ at step j of this hybrid argument, where $\mathbf{t}_j = t^+(1, \dots, 1, 0, \dots, 0)$, i.e., the vector with the first j entries equal to t^+ and 0 elsewhere. We proceed iteratively, defining a hybrid argument. We thoroughly track the loss of each game hop. Let $j \in [0, dk_H - 1]$. We now define games $G_{j,0}$ up to $G_{j,9}$ so that $G_{j,0}$ features $\mathbf{t}_j$ in the public key, while $G_{j,9}$ uses $\mathbf{t}_{j+1}$ instead. We define the games so that $G_{0,0} = G_8$, and $G_{j,9} = G_{j+1,0}$. We note that each hybrid change applies to both branches.

Game $G_{j,0}$. In this game, the challenger samples all the tags as in G_1 , samples ρ and then t^+ as described in G_3 depending on the value of ρ (and aborts in the end based on the type of forgery), samples $\mathbf{A}, \mathbf{a}_3$ uniformly, $\mathbf{d}, \mathbf{u}$ as described in games G_4 and G_6 depending on ρ . It then samples $\mathbf{R}_1, \mathbf{R}_2$ from $\mathcal{B}_1^{d \times dk_H}$ conditioned on $\|\mathbf{R}_i\|_2 \leq B_R$, before defining $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 + q_L \mathbf{G}_{H, \mathbf{t}_j}$ where

$$\mathbf{t}_j = t^+ \sum_{k=0}^{j-1} \mathbf{e}_k = t^+ (\underbrace{1, \dots, 1}_{j-1 \text{ times}}, 0, \dots, 0).$$

It uses these materials to construct pp, pk which it sends to $\mathcal{A}$. When receiving the i -th signature query on a message m_i , it proceeds as follows. If $\rho = 2$ and $i = i^+$, the challenger proceeds as in G_8 by using the preimage hidden in $\mathbf{u}$. In all other cases, i.e., $(\rho, i) \neq (2, i^+)$, it samples v_3^i from $\mathcal{D}_{\mathcal{R}, s_4}$, computes $\mathbf{u}_i = \mathbf{u} + m_i \mathbf{d} - v_3^i \mathbf{a}_3 \bmod q\mathcal{R}$, and $(\mathbf{v}_{1,1}^i, \mathbf{v}_{1,2}^i, \mathbf{v}_2^i) = \text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u}_i, t_i \mathbf{1} - \mathbf{t}_j, s_1, s_2, s_3, s_4, s_L, s_H)$. It then checks $\mathbf{v}_{1,1}^i, \mathbf{v}_{1,2}^i, \mathbf{v}_2^i, v_3^i$ verify the correct bounds and outputs the signature $(t_i, \mathbf{v}_{1,2}^i, \mathbf{v}_2^i, v_3^i)$ to $\mathcal{A}$. For these queries, everything is done legitimately except the effective tag which is $t_i \mathbf{1} - \mathbf{t}_j$. We then only need to check that the effective tag $\mathbf{t} = t_i \mathbf{1} - \mathbf{t}_j$ lies in $(\mathcal{R}_{b_H}^\times)^{dk_H}$.

For that, observe that the entries of $\mathbf{t}$ are either t_i or $t_i - t^+$, which are in S_1 either way, i.e., have infinity norm bounded by 1. They are also all non zero because when $(\rho, i) \neq (2, i^+)$, we have $t_i \neq t$ (and $t_i \neq 0$). Because b_H is prime

such that $b_H = 2\kappa + 1 \bmod 4\kappa$ and $b_H^{1/\kappa}/\sqrt{\kappa} > 1$, Lemma 2.2 ensures all the entries of $\mathbf{t}$ are then units in $\mathcal{R}_{b_H}$.

Note that when $j = 0$, $\mathbf{t}_j = \mathbf{0}$ and therefore $G_{0,0} = G_8$.

Game $G_{j,1}$. This game is the same as $G_{j,0}$ except that the challenger hides a partial gadget in $\mathbf{a}_3$. Concretely, the challenger samples $\mathbf{a}'_3 \leftarrow U(\mathcal{R}_q^d)$ and defines $\mathbf{a}_3 = q_L b_H^{\lfloor j/d \rfloor} \mathbf{e}_{(j \bmod d)} - \mathbf{a}'_3 \bmod q\mathcal{R}$. Because $\mathbf{a}'_3$ is uniform and independent of this partial gadget, we have

$$\text{Adv}_{j,0}^{\mathbf{1}} = \text{Adv}_{j,1}^{\mathbf{1}}, \text{ and } \text{Adv}_{j,0}^{\mathbf{2}} = \text{Adv}_{j,1}^{\mathbf{2}}.$$

Game $G_{j,2}$. We change $G_{j,1}$ by hiding an M-LWE secret in $\mathbf{a}'_3$. The challenger samples $\mathbf{r}'_{1,j}, \mathbf{r}'_{2,j}$ from $\mathcal{B}_1^d$ such that when defining

$$\mathbf{R}'_\ell = [\mathbf{r}_{\ell,0} | \dots | \mathbf{r}_{\ell,j-1} | \mathbf{r}'_{\ell,j} | \mathbf{r}_{\ell,j+1} | \dots | \mathbf{r}_{\ell,dk_H-1}],$$

it satisfies $\|\mathbf{R}'_\ell\|_2 \leq B_R$, for $\ell \in \{1, 2\}$, and where the $\mathbf{r}_{\ell,0}, \dots, \mathbf{r}_{\ell,dk_H-1}$ are the columns of $\mathbf{R}_\ell$. It then defines $\mathbf{a}'_3 = \mathbf{r}'_{1,j} + \mathbf{A}\mathbf{r}'_{2,j} \bmod q\mathcal{R}$ and the final vector $\mathbf{a}_3 = q_L b_H^{\lfloor j/d \rfloor} \mathbf{e}_{(j \bmod d)} - \mathbf{a}'_3 \bmod q\mathcal{R}$. Distinguishing the views of the adversary between $G_{j,1}$ and $G_{j,2}$ is exactly solving $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1}$. Hence, it holds that

$$\text{Adv}_{j,1}^{\mathbf{1}} \leq \text{Adv}_{j,2}^{\mathbf{1}} + \varepsilon_{\text{M-LWE}}, \text{ and } \text{Adv}_{j,1}^{\mathbf{2}} \leq \text{Adv}_{j,2}^{\mathbf{2}} + \varepsilon_{\text{M-LWE}},$$

where $\varepsilon_{\text{M-LWE}}$ is the hardness bound of $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1}$.

Game $G_{j,3}$. We now switch trapdoors to use $(\mathbf{R}'_1, \mathbf{R}'_2)$ instead of $(\mathbf{R}_1, \mathbf{R}_2)$ within the sampling procedure. As they both satisfy the spectral norm bound B_R , we can use either to have the same security guarantees. Concretely, we change the way signing queries are answered as follows. When $\rho = 2$ and $i = i^+$, we keep answering the signing query using the preimage in $\mathbf{u}$ as in $G_{j,2}$. When $(\rho, i) \neq (2, i^+)$, the challenger samples $v_{2,j}^i$ from $\mathcal{D}_{\mathcal{R},s_4}$ and computes $\mathbf{u}_i = \mathbf{u} + m_i \mathbf{d} - (q_L t_i b_H^{\lfloor j/d \rfloor} \mathbf{e}_{(j \bmod d)} - \mathbf{b}_j) v_{2,j}^i \bmod q\mathcal{R}$, where $\mathbf{b}_j$ is the j -th column of $\mathbf{B}$. Note that the vector in front of $v_{2,j}^i$ is the j -th column of $q_L \mathbf{G}_{H,t_i} \mathbf{1} - \mathbf{B}$. Then, the challenger samples $(\mathbf{v}_{1,1}^i, \mathbf{v}_{1,2}^i, \mathbf{v}_2^i)$ from $\text{TSampler}(\mathbf{R}'_1, \mathbf{R}'_2, \mathbf{A}, \mathbf{u}_i, \mathbf{t}, s_1, s_2, s_3, s_4, s_L, s_H)$ where

$$\mathbf{t} = t_i \mathbf{1} - \mathbf{t}_j + (1 - t_i) \mathbf{e}_j = \underbrace{(t_i - t^+, \dots, t_i - t^+)}_{j \text{ times}}, 1, t_i, \dots, t_i$$

It then parses $\mathbf{v}_2^i$ as $[v_{2,0}^i, \dots, v_{2,j-1}^i, v_3^i, v_{2,j+1}^i, \dots, v_{2,dk_H-1}^i]^T$. Finally, it defines $\mathbf{v}_2^i = [v_{2,\ell}^i]_{\ell \in [0, dk_H-1]}$ and sends $(t_i, \mathbf{v}_{1,2}^i, \mathbf{v}_2^i, v_3^i)$ as the signature after verifying it passes the verification bounds.

Based on how s_x is defined, we have $s_x = b_x \eta_\varepsilon(\mathbb{Z}^{ndk_x}) \geq b_x \eta_{\varepsilon/dk_x}(\mathbb{Z}^n)$. We can use Theorem 3.1 to have the marginal distance δ of TSampler with the ideal distribution, with

$$\delta = \left(\frac{1 + \frac{\varepsilon}{dk_L}}{1 - \frac{\varepsilon}{dk_L}} \right)^{dk_L} \left(\frac{1 + \frac{\varepsilon}{dk_H}}{1 - \frac{\varepsilon}{dk_H}} \right)^{dk_H} \left(\frac{1 + \varepsilon}{1 - \varepsilon} \right)^{6d(n-1)+3}.$$

The latter is obtained from Equation (5) by selecting $\varepsilon_L = \varepsilon/dk_L$ and $\varepsilon_H = \varepsilon/dk_H$. By the trapdoor switching of Lemma 3.4, it holds that the distribution of each query is within marginal distance δ^2 of that in the previous game. Using Lemma 2.3 with $a = 2\lambda$ and the multiplicativity of the Rényi divergence, we have

$$\begin{aligned} \text{Adv}_{j,2}^{\bullet} &\leq \left(1 + \frac{\lambda(2\lambda - 1)(\delta^2 - 1)^2}{(2 - \delta^2)^{2\lambda+1}}\right)^{Q/2\lambda} \text{Adv}_{j,3}^{\bullet} 1^{-1/2\lambda} \\ \text{and } \text{Adv}_{j,2}^{\circ} &\leq \left(1 + \frac{\lambda(2\lambda - 1)(\delta^2 - 1)^2}{(2 - \delta^2)^{2\lambda+1}}\right)^{(Q-1)/2\lambda} \text{Adv}_{j,3}^{\circ} 1^{-1/2\lambda}. \end{aligned}$$

The first slack, which we call μ , is equivalent to $1 + Q(\lambda - 1/2)(\delta^2 - 1)^2$ which is itself equivalent to $1 + Q(\lambda - 1/2)(20 + 24d(n - 1))^2 \varepsilon^2$. The second slack can actually be bounded by μ as well so we use the same slack for both equations from now on. To have an acceptable loss, one then needs to choose $\varepsilon \leq ((20 + 24d(n - 1))\sqrt{Q(\lambda - 1/2)})^{-1}$ which is the case for our parameters. These equivalents are only there to derive a rough condition on ε . In the concrete parameter selection and security estimation, we use the actual formula from the equation above.

Game $G_{j,4}$. As the j -th column of $\mathbf{R}_1$ and $\mathbf{R}_2$ are no longer used to answer signing queries, we can now simulate the j -th column of $\mathbf{B}$. The challenger then samples $\mathbf{b}_j$ uniformly in $\mathcal{R}_q^d$. At this stage, $\mathbf{R}_1, \mathbf{R}_2$ does not have to satisfy the spectral norm bound of the key generation, only $\mathbf{R}'_1, \mathbf{R}'_2$ should. As in $G_{j,2}$, we then get

$$\text{Adv}_{j,3}^{\bullet} \leq \text{Adv}_{j,4}^{\bullet} + \varepsilon_{\text{M-LWE}}, \text{ and } \text{Adv}_{j,3}^{\circ} \leq \text{Adv}_{j,4}^{\circ} + \varepsilon_{\text{M-LWE}},$$

Game $G_{j,5}$. We now hide the tag t^+ in the j -th column of $\mathbf{B}$. More precisely, the challenger samples $\mathbf{b}'_j \leftarrow U(\mathcal{R}_q^d)$, and defines $\mathbf{b}_j = \mathbf{b}'_j + t^+ q_L b_H^{\lfloor j/d \rfloor} \mathbf{e}_{(j \bmod d) \bmod q\mathcal{R}}$. As $\mathbf{b}'_j$ is uniform and independent of the shift, we have

$$\text{Adv}_{j,4}^{\bullet} = \text{Adv}_{j,5}^{\bullet}, \text{ and } \text{Adv}_{j,4}^{\circ} = \text{Adv}_{j,5}^{\circ},$$

Game $G_{j,6}$. We reintroduce an M-LWE secret in the j -th column by changing $\mathbf{b}'_j$. The challenger goes back to generating $\mathbf{R}_1, \mathbf{R}_2$ satisfying the spectral bound and uses their j -th column $\mathbf{r}_{1,j}, \mathbf{r}_{2,j}$ to define $\mathbf{b}'_j = \mathbf{r}_{1,j} + \mathbf{A}\mathbf{r}_{2,j} \bmod q\mathcal{R}$. Hence, at this stage, we now have $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 + q_L \mathbf{G}_{H,t_{j+1}}$. Again, under $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1}$ we have

$$\text{Adv}_{j,5}^{\bullet} \leq \text{Adv}_{j,6}^{\bullet} + \varepsilon_{\text{M-LWE}}, \text{ and } \text{Adv}_{j,5}^{\circ} \leq \text{Adv}_{j,6}^{\circ} + \varepsilon_{\text{M-LWE}},$$

Game $G_{j,7}$. We now switch back from $(\mathbf{R}'_1, \mathbf{R}'_2)$ to $(\mathbf{R}_1, \mathbf{R}_2)$ to answer signing queries. As we have changed the public key to go from $\mathbf{t}_j$ to $\mathbf{t}_{j+1}$ we just need to update the effective tag in the sampling procedure. More precisely, when $(\rho, i) \neq (2, i^+)$, the challenger samples $v_3^i \leftarrow \mathcal{D}_{\mathcal{R},s_4}$, computes $\mathbf{u}_i = \mathbf{u} + m_i \mathbf{d} -$

$v_3^i \mathbf{a}_3 \bmod q\mathcal{R}$ and finally samples $(\mathbf{v}_{1,1}^i, \mathbf{v}_{1,2}^i, \mathbf{v}_2^i)$ as

$$\begin{bmatrix} \mathbf{v}_{1,1}^i \\ \mathbf{v}_{1,2}^i \\ \mathbf{v}_2^i \end{bmatrix} = \text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u}_i, t_i \mathbf{1} - \mathbf{t}_{j+1}, s_1, s_2, s_3, s_4, s_L, s_H).$$

It then checks the verifications bounds and sends the signature $(t_i, \mathbf{v}_{1,2}^i, \mathbf{v}_2^i, v_3^i)$ to the adversary. Just like in $G_{j,3}$, using the trapdoor switching of Lemma 3.4, the relative error lemma of Lemma 2.3 and properties of the Rényi divergence, we have

$$\text{Adv}_{j,6}^{\bullet} \leq \mu \text{Adv}_{j,7}^{\bullet}^{1-1/2\lambda}, \text{ and } \text{Adv}_{j,6}^{\circ} \leq \mu \text{Adv}_{j,7}^{\circ}^{1-1/2\lambda},$$

Game $G_{j,8}$. Since we are no longer using $\mathbf{r}'_{1,j}, \mathbf{r}'_{2,j}$ to produce preimages, we can simulate $\mathbf{a}'_3$ under M-LWE. The challenger thus samples $\mathbf{a}'_3$ from $U(\mathcal{R}_q^d)$ and still sets $\mathbf{a}_3 = q_L b_H^{\lfloor j/d \rfloor} \mathbf{e}_{(j \bmod d)} - \mathbf{a}'_3 \bmod q\mathcal{R}$. As in $G_{j,2}$, we have

$$\text{Adv}_{j,7}^{\bullet} \leq \text{Adv}_{j,8}^{\bullet} + \varepsilon_{\text{M-LWE}}, \text{ and } \text{Adv}_{j,7}^{\circ} \leq \text{Adv}_{j,8}^{\circ} + \varepsilon_{\text{M-LWE}},$$

Game $G_{j,9}$. We finally remove the partial gadget from $\mathbf{a}_3$, i.e., the challenger samples $\mathbf{a}_3$ from $U(\mathcal{R}_q^d)$ directly. Because in $G_{j,8}$, $\mathbf{a}'_3$ is uniform and independent of the partial gadget, we directly obtain

$$\text{Adv}_{j,8}^{\bullet} = \text{Adv}_{j,9}^{\bullet}, \text{ and } \text{Adv}_{j,8}^{\circ} = \text{Adv}_{j,9}^{\circ},$$

Combining these hybrids, we then obtain

$$\text{Adv}_{j,0}^x \leq \varepsilon_{\text{M-LWE}} + \mu \left(2\varepsilon_{\text{M-LWE}} + \mu (\varepsilon_{\text{M-LWE}} + \text{Adv}_{j,9}^x)^{\frac{2\lambda-1}{2\lambda}} \right)^{\frac{2\lambda-1}{2\lambda}}$$

for $x \in \{\bullet, \circ\}$, which can be written as $\text{Adv}_{j,0}^x \leq h(\text{Adv}_{j,9}^x)$ for the function h defined in the theorem statement. Looping over j from 0 to $dk_H - 1$, we get $\text{Adv}_8^x = \text{Adv}_{0,0}^x \leq h^{\circ dk_H}(\text{Adv}_{dk_H-1,9}^x)$, where $h^{\circ dk_H}$ is the dk_H -th composition power of h . Notice that in $G_{dk_H-1,9}$, the public key is exactly $\mathbf{B} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 + q_L \mathbf{G}_{H,t+1} = \mathbf{R}_1 + \mathbf{A}\mathbf{R}_2 + q_L t^+ \mathbf{G}_H$ as desired. We now change the game slightly before reducing to M-SIS.

Game G_9 . We now change the winning conditions of the game by essentially rejecting the forgery when the tag guess is incorrect. More precisely, the challenger rejects if the tag t^* included in the forgery differs from t^+ . As t^+ is set differently in each branch, we have two different loss factor. But the key is that t^+ is hidden to the view of the adversary. In branch $\rho = 1$, recall that by the change in G_2 , the adversary must return $t^* \notin \{t_i; i \in [Q]\}$. So the guess is correct with probability $1/(|\mathcal{T}_w| - Q) = 1/(\binom{n}{w} - Q)$. In branch $\rho = 2$, the adversary must return $t^* \in \{t_i; i \in [Q]\}$, so the guess is correct with probability $1/Q$. In the end, we have

$$\text{Adv}_{dk_H-1,9}^{\bullet} = \left(\binom{n}{w} - Q \right) \text{Adv}_9^{\bullet}, \text{ and } \text{Adv}_{dk_H-1,9}^{\circ} = Q \text{Adv}_9^{\circ},$$

Game G_{10} . We now enforce certain bounds on the forgery to be able to use it to solve M-SIS. We define $\mathbf{R} = [\mathbf{R}_1^T | \mathbf{R}_2^T]^T$ in both branches and $\mathbf{s} = [\mathbf{s}_1^T | \mathbf{s}_2^T]^T$ in branch $\rho = 2$. When receiving the forgery $(t^*, \mathbf{v}_{1,2}^*, \mathbf{v}_2^*, v_3^*)$, the challenger proceeds as follows. In branch $\rho = 1$, the challenger aborts if $\|-\mathbf{R}\mathbf{v}_2^*\|_2 > \sqrt{nd}\|\mathbf{v}_2^*\|_2$. By Lemma 2.8, it holds that $\|\mathbf{R}\|_{2,\mathbb{Z}}$ is heuristically bounded by $\sqrt{nd}$ with probability $1/C$ for $C \approx 2$ (as $\mathbf{R}$ is indeed drawn from $\mathcal{B}_1$). Similarly, in branch $\rho = 2$, the challenger aborts if $\|[\mathbf{R}|\mathbf{s}]\Delta\mathbf{v}_{2m}\| > \sqrt{nd}\|\Delta\mathbf{v}_{2m}\|_2$ where $\Delta\mathbf{v}_{2m} = [(\mathbf{v}_2^* - \mathbf{v}_2^+)^T | m^* - m^+]^T$, with m^+ the message of the i^+ -th query and $\mathbf{v}_2^+$ the corresponding (partial) signature. Again, by Lemma 2.8, it holds that $\|[\mathbf{R}|\mathbf{s}]\|_{2,\mathbb{Z}}$ is heuristically bounded by $\sqrt{nd}$ with probability $1/C$ for $C \approx 2$. We then have

$$\text{Adv}_9^\bullet = C\text{Adv}_{10}^\bullet, \text{ and } \text{Adv}_9^\circ = C\text{Adv}_{10}^\circ,$$

Exploiting the Forgery. We now bound Adv_{10} by proving that one can use a successful adversary in G_{10} to solve a given instance of M-SIS. This will naturally bound Adv_{10} by the hardness bound for M-SIS. We deal with the two branches separately, leading to two different M-SIS instance. The challenger receives two instances $\mathbf{A}_1 = [\mathbf{I}_d | \mathbf{A} | \mathbf{a}_3 | \mathbf{d} | \mathbf{u}] \in \mathcal{R}_q^{d \times (2d+3)}$ and $\mathbf{A}_2 = [\mathbf{I}_d | \mathbf{A} | \mathbf{a}_3] \in \mathcal{R}_q^{d \times (2d+1)}$. It then uses a winning adversary to solve either of these two instances. The challenger will use $\mathbf{A}_\rho$ to define the cryptographic materials.

Branch $\rho = 1$. If $\rho = 1$, the challenger discards $\mathbf{A}_2$ and uses $\mathbf{A}_1$ to define the public parameters $\mathbf{A}, \mathbf{a}_3, \mathbf{d}, \mathbf{u}$. It then follows game G_{10} in the branch $\rho = 1$ and eventually receives a valid forgery $(m^*, t^*, \mathbf{v}_{1,2}^*, \mathbf{v}_2^*, v_3^*)$. Because the forgery wins G_{10} , it holds that $t^* = t^+$, and it also means that $\text{Verify}(\text{pk}, m^*, \text{sig}^*, \text{pp}) = 1$. By the change in G_{10} , we also have $\|\mathbf{R}\mathbf{v}_2^*\| \leq \sqrt{nd}\|\mathbf{v}_2^*\|_2 \leq \sqrt{nd}B_2$.

Because $t^* = t^+$ the matrix $\mathbf{A}_{t^* \mathbf{I}_d}$ used in the signature is $[\mathbf{I}_d | \mathbf{A}] - \mathbf{R}_1 - \mathbf{A}\mathbf{R}_2 \bmod q\mathcal{R}$. We then rewrite the recovery of $\mathbf{v}_{1,1}^*$ in Verify as

$$(\mathbf{v}_{1,1}^* - \mathbf{R}_1\mathbf{v}_2^*) + \mathbf{A}(\mathbf{v}_{1,2}^* - \mathbf{R}_2\mathbf{v}_2^*) + v_3^*\mathbf{a}_3 - m^*\mathbf{d} - \mathbf{u} = \mathbf{0} \bmod q\mathcal{R},$$

which means $\mathbf{A}_1\mathbf{x}^* = \mathbf{0} \bmod q\mathcal{R}$ for

$$\mathbf{x}^* = \begin{bmatrix} \begin{bmatrix} \mathbf{v}_{1,1}^* \\ \mathbf{v}_{1,2}^* \end{bmatrix} - \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \end{bmatrix} \mathbf{v}_2^* \\ v_3^* \\ -m^* \\ -1 \end{bmatrix},$$

The last coefficient ensures $\mathbf{x}^* \neq \mathbf{0}$, and based on the verification bounds and the one enforced in G_{10} , we have

$$\|\mathbf{x}^*\|_2 \leq \sqrt{\left(\sqrt{B_{1,1}^2 + B_{1,2}^2} + \sqrt{nd}B_2\right)^2 + B_3^2 + n + 1} = \beta_\bullet.$$

It shows that $\mathbf{x}^*$ is a valid solution of M-SIS $_{n,d,2d+3,q,\beta_\bullet}$ for the instance $\mathbf{A}_1$. As a result, we have

$$\text{Adv}_{10}^\bullet \leq \varepsilon_{\text{M-SIS}}^\bullet,$$

where $\varepsilon_{\text{M-SIS}}^{\bullet}$ is the hardness bound of $\text{M-SIS}_{n,d,2d+3,q,\beta_{\bullet}}$.

Branch $\rho = 2$. When $\rho = 2$, the challenger discards $\mathbf{A}_1$ and uses $\mathbf{A}_2$ to define $\mathbf{A}, \mathbf{a}_3$. It follows G_{10} in the branch $\rho = 2$ until it eventually receives a type $\bullet$ forgery (as per the winning conditions we changed in the game hops). We now again have $t^* = t^+$ and $\text{Verify}(\text{pk}, m^*, \text{sig}^*, \text{pp}) = 1$. Because $\mathbf{u} = \mathbf{A}_{t+\mathbf{I}_d}[\mathbf{v}_{1,1}^+ | \mathbf{v}_{1,2}^+ | \mathbf{v}_2^+] + \mathbf{a}_3 v_3^+ - m^+ \mathbf{d} \bmod q\mathcal{R}$, we can rewrite the definition of $\mathbf{v}_{1,1}^*$ from Verify as $\mathbf{A}_2 \mathbf{x}^* = \mathbf{0} \bmod q\mathcal{R}$ with

$$\mathbf{x}^* = \begin{bmatrix} \begin{bmatrix} \mathbf{v}_{1,1}^* - \mathbf{v}_{1,1}^+ \\ \mathbf{v}_{1,2}^* - \mathbf{v}_{1,2}^+ \end{bmatrix} - \begin{bmatrix} \mathbf{R}_1 \mathbf{s}_1 \\ \mathbf{R}_2 \mathbf{s}_2 \end{bmatrix} \begin{bmatrix} \mathbf{v}_2^* - \mathbf{v}_2^+ \\ m^* - m^+ \end{bmatrix} \\ v_3^* - v_3^+ \end{bmatrix},$$

To show $\mathbf{x}^* \neq \mathbf{0}$, we use the same unpredictability argument from prior works, e.g., [LLM⁺16, LNPS21, LNP22, JRS23, AGJ⁺24, JS25a, JS25b]. Because $m^* \neq m^+$, $\mathbf{x}^*$ depends on $(\mathbf{s}_1, \mathbf{s}_2)$. Yet, because it is hidden from the adversary's view, the unpredictability of $(\mathbf{s}_1, \mathbf{s}_2)$ ensures that $\mathbf{x}^* \neq \mathbf{0}$ with overwhelming probability. Then, based on the verification bounds of the forgery, the bounds on the i^+ -th query enforced in G_9 , and the bound enforced in G_{10} , it holds that

$$\|\mathbf{x}^*\|_2 \leq \sqrt{\left(2\sqrt{B_{1,1}^2 + B_{1,2}^2} + \sqrt{nd}\sqrt{4B_2^2 + n}\right)^2 + 4B_3^2} = \beta_{\bullet}.$$

It then holds $\mathbf{x}^*$ is a valid solution of $\text{M-SIS}_{n,d,2d+1,q,\beta_{\bullet}}$, yielding

$$\text{Adv}_{10}^{\bullet} \leq \varepsilon_{\text{M-SIS}}^{\bullet} + \text{negl}(\lambda),$$

where $\varepsilon_{\text{M-SIS}}^{\bullet}$ is the hardness bound of $\text{M-SIS}_{n,d,2d+1,q,\beta_{\bullet}}$. Finally combining all the advantage equations, we then obtain the final security as claimed in the theorem statement. $\square$

4.1.2 Size Performance. The signature scheme above implementing our sampler yields signatures of 3607 B and public keys of 19.03 KB, which are thus respectively 27% and 33% smaller than those of the signature scheme presented in [JS25b]. Our scheme reaches a similar security as theirs with 119 bits of security in the Core-SVP model (approximately NIST-II level). Note that in more realistic sieving cost models, this provides over 128 bits of security. The security is assessed using Theorem 4.1, and an example parameter set is given in Table C.1. We compare in Table 4.1 our scheme to that of [AGJ⁺24] and [JS25b]. The scheme of [AGJ⁺24] essentially uses the full sampler with $k_H = k = 5$, [JS25b] uses their truncated sampler with $k_H = 3$, and ours allows for choosing a smaller value $k_H = 2$.

4.1.3 Implementation Performance. More than the sole size metric, our sampler (and a fortiori the constructions using it) is also faster. The sampler of [JS25b] indeed required sampling a perturbation $\mathbf{p}$ from a distribution that

	$ pk $	$ t $	$ v_{1,2} $	$ v_2 $	$ v_3 $	Total	Sec.
[AGJ ⁺ 24]	48672	32	1666	4152	1038	6889 B	126
[JS25b]	29216	32	1653	2603	651	4939 B	122
Ours	19488	32	1629	1730	216	3607 B	119

Table 4.1. Comparison of the state-of-the-art standard model signatures based on well-known lattice assumptions. All sizes are given in bytes (B). The “Sec.” column gives the achieved security in the Core-SVP model. All schemes reach roughly NIST-II level. The scheme of [AGJ⁺24] essentially uses the full sampler with $k_H = k = 5$, [JS25b] uses their truncated sampler with $k_H = 3$, and ours uses $k_H = 2$.

depends on the tag. Beyond having to take larger parameters leading to less compact signatures, this dependency also impacts the sampling efficiency, and the signing time as a result. Indeed, because the tag changes at each signature, one needs to compute sampling materials online as they depended on both the secret key and the tag. The dependency on the tag prevents their precomputations at key generation.

Our sampler does not suffer from this limitation because of our enhanced tag management structure. In particular, all the sampling materials that depend on the secret key can be precomputed at key generation similarly to the full gadget sampler used in [AGJ⁺24] and prior works. We thus remove the almost 50 % overhead observed in the signing time in [JS25b] compared to [AGJ⁺24], which was mostly due to these online Schur complements computations. Additionally, as we have a smaller overall preimage dimension, our verification is even faster than [JS25b]. To assess concretely the gains of our sampler over [JS25b], we give a full implementation in C³ of our sampler in the context of our signature scheme. Our implementation is built upon that of [AGJ⁺24,JS25b] and uses the same libraries for the arithmetic backend for a fair comparison. We benchmark our implementation and the one of [JS25b] on a laptop with an Intel i7 12800H CPU running at 4.6 GHz, and use the same compilation options as [JS25b]. More precisely, we use `-O3 -march=native` using `gcc 11.4.0` with `pthread` disabled when building the Flint library. Note that we use Flint only for comparison purposes. Implementing from scratch the arithmetic backend would likely lead to vast timing improvements as it would make use of stack allocations instead of the (many) heap allocations made by Flint which slow down the implementation. The timing comparison is reported in Table 4.2.

4.2 Anonymous Credentials

The standard model signature provided in Section 4.1 thus improves upon that of [AGJ⁺24,JS25b], but it does so while retaining its structure and functionalities. In particular, the signature is designed to interact well with commit-

³ https://github.com/tagfriendly-sampler/tag-friendly_sampler

Procedure		Time (ms)			
		mean	med	min	max
Perturbation Sampler	[AGJ ⁺ 24]	51.983	51.838	50.883	53.636
	[JS25b]	80.234	80.280	77.287	82.914
	Alg. 3.3	51.937	51.826	50.309	52.884
Gadget Sampling	[AGJ ⁺ 24]	1.821	1.818	1.763	1.906
	[JS25b]	1.822	1.820	1.761	1.897
	Alg. 3.2, steps 3-14	1.669	1.651	1.597	1.870
Full sampler	[AGJ ⁺ 24]	56.526	56.534	55.349	57.931
	[JS25b]	83.931	83.891	79.302	85.769
	Alg. 3.2	55.254	55.247	54.286	55.606
Signature	[AGJ ⁺ 24]	56.949	56.953	55.651	58.583
	[JS25b]	84.255	84.168	79.898	86.821
	Alg. 4.3	55.308	55.322	54.426	56.088
Verification	[AGJ ⁺ 24]	1.127	1.126	1.104	1.191
	[JS25b]	0.771	0.771	0.746	0.802
	Alg. 4.4	0.563	0.564	0.549	0.595

Table 4.2. Benchmark results in milliseconds (ms). Statistics over 100 executions. Where applicable, the key and message were randomized. The timings for [AGJ⁺24] are those reported in [JS25b] from their parameter-optimized standalone signature implementation of [AGJ⁺24] with a full gadget.

ments and its verification equation can be efficiently proven using standard lattice zero-knowledge frameworks [LNP22,BS23]. It can then readily be plugged in many privacy-enhanced applications. We start by the anonymous credentials of [AGJ⁺24] which is a lattice version of the so-called *signature with efficient protocols* paradigm coined by Camenisch and Lysyanskaya [CL02].

Their anonymous credentials can be straightforwardly obtained from the underlying standard model signature. First, we enhance the commitment $\mathbf{d}m$ to the message $m \in T_1$ into $\mathbf{c} = \mathbf{r}_1 + \mathbf{A}\mathbf{r}_2 + \mathbf{D}\mathbf{m} \bmod q\mathcal{R}$, where $(\mathbf{r}_1, \mathbf{r}_2)$ is a very short (typically binary) commitment randomness to hide the message and provide anonymity, and we also tackle higher-dimensional messages $\mathbf{m} \in T_1^{2d+m'}$ to encode more attributes. The actual message $\mathbf{m}$ contains the user secret key $\mathbf{s} \in T_1^{2d}$ (associated to its public key $\text{upk} = [\mathbf{D}]_{:2d}\mathbf{s} \bmod q\mathcal{R}$), and m' attribute slots encoded in T_1 . A single attribute can be encoded within a single slot or several slots. The user then sends the commitment $\mathbf{c}$, as well as a proof π_1 of well-formedness, to the signer. The latter then proceeds as in the standard model signature except that it skips the commitment phase and uses the $\mathbf{c}$ it received from the user. It thus sends $(t, \mathbf{v}_{1,2}, \mathbf{v}_2, v_3)$ to the user such that t is in the proper tag space, and $\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2, v_3$ are short where $\mathbf{v}_{1,1}$ is uniquely recovered from the rest by the user and verifies

$$\mathbf{v}_{1,1} + \mathbf{A}\mathbf{v}_{1,2} + (q_L t \mathbf{G}_H - \mathbf{B})\mathbf{v}_2 + \mathbf{a}_3 v_3 = \mathbf{u} + \mathbf{c} \bmod q\mathcal{R}.$$

The latter can be re-written by the user as $\mathbf{v}'_{1,1} + \mathbf{A}\mathbf{v}'_{1,2} + (q_L t \mathbf{G}_H - \mathbf{B})\mathbf{v}_2 + \mathbf{a}_3 v_3 = \mathbf{u} + \mathbf{D}\mathbf{m} \bmod q\mathcal{R}$, where $\mathbf{v}'_{1,i} = \mathbf{v}_{1,i} - \mathbf{r}_i$ is obtained by merging the commitment randomness. The credential is $\text{cred} = (t, \mathbf{v}'_{1,2}, \mathbf{v}_2, v_3)$. For obvious privacy reasons, the users never show their credential directly but generate instead a zero-knowledge proof π_2 of the extended credential $(t, \mathbf{v}'_{1,1}, \mathbf{v}'_{1,2}, \mathbf{v}_2, v_3)$ and its hidden attributes $\mathbf{m}$ such that this equation is verified with all the elements being short, t in the proper tag space, and $\mathbf{m} \in T_1^{2d+m}$. The user also has the freedom of publishing certain attributes (except its secret key) from $\mathbf{m}$, then enabling selective disclosure.

The transcript of the issuance of credentials is then constituted of $(\mathbf{c}, \pi_1, \text{sig})$, while the showing protocol of the anonymous credentials system simply consists in the proof π_2 .

For completeness, we mention a very technical issue regarding the relation between the sampler modulus q and the proof modulus q_π for the zero-knowledge framework of [LNP22]. Indeed, since we require $q = q_L b_H^{k_H}$, having q divide the proof modulus q_π in π_1, π_2 is no longer the best option as it would require to increase the soundness parameter ℓ and thus increase the number of garbage commitments in the protocol of [LNP22]. Instead, we lift the modulo q equation over $\mathcal{R}$, prove it holds modulo q_π , and use the approximate range proof component to argue that q_π is sufficiently large so that it holds in $\mathcal{R}$ and then modulo q . This requires choosing q_π slightly larger than in [AGJ+24, JS25b], but we can now choose it to be prime, which then reduces the soundness parameter ℓ (as it depends on the smallest prime factor of q_π). All in all, this reduction of ℓ compensate for the increase of q_π . This lifting approach has been used in prior works, typically for verifiable encryption, e.g., [LNP22, Sec. 6.3][JS25a], or when the relation modulus q does not have the desired properties for the proof framework of [LNP22].

	Transcript $\mathbf{c}, \pi_1, \text{sig}$	Credential cred	Show π_2	Sec. $\lambda_{\text{anon}}, \lambda_{\text{uf}}$
[JS25b]	32.15 KB	4.87 KB	53.98 KB	135, 125
Ours	31.18 KB	3.52 KB	46.89 KB	132, 125

Table 4.3. Size comparison of the issuance transcript, credential, credential proof size, and security of our anonymous credentials and that of [JS25b]. The λ_{anon} and λ_{uf} are the bit-security for the anonymity and unforgeability of the anonymous credentials system.

Overall, our reduced preimage dimension improves the size of π_2 for similar security levels, as illustrated in Table 4.3. This size is indeed reduced to 46.89 KB, which is 13% smaller than the anonymous credentials of [AGJ+24] with the sampler of [JS25b], which was the most compact to date based on standard assumptions. These figures (both in absolute and relative values) might seem inconsistent with the ones provided for the signatures. However, we recall that

the framework of [LNP22] is very sensitive to the witness dimension, explaining why the 1.35 KB improvement of the signature stemming from our sampler is amplified to 7.09 KB in the resulting proof π_2 . Unfortunately, the framework of [LNP22] also incurs some overheads, which mechanically mitigate the relative improvement (13% instead of 27%) induced by our sampler.

We also expect improvements on the proof generation/verification timings due to the smaller witness dimension, but we leave the full implementation for future work.

For completeness, we provide the full description and security proof of our anonymous credentials in Appendix A.

4.3 Blind Signature

Our sampler and signature also improve upon the blind signature construction of [JS25a]. When thinking of blind signatures as one-show anonymous credentials, their construction is naturally very close to the anonymous credentials construction of [AGJ+24]. They however leverage the fact that each blind signature is only shown once to introduce a new technique called entropy recycling. Their idea is to inject more randomness than necessary in the commitment of the anonymous credentials, and use this extra entropy later on to mask part of the preimage allowing for giving this part in clear and minimize the impact of the zero-knowledge proof.

With a little bit more details, we can describe their construction and ours (which only differ by the sampler they use, and thus the concrete parameter selection) as follows. The issuance of the blind signature starts by the signer sending the tag to the user. The user then commits to message m as $\mathbf{c} = \mathbf{r}_{1,1} + \mathbf{A}\mathbf{r}_{1,2} + (q_L t \mathbf{G}_H - \mathbf{B})\mathbf{r}_2 + \mathbf{A}_3\mathbf{r}_3 + \mathbf{d}m \bmod q\mathcal{R}$. The randomness $\mathbf{r}_{1,i}$ features two components and is roughly defined as $\mathbf{r}_{1,i} = \mathbf{r}_{1,i,\text{mask}} + b_{1,i}\mathbf{r}_{1,i,\text{hid}}$, where $b_{1,i}$ is an integer decomposition base. The role of $\mathbf{r}_{1,1,\text{mask}}, \mathbf{r}_{1,2,\text{mask}}, \mathbf{r}_2, \mathbf{r}_3$ is to hide the low bits of $\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2, \mathbf{v}_3$ respectively later on. As such we choose them uniform modulo $b_{1,1}, b_{1,2}, b_2, b_3$ respectively so they can act as perfect masks of the low bits (where we truncate at these decomposition bases). On the other hand, $\mathbf{r}_{1,1,\text{hid}}, \mathbf{r}_{1,2,\text{hid}}$ can be very small as in the anonymous credentials case because their sole purpose is to ensure the commitment is hiding. In addition to this commitment, the user provides a so-called verifiable encryption-to-the-sky (i.e., verifiable encryption where the public key is generated as a ROM-output) of the message m , and proves the commitment and ciphertext are well-formed using the zero-knowledge proof π_1 .

If π_1 verifies, the signer produces a preimage of $\mathbf{u} + \mathbf{c}$ using the given sampler at hand, which it sends back to the user. The user can then, at a high-level, compute $\mathbf{w}_x = \mathbf{v}_x - \mathbf{r}_x$ and decompose it as $\mathbf{w}_{x,L} + b_x\mathbf{w}_{x,H}$. Because the $\mathbf{r}_x$ were chosen uniformly modulo b_x , the low bits $\mathbf{w}_{x,L}$ do not leak any information on $\mathbf{v}_x$ and can then be given in clear. Hence, the zero-knowledge proof π_2 only needs to prove knowledge of the $\mathbf{w}_{x,H}$. As $\mathbf{w}_{x,H}$ is a factor b_x smaller than $\mathbf{w}_x$, it saves a factor of b_x^2 on the zero-knowledge proof modulus, which improves its size.

Just as reported in [JS25b], we can plug TSampler in this framework for the generation of $(\mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2, \mathbf{v}_3)$. In the case of [JS25b], the norm overhead incurred by $\mathbf{r}_{1,1,\text{mask}}, \mathbf{r}_{1,2,\text{mask}}, \mathbf{r}_2, \mathbf{r}_3$, and the large truncation error imposed certain constraints on the parameter to keep a meaningful security while having the most compact sizes. In particular, they reached $k_H = k - \ell = 2$ but needed to increase the module rank to $d = 5$ (compared to $d = 4$ for the anonymous credentials) to compensate for the truncation error and the masks. In our case, we can reach the same security with $k_H = 2$ while keeping $d = 4$. Hence, although the final gadget dimension k_H is similar, our sampler allows for exploring different parameters and in particular reducing d to decrease the witness dimension for the zero-knowledge proof. Finally, we can reach the same level of security with a smaller modulus q overall. In the end, it leads to a blind signature of 26.19 KB, which represents a 27% improvement, while keeping a similar security level. We give in Table 4.4 the sizes of the issuance transcript and the blind signature.

	Transcript c, ct, π_1 , sig	Blind Signature $\mathbf{w}_L, \pi_2$	Sec. $\lambda_{\text{anon}}, \lambda_{\text{omuf}}$
[JS25b]	53.21 KB	36.28 KB	125, 126
Ours	36.15 KB	26.19 KB	128, 122

Table 4.4. Size comparison of the issuance transcript and blind signature ($\text{bsig} = (\mathbf{w}_L, \pi_2)$) of our blind signature scheme and that of [JS25b]. The λ_{anon} and λ_{omuf} are the bit-security for the anonymity and one-more unforgeability of the blind signature scheme.

On Round-Optimality. As our sampling parameters are relatively different, it also leads to a different choice for the decomposition bases $b_{1,1}, b_{1,2}, b_2, b_3$ to be optimal for security/efficiency. In particular, we observe that the optimal base for b_2, b_3 is actually 0 in our case, meaning that we do not need $\mathbf{r}_2, \mathbf{r}_3$. It thus leads to a smaller witness dimension for π_1 (hence considerably reducing the transcript size), but most importantly it means we do not need the signer to send the tag t in advance as it is not used in the commitment anymore. Hence, when $b_2 = 0$, the blind signature is actually round-optimal, as opposed to 3-round as presented in [JS25a]. For completeness, we provide the full description and security proof in Appendix B.

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A Anonymous Credentials

We now give the full description of the anonymous credentials system based on our signature and sampler. The construction follows the *signature with efficient protocols* framework of [JRS23,AGJ⁺24], and is proven secure in the security model of [FHS19], on well-known lattice assumptions. We note that our construction straightforwardly follows by plugging our standard model signature of Section 4.1 within the framework of [AGJ⁺24]. It only requires minor adjustments to make the commitment hiding and within the security proof, adjustments which are fully described in [AGJ⁺24], and recalled in Section 4.2. We however detail the construction and updated security proof for the sake of completeness and to ease future reference.

A.1 Syntax and Model

Let us first recall the syntax and security model of anonymous credentials system as defined in [FHS19]. Anonymous credentials, sometimes referred to as attribute-based credentials, is a primitive covering many privacy-enhanced use-cases. At a high-level, it allows users to ask organizations to emit credentials on their (possibly concealed) attributes, and later show their credentials to verifiers in a privacy-preserving manner, that is while revealing only the necessary information and in particular keeping anonymity and untraceability.

Definition A.1. *An anonymous credentials system is defined as a set of algorithms and protocols (Setup, OKeyGen, UKeyGen, Issue, Show) where*

- Setup: takes the security parameter λ and generates the public parameters pp (which includes λ)
- OKeyGen: takes pp and generates the organization’s key pair (opk, osk)
- UKeyGen: takes pp and generates the user’s key pair (upk, usk)
- Issue_{O,U}: is a protocol between an organization O , holding $(\text{osk}, \text{opk}, \text{upk}, \text{pp})$ and the user’s disclosed attributes $\mathbf{m}(\mathcal{I}) = ([\mathbf{m}]_i)_{i \in \mathcal{I}}$, and a user U holding $(\text{usk}, \text{upk}, \text{opk}, \text{pp})$ and their attributes $\mathbf{m}$. The user U either gets a credential cred on its attributes or $\perp$ if the protocol aborted, while O only gets a notification of whether the execution was successful.
- Show_{U,V}: is a protocol between a user U holding $(\text{usk}, \text{upk}, \text{opk}, \text{pp}, \mathbf{m}, \text{cred})$ and a verifier V holding $(\text{opk}, \text{pp}, \mathbf{m}(\mathcal{I}'))$ where $\mathbf{m}(\mathcal{I}') = ([\mathbf{m}]_i)_{i \in \mathcal{I}'}$ are the attributes disclosed by the user. The protocol outputs 1 to V if the credential cred is valid with respect to the disclosed attributes, and 0 otherwise.

Although there is no consensus on the security model of anonymous credentials, the popular model of [FHS19] has been used for many constructions in the classical and post-quantum setting. The only peculiarity is that they assume all the attributes (except usk) are revealed to the organization in the Issue protocol. This is only to formalize the unforgeability notion in an easier way that is agnostic to the specific tools used in each construction. Some works, e.g., [BLNS23], have also proposed to extend the notion of one-more unforgeability, originally

introduced for blind signatures, to anonymous credentials. In this work, we decide to keep the model of [FHS19] but notice that our construction enables hiding all (or part of) the attributes at will in the issuance protocol. The model of [FHS19] requires that an anonymous credentials system as defined above must be *anonymous* and *unforgeable*.

Security Notations. They formalize these notions with security games, which require the following security notations for variables and oracles.

- HU: Set of honest user indices (empty at the beginning)
- CU: Set of corrupt user indices (empty at the beginning)
- ctr: Issuance counter (0 at the beginning)
- A: Set of $(j, j', \mathbf{m})$ filled after each successful issuance of a credential for user j with attributes $\mathbf{m}$ and issuance index j' ($\mathcal{O}_{\text{ObtIss}}$ or $\mathcal{O}_{\text{Issue}}$)
- $\mathcal{O}_{\text{HU}}(j)$: Given a user index j , it returns $\perp$ if $j \in \text{HU} \cup \text{CU}$. Otherwise, it performs $(\text{upk}_j, \text{usk}_j) = \text{UKeyGen}(\text{pp})$, updates $\text{HU} = \text{HU} \cup \{j\}$ and returns upk_j .
- $\mathcal{O}_{\text{CU}}(j, \text{upk})$: Given a user index j and optionally a public key upk , it registers a new user with public key upk if $j \notin \text{HU}$. Otherwise, if $j \in \text{HU}$, it updates $\text{HU} = \text{HU} \setminus \{j\}$ and $\text{CU} = \text{CU} \cup \{j\}$ and returns usk_j . The former case models the ability to register users with malformed keys, i.e., who do not know the associated secret key.
- $\mathcal{O}_{\text{ObtIss}}(j, \mathbf{m})$: Given a user index $j \in \text{HU}$ and attributes $\mathbf{m}$, it then runs $\text{Issue}_{\mathcal{O}, U_j}((\text{osk}, \text{opk}, \text{upk}_j, \text{pp}, \mathbf{m}); (\text{usk}_j, \text{upk}_j, \text{opk}, \text{pp}, \mathbf{m}))$ assuming the roles of both $\mathcal{O}$ and user j . If successful, it increments ctr, stores the resulting credential, and stores $(j, \text{ctr}, \mathbf{m})$ in A. It returns $\top$ if the execution succeeded. If $j \notin \text{HU}$, it simply returns $\perp$.
- $\mathcal{O}_{\text{Obtain}}(j, \mathbf{m})$: Given a user index j and attributes $\mathbf{m}$, it returns $\perp$ if $j \notin \text{HU}$. Otherwise, it runs $\text{Issue}_{\mathcal{A}, U}(\cdot, (\text{usk}_j, \text{upk}_j, \text{opk}, \text{pp}, \mathbf{m}))$ with the adversary $\mathcal{A}$ posing as the organization.
- $\mathcal{O}_{\text{Issue}}(j, \mathbf{m})$: Given a user index j and attributes $\mathbf{m}$, it returns $\perp$ if $j \notin \text{CU}$. Else, it runs $\text{Issue}_{\mathcal{O}, \mathcal{A}}((\text{osk}, \text{opk}, \text{upk}_j, \text{pp}, \mathbf{m}), \cdot)$ where the adversary plays the user. If successful, it increments the issuance counter ctr, stores the credential, and adds $(j, \text{ctr}, \mathbf{m})$ in A.
- $\mathcal{O}_{\text{Show}}(j', \mathbf{m}_{\mathcal{I}}^{(j')})$: It takes an issuance index j' and disclosed attributes $\mathbf{m}_{\mathcal{I}}^{(j')}$. The issuance index corresponds to a successfully issued credential $\text{cred}^{(j')}$ on $\mathbf{m}_{\mathcal{I}}^{(j')}$ for a user j during the j' -th query to $\mathcal{O}_{\text{ObtIss}}$ or $\mathcal{O}_{\text{Obtain}}$. If $j \in \text{HU}$, it runs $\text{Show}_{U, \mathcal{A}}((\text{usk}_j, \text{opk}, \text{pp}, \mathbf{m}^{(j')}(\mathcal{I}), \text{cred}^{(j')}(\mathcal{I}), \cdot)$ with the adversary posing as the verifier, and returns $\perp$ if $j \notin \text{HU}$.

Anonymity. The anonymity captures the fact that a user showing cred while disclosing $\mathbf{m}(\mathcal{I})$ remains anonymous among all the users who have the same disclosed attributes, i.e., those for which $\mathbf{m}'(\mathcal{I}) = \mathbf{m}(\mathcal{I})$. It prevents identifying the user interacting Show, as no information is leaked on cred or $\mathbf{m}(\mathcal{I})$, and it also prevents tracing the user across different showings.

Definition A.2. An anonymous credentials system is anonymous if, for any PPT adversary $\mathcal{A}$, $\text{Adv}_{an}[\mathcal{A}] = |\mathbb{P}[\text{Game}_{an-0}(\mathcal{A}) = 1] - \mathbb{P}[\text{Game}_{an-1}(\mathcal{A}) = 1]|$ is negligible in λ , where $\text{Game}_{an-b}(\mathcal{A})$ is defined below for $b \in \{0, 1\}$.

```

Gamean-b( $\mathcal{A}$ ):
1  $\text{pp} \leftarrow \text{Setup}(1^\lambda)$ 
2  $(\text{osk}, \text{opk}) \leftarrow \text{OKeyGen}(\text{pp})$ 
3  $(j'_0, j'_1, \mathbf{m}(\mathcal{I})) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{HU}}, \mathcal{O}_{\text{CU}}, \mathcal{O}_{\text{Obtain}}, \mathcal{O}_{\text{Show}}}(\text{opk}, \text{osk})$ 
4 if  $\text{ctr} < \min(j'_0, j'_1)$ , return 0
5 Look up  $\mathbf{A}$  for  $(j_0, j'_0, \mathbf{m}^{(j'_0)})$  and  $(j_1, j'_1, \mathbf{m}^{(j'_1)})$ 
6 if  $\mathbf{m}^{(j'_0)}(\mathcal{I}) \neq \mathbf{m}(\mathcal{I}) \vee \mathbf{m}^{(j'_1)}(\mathcal{I}) \neq \mathbf{m}(\mathcal{I})$ , return 0
7 Execute  $(\emptyset, 1) \leftarrow \text{Show}_{\mathcal{C}, \mathcal{A}}((\text{usk}_{j_b}, \text{opk}, \text{pp}, \mathbf{m}^{(j'_b)}, \text{cred}^{(j'_b)}, \mathcal{I}), \cdot)$ 
8  $b^* \leftarrow \mathcal{A}^{\mathcal{O}_{\text{HU}}, \mathcal{O}_{\text{CU}}, \mathcal{O}_{\text{Obtain}}, \mathcal{O}_{\text{Show}}}(\text{opk}, \text{osk})$ 
9 if  $\mathcal{O}_{\text{CU}}$  has been queried on  $j_0$  or  $j_1$ , return 0
10 return  $b^*$ 

```

Unforgeability. The unforgeability notion ensures that a user cannot successfully show a credential that it did not receive from the legitimate issuer. It means that it cannot impersonate an honest user, and cannot forge fresh credentials that have not been issued in the `Issue` protocol. It therefore encompasses forgeries where an adversary would (1) impersonate an honest user, (2) trick the verifier with a falsified proof, and (3) forge a fresh credential, that is a signature.

Definition A.3. An anonymous credentials system is unforgeable if, for any PPT adversary $\mathcal{A}$, $\text{Adv}_{uf}[\mathcal{A}] = \mathbb{P}[\text{Game}_{uf}(\mathcal{A}) = 1]$ is negligible in λ , where $\text{Game}_{uf}(\mathcal{A})$ is defined below.

```

Gameuf( $\mathcal{A}$ ):
1  $\text{pp} \leftarrow \text{Setup}(1^\lambda)$ 
2  $(\text{osk}, \text{opk}) \leftarrow \text{OKeyGen}(\text{pp})$ 
3  $\mathbf{m}(\mathcal{I}) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{HU}}, \mathcal{O}_{\text{CU}}, \mathcal{O}_{\text{Obtain}}, \mathcal{O}_{\text{Issue}}, \mathcal{O}_{\text{Show}}}(\text{opk})$ 
4  $(\emptyset, b) \leftarrow \text{Show}_{\mathcal{A}, \mathcal{C}}(\cdot, (\text{opk}, \text{pp}, \mathbf{m}(\mathcal{I})))$ 
5 if  $\exists(j, j', \mathbf{m}^{(j')}) \in \mathbf{A}$  s.t.  $\mathbf{m}^{(j')}(\mathcal{I}) = \mathbf{m}(\mathcal{I})$  and  $j \in \text{CU}$ , return 0
6 return  $b$ 

```

A.2 Our Anonymous Credentials Construction

We now give the full description of `Setup`, `OKeyGen`, `UKeyGen`, `Issue`, and `Show`. They follow the same blueprint as [AGJ⁺24] but based on our new signature scheme of Section 4.1. The setup of parameters is the same as Algorithm 4.1 with minor tweaks to allow for more attributes and for the user key generation. Indeed, the signature of Section 4.1 can be seen as signing a message of dimension 1, while in the anonymous credentials, we effectively sign $(\text{usk}, \mathbf{m})$ which is a message of dimension $m = 2d + m'$. Also, in the unforgeability proof, we need to slightly adjust the rejection sampling argument to account for the commitment randomness in addition to the message. We highlight the lines that change.

Algorithm A.1: AC.Setup(λ)

1. Choose $n \geq \kappa$ to be powers of two.
2. Choose $d \in \mathbb{N}^\times$.
3. Choose $m' \in \mathbb{N}$ and set $m \leftarrow 2d + m'$. $\triangleright m'$ number of attribute slots
4. Choose $b_H \in \mathbb{N}^\times$ a small prime s.t. $b_H = 2\kappa + 1 \pmod{4\kappa}$ and $b_H > \sqrt{\kappa}^\kappa$.
5. Choose $k_H, b_L, k_L \in \mathbb{N}^\times$
6. $q_L \leftarrow b_L^{k_L}$
7. $q_H \leftarrow b_H^{k_H}$
8. $q \leftarrow q_L q_H$
9. Choose $w \in \mathbb{N}^\times$ st $\binom{n}{w} \geq Q$.
10. $\mathcal{T}_w \leftarrow \{t \in T_1 : \|t\|_1 = w\}$.
11. $B_R \leftarrow \frac{3}{4}(\sqrt{nd} + \sqrt{ndk_H} + 6)$
12. For $x \in \{L, H\}$, define $\mathbf{G}_x = [1 \dots |b_x^{k_x-1}|] \otimes \mathbf{I}_d \in \mathcal{R}^{d \times dk_x}$
13. Choose $\varepsilon \in (0, 1)$ $\triangleright$ typically $\varepsilon \approx 2^{-40}$
14. $s_L \leftarrow b_L \cdot \sqrt{\ln(2ndk_L/\varepsilon)/\pi}$ $\triangleright s_L \approx b_L \eta_\varepsilon(\mathbb{Z}^{ndk_L}) \geq b_L \eta_{\varepsilon/dk_L}(\mathbb{Z}^n)$ by Lemma 2.4
15. $s_H \leftarrow b_H \cdot \sqrt{\ln(2ndk_H/\varepsilon)/\pi}$ $\triangleright s_H \approx b_H \eta_\varepsilon(\mathbb{Z}^{ndk_H}) \geq b_H \eta_{\varepsilon/dk_H}(\mathbb{Z}^n)$ by Lemma 2.4
16. $\eta \leftarrow \sqrt{\ln(2nd(k_L + k_H)/\varepsilon)/\pi}$. $\triangleright \eta \approx \eta_\varepsilon(\mathbb{Z}^{nd(k_L+k_H)})$ by Lemma 2.4
17. $s_1 \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$.
18. $s_2 \leftarrow s_L^2/\sqrt{s_L^2 - \eta^2}$.
19. $s_{1,1} \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) \cdot (b_L^{2k_L} - 1)/(b_L^2 - 1) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$.
20. $s_3 \leftarrow \sqrt{3s_H^2 B_R/\sqrt{s_H^2 - \eta^2}}$.
21. $s_4 \leftarrow \sqrt{3s_H^2/\sqrt{s_H^2 - \eta^2}}$.
22. $(\alpha_{1,1}, \alpha_{1,2}) \leftarrow (s_{1,1}/(n\sqrt{md/2} + \sqrt{nd}), s_3/(n\sqrt{md/2} + \sqrt{nd}))$.
23. $M_{1,i} \leftarrow \exp(\pi/\alpha_{1,i}^2)$.
24. $\mathbf{D}_s \leftarrow U(\mathcal{R}_q^{d \times 2d})$.
25. $\mathbf{D} \leftarrow U(\mathcal{R}_q^{d \times m'})$.
26. $\mathbf{A} \leftarrow U(\mathcal{R}_q^{d \times d})$.
27. $\mathbf{a}_3 \leftarrow U(\mathcal{R}_q^d)$.
28. $\mathbf{u} \leftarrow U(\mathcal{R}_q^d)$.
29. **params** $\leftarrow (n, d, m', m, b_H, k_H, q_H, b_L, k_L, q_L, q, w, B_R, \varepsilon, s_L, s_H, \eta, s_1, s_2, s_3, s_4, s_{1,1})$

Output: $\text{pp} = (\text{params}, \mathbf{A}, \mathbf{D}_s, \mathbf{D}, \mathbf{a}_3, \mathbf{u})$

Algorithm A.2: AC.OKeYGen(pp)

1. $(\mathbf{B}, (\mathbf{R}_1, \mathbf{R}_2)) \leftarrow \text{KeyGen}(\text{pp})$ $\triangleright$ Algorithm 4.2

Output: $\text{opk} = \mathbf{B}, \text{osk} = (\mathbf{R}_1, \mathbf{R}_2)$

Algorithm A.3: AC.UKeyGen(pp)

1. $\mathbf{s} \leftarrow U(T_1^{2d})$
2. $\mathbf{b} \leftarrow \mathbf{D}_s \mathbf{s} \pmod{q\mathcal{R}}$

Output: $\text{upk} = \mathbf{b}, \text{usk} = \mathbf{s}$

Algorithm A.4: AC.Issue $_{O,U}((\text{osk}, \text{opk}, \text{upk}, \text{pp}, \mathbf{m}(\mathcal{I}), \text{st}); (\text{usk}, \text{upk}, \text{opk}, \text{pp}, \mathbf{m}))$

- User U .
1. $\mathbf{r}_1, \mathbf{r}_2 \leftarrow U(T_1^d)$.
2. $\mathbf{c} \leftarrow \mathbf{r}_1 + \mathbf{A}\mathbf{r}_2 + \mathbf{D}_s \text{usk} + \mathbf{D}\mathbf{m} \pmod{q\mathcal{R}}$.
3. $\pi_1 \leftarrow \text{Prove}_1(\text{statement}_1, \mathbf{r}_1, \mathbf{r}_2, \text{usk}, \mathbf{m}(\overline{\mathcal{I}}))$ $\triangleright$ See Sect. A.3
4. Send $(\mathbf{c}, \pi_1)$ to O .
- Organization O .
5. **if** $\text{Verify}_1(\pi_1, \text{statement}_1) = 0$, **abort**.
6. $t \leftarrow F(\text{st}) \in \mathcal{T}_w$. $\triangleright t \in T_1, \|t\|_1 = w$
7. $v_3 \leftarrow \mathcal{D}_{\mathcal{R}, s_4}$.
8. $(\mathbf{v}'_{1,1}, \mathbf{v}'_{1,2}, \mathbf{v}_2) \leftarrow \text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u} + \mathbf{c} - v_3 \mathbf{a}_3 \pmod{q\mathcal{R}}, t\mathbf{1}, s_1, s_2, s_3, s_4, s_L, s_H)$


```

9. if  $\|\mathbf{v}'_{1,1}\|_2 > B_{1,1} \vee \|\mathbf{v}'_{1,2}\|_2 > B_{1,2} \vee \|\mathbf{v}_2\|_2 > B_2 \vee \|\mathbf{v}_3\|_2 > B_3$  goto 7).
10. Send  $(t, \mathbf{v}'_{1,2}, \mathbf{v}_2, v_3)$  to  $U$ 
11.  $\mathbf{st} \leftarrow \mathbf{st} + 1$ .
    User  $U$ .
12. if  $\text{Verify}(\text{opk}, \mathbf{m}, (t, \mathbf{v}'_{1,2}, \mathbf{v}_2, v_3), \text{pp}) = 0$ , abort
13.  $\mathbf{v}_{1,2} \leftarrow \mathbf{v}'_{1,2} - \mathbf{r}_2$ 

```

▷ Algorithm 4.4

Output: U gets $\text{cred} = (t, \mathbf{v}_{1,2}, \mathbf{v}_2, v_3)$

Algorithm A.5: $\text{AC.Show}_{U,V}((\text{usk}, \text{upk}, \text{opk}, \text{pp}, \mathbf{m}, \text{cred}); (\text{opk}, \text{pp}, \mathbf{m}(\mathcal{I}'))$

```

User  $U$ .
1.  $\mathbf{v}_{1,1} \leftarrow \mathbf{u} + \mathbf{D}_s \text{usk} + \mathbf{Dm} - \mathbf{A}\mathbf{v}_{1,2} - (q_L \mathbf{G}_{H,t1} - \mathbf{B})\mathbf{v}_2 - v_3 \mathbf{a}_3 \bmod q\mathcal{R} \in \mathcal{R}^d$ .
2.  $\pi_2 \leftarrow \text{Prove}_2(\text{statement}_2, t, \mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2, v_3, \text{usk}, \mathbf{m}(\mathcal{I}'))$ 
3. Send  $\pi_2$  to  $V$ 
    Verifier  $V$ .
4.  $b \leftarrow \text{Verify}_2(\pi_2, \text{statement}_2)$ 

```

▷ See Sect. A.3

Output: V gets b (1 if valid, 0 otherwise)

A.3 Zero-Knowledge Arguments

Our anonymous credentials system resorts to two different zero-knowledge arguments, both of which can be made non-interactive and can be instantiated with the framework of [LNP22]. The first one occurs in `Issue` where the user needs to prove the well-formedness of $\mathbf{c}$ and that it knows the secret key usk associated to upk . Within the issuance phase, we give the freedom to hide certain attributes from the organization. In that case, for a set of indices $\mathcal{I}$ designating the disclosed attribute slots, we denote by $\bar{\mathcal{I}} = [m'] \setminus \mathcal{I}$, and $\mathbf{D}_{\mathcal{I}}$ and $\mathbf{D}_{\bar{\mathcal{I}}}$ the submatrices of $\mathbf{D}$ such that $\mathbf{Dm} = \mathbf{D}_{\mathcal{I}}\mathbf{m}(\mathcal{I}) + \mathbf{D}_{\bar{\mathcal{I}}}\mathbf{m}(\bar{\mathcal{I}})$. The user then proves knowledge of $(\mathbf{r}_1, \mathbf{r}_2, \text{usk}, \mathbf{m}(\bar{\mathcal{I}})) \in \mathcal{R}^{4d+m'-|\mathcal{I}|}$ such that

$$\begin{aligned}
\mathbf{r}_1 + \mathbf{A}\mathbf{r}_2 + \text{upk} + \mathbf{D}_{\bar{\mathcal{I}}}\mathbf{m}(\bar{\mathcal{I}}) &= \mathbf{c} - \mathbf{D}_{\mathcal{I}}\mathbf{m}(\mathcal{I}) \bmod q\mathcal{R}, \\
\mathbf{D}_s \text{usk} &= \text{upk} \bmod q\mathcal{R}, \\
\mathbf{r}_1, \mathbf{r}_2 &\in T_1^d, \quad \mathbf{m}(\bar{\mathcal{I}}) \in T_1^{m'-|\mathcal{I}|}, \quad \text{usk} \in T_1^{2d}
\end{aligned}$$

The statement statement_1 contains all the public elements of the above relation, those not in red.

The second argument required in the `Show` protocol essentially proves knowledge of the verification of credential over the hidden attributes. Hence, the user proves knowledge of $(t, \mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2, v_3, \text{usk}, \mathbf{m}(\bar{\mathcal{I}}')) \in \mathcal{R}^{2+d(4+k_H)+m'-|\mathcal{I}'|}$ such that

$$\begin{aligned}
\mathbf{v}_{1,1} + \mathbf{A}\mathbf{v}_{1,2} + (q_L t \mathbf{G}_H - \mathbf{B})\mathbf{v}_2 + \mathbf{a}_3 v_3 - \mathbf{D}_s \text{usk} - \mathbf{D}_{\bar{\mathcal{I}'}}\mathbf{m}(\mathcal{I}') &= \mathbf{u}_{\mathcal{I}'} \bmod q\mathcal{R}, \\
\|\mathbf{v}_{1,1}\|_2 \leq B'_{1,1}, \|\mathbf{v}_{1,2}\|_2 \leq B'_{1,2}, \|\mathbf{v}_2\|_2 \leq B_2, \|\mathbf{v}_3\|_2 \leq B_3 \\
t \in T_1, \quad \|t\|_2 = \sqrt{w}, \quad \mathbf{m}(\bar{\mathcal{I}}) &\in T_1^{m'-|\mathcal{I}|}, \quad \text{usk} \in T_1^{2d}
\end{aligned}$$

where $\mathbf{u}_{\mathcal{I}'} = \mathbf{u} + \mathbf{D}_{\mathcal{I}'}\mathbf{m}(\mathcal{I}')$, and $B'_{1,i} = B_{1,i} + \sqrt{nd}$ due to the fact that we subtracted $\mathbf{r}_i$ to $\mathbf{v}'_{1,i}$ to obtain the final credential. Also, note that because we deal with tags of the form $t\mathbf{1}$, we have $\mathbf{G}_{H,t1} = t\mathbf{G}_{H,1} = t\mathbf{G}_H$ where $\mathbf{G}_H =$

$[1|b_H|\dots|b_H^{k_H-1}]\otimes\mathbf{I}_d$. The statement `statement2` contains all the public elements of the above relation, as before.

Instantiating [LNP22]. As mentioned, these two relations can be efficiently proven using the framework of [LNP22] with the optimizations of [LN22]. The only subtlety is that this framework requires a careful choice of modulus q_π . Yet, our modulus q is now of the form $q = q_L b_H^{k_H}$. As for the construction of [AGJ⁺24], we could use $q_\pi = q \cdot q_1$ where q_1 is a slack to virtually increase the proof modulus, which also requires multiplying the modulo q equations by q_1 . However, doing so would place constraints on q_L and b_H . Typically, they should be prime that split in 2 prime ideal factors (i.e., $q_L, b_H = 5 \bmod 8$). Even so, the soundness parameter ℓ in [LNP22] depends on the smallest prime factor of q_π , which here would likely be b_H , which is itself relatively small. To ensure a sufficient soundness, one would require choosing a much larger ℓ , thus increasing the number of garbage commitments and deteriorating the proof size.

Instead, we opt for the lifting approach. The idea is to choose q_π to be a large prime and lift all the modulo q equations mod q_π , and prove the wraparound terms are small enough so that the equation holds over $\mathcal{R}$. Concretely, to prove $f(\mathbf{w}) = \mathbf{0} \bmod q\mathcal{R}$, with f being a quadratic function of the witness $\mathbf{w}$, we first write the equation in $\mathcal{R}$ as $f(\mathbf{w}) + q\mathbf{j} = \mathbf{0}$. To prove this equation, we first prove it modulo q_π , and we also prove that $\|f(\mathbf{w}) + q\mathbf{j}\|_\infty < q_\pi$ so as to show there is no wraparound modulo q_π and that the equation indeed holds in $\mathcal{R}$. As $\mathbf{w}$ is proven small in our relations with exact bounds, and that the public matrices and vectors have coefficients bounded by $q/2$ at worst, we can easily derive a bound on $f(\mathbf{w})$. We then need to prove $\mathbf{j}$ is small enough. For that, observe that here we do not need an exact norm bound but only a relaxed one. So we can use the approximate range proof component to prove a bound on $\mathbf{j}$ that should be small enough for the above condition on q_π to be verified. Also, we can avoid committing to $\mathbf{j}$ by expressing it as $\mathbf{j} = -q^{-1}f(\mathbf{w})$, which is a quadratic relation of the witness.

This lifting approach increases the bound proven in the approximate range proof because of the presence of $\mathbf{j}$, which then requires taking a larger q_π . However, as q_π is now a large prime, the soundness parameter ℓ can be taken very small, thus reducing the contribution of the garbage commitments in the proof size. We therefore adopt this lifting strategy in our instantiation of [LNP22] to prove the two relations above. It thus avoids placing further restrictions on the signature modulus q and its factors q_L, b_H . We note that this lifting strategy has already been described and used in [LNP22], and subsequent works like [AKSY22, JS25a] for verifiable encryption proofs.

As this framework and lifting strategy has already been covered at length, we only give the security statements for both relations. Note that we also include the bimodal optimizations of [LN22] for better compactness, and include the bimodal bit $\mathbf{b}$ in the witness. This approach has been fully described in [JS25a]. Recall that c_N is the Gaussian tailcut in dimension N , but we now take a tailcut that yields a bound verified with overwhelming probability instead of p . To avoid introducing different notations for the two different instantiations of the frame-

work of [LNP22], we use the same parameter names ($n', \rho, \tilde{\eta}, m_1, d_2, m_2, \ell, D, \gamma$, etc.) for both. We however note that their actual values are different depending on the relation to prove, as shown in Tables C.2 and C.3.

Lemma A.1 (Security of Prove₁). *Let $\mathcal{R}'$ be the power-of-two cyclotomic ring of degree $n' \leq n$ and let $k' = n/n'$. We denote by τ the coefficient embedding in $\mathcal{R}'$. Let $\rho, \tilde{\eta}$ define the challenge space $\mathcal{C}_{\rho, \tilde{\eta}} = \{c \in \tau^{-1}([- \rho, \rho]^{n'}) : c^* = c \wedge \sqrt[64]{\|c^{64}\|_1} \leq \tilde{\eta}\}$. Let M_1, M_2, M_3 be in $(1, \infty)$, and let $\alpha_j = \sqrt{\pi / \ln M_j}$ for $j \in [3]$. Define the witness $\mathbf{s}_1 = (\mathbf{r}_1, \mathbf{r}_2, \text{usk}, \mathbf{m}(\overline{\mathcal{I}}), \mathbf{b})$ of dimension $m_1 = k'(4d + m' - |\mathcal{I}|) + 1$ when embedded in $\mathcal{R}'$. Then, define $B = \sqrt{n'm_1}$ and $B_j = \sqrt{nd}(1/2 + 1/q + nd/2 + nm/2)$. We then define $\sigma_1 = \alpha_1 \tilde{\eta} B$, $\sigma_2 = \alpha_2 \tilde{\eta} \sqrt{n'm_2}$, and $\sigma_3 = \alpha_3 \sqrt{337} \sqrt{B^2 + B_j^2}$. We then define $B_{arp} = \frac{2}{\sqrt{26}} c_{256} \sigma_3 \sqrt{256}$. Then, let $d_2, m_2, \ell, q_{\pi_1}$ be positive integers such that*

q_{π_1} prime s.t. $q_{\pi_1} = 5 \bmod 8$,

$$q_{\pi_1} > \max \left(41n'm_1 B_{arp}, B_{arp}^2 + B_{arp} \sqrt{n'm_1}, 1 + q \left(\frac{1 + nd + nm}{2} + B_{arp} \right) \right),$$

We also let γ be an even divisor of $q_{\pi_1} - 1$, and D be the largest integer such that $2^{D-1} \rho n' < \gamma$. Then, (Prove₁, Verify₁) is knowledge sound with an extractor running in expected polynomial time and soundness error

$$\varepsilon_{snd,1} = \frac{2}{|\mathcal{C}_{\rho, \tilde{\eta}}|} + q_{\pi_1}^{-n'/2} + q_{\pi_1}^{-2\ell} + 2^{-128} + \varepsilon_{\text{M-SIS}}^{\text{sound}}$$

and zero-knowledge with loss $\varepsilon_{zk,1} = \varepsilon_{\text{M-LWE}}^{\text{zk}} + \text{negl}(\lambda)$. The term $\varepsilon_{\text{M-SIS}}^{\text{sound}}$ is the hardness bound of M-SIS _{$n', d_2, m_1 + m_2, q_{\pi_1}, \beta_{\pi_1}$} where

$$\beta_{\pi_1} = 4\tilde{\eta} \sqrt{4c_{n'm_1}^2 \sigma_1^2 n'm_1 + \left(2c_{n'm_2} \sigma_2 \sqrt{n'm_2} + (2^D \tilde{\eta} + \gamma) \sqrt{n'd_2} \right)^2},$$

while $\varepsilon_{\text{M-LWE}}^{\text{zk}}$ is the hardness bound of M-LWE _{$n', m_2 - d_2 - \lfloor 256/n' \rfloor - \ell - 1, m_2, q_{\pi_1}, \mathcal{B}_1, \mathcal{B}_1$} (with $\mathcal{B}_1$ the centered binomial distribution of parameter 1 over $\mathcal{R}'$).

The size of π_1 is computed from the parameters of the proof system introduced in Lemma A.1 by

$$\begin{aligned} |\pi_1| = & n'd_2 (\lceil \log_2 q_{\pi_1} \rceil - D + 2.25) + \left(\frac{256}{n'} + 2\ell + 1 \right) \lceil \log_2 q_{\pi_1} \rceil \\ & + n' \lceil \log_2 (2\rho + 1) \rceil + n'm_1 \left(\frac{1}{2} + \log_2 \sigma_1 \right) \\ & + n'(m_2 - d_2) \left(\frac{1}{2} + \log_2 \sigma_2 \right) + 256 \left(\frac{1}{2} + \log_2 \sigma_3 \right). \end{aligned}$$

Lemma A.2 (Security of Prove₂). *Let $\mathcal{R}'$ be the power-of-two cyclotomic ring of degree $n' \leq n$ and let $k' = n/n'$. We denote by τ the coefficient embedding*

in $\mathcal{R}'$. Let $\rho, \tilde{\eta}$ define the challenge space $\mathcal{C}_{\rho, \tilde{\eta}} = \{c \in \tau^{-1}([- \rho, \rho]^{n'}) : c^* = c \wedge \sqrt[64]{\|c^{64}\|_1} \leq \tilde{\eta}\}$. Let M_1, M_2, M_3 be in $(1, \infty)$, and let $\alpha_j = \sqrt{\pi / \ln M_j}$ for $j \in [3]$. Define the witness $\mathbf{s}_1 = (t, \mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_2, v_3, \mathbf{usk}, \mathbf{m}(\overline{\mathcal{I}'}), \mathbf{b})$ of dimension $m_1 = k'(2 + d(4 + k_H) + m' - |\mathcal{I}'|) + 1$ when embedded in $\mathcal{R}'$. Then, define $B = \sqrt{w + B'_{1,1}{}^2 + B'_{1,2}{}^2 + B_2^2 + B_3^2 + nm + 1}$ and

$$B_j = \frac{B'_{1,1}}{q} + \sqrt{nd} \left(\frac{1 + nm + \sqrt{nd}B'_{1,2} + \sqrt{ndk_H}B_2 + \sqrt{n}B_3}{2} + wB_2 \frac{1 - 1/q_H}{b_H - 1} \right)$$

We then define $\sigma_1 = \alpha_1 \tilde{\eta} B$, $\sigma_2 = \alpha_2 \tilde{\eta} \sqrt{n'm_2}$, and $\sigma_3 = \alpha_3 \sqrt{337} \sqrt{B^2 + B_j^2}$. We then define $B_{arp} = \frac{2}{\sqrt{26}} c_{256} \sigma_3 \sqrt{256}$. Then, let $d_2, m_2, \ell, q_{\pi_2}$ be positive integers such that

$$\begin{aligned} q_{\pi_2} & \text{ prime s.t. } q_{\pi_2} = 5 \bmod 8, \\ q_{\pi_2} & > \max(B'_{1,1}{}^2, B'_{1,2}{}^2, B_2^2, B_3^2, w) \\ q_{\pi_2} & > B_{arp}^2 - \min(B'_{1,1}{}^2, B'_{1,2}{}^2, B_2^2, B_3^2, w) \\ q_{\pi_2} & > \max(41n'm_1 B_{arp}, w + \sqrt{nw}, B_{arp}^2 + B_{arp} \sqrt{nm}) \\ q_{\pi_2} & > B'_{1,1} + q \left(\frac{1 + nm + \sqrt{nd}B'_{1,2} + \sqrt{ndk_H}B_2 + \sqrt{n}B_3}{2} + wB_2 + B_{arp} \right), \end{aligned}$$

We also let γ be an even divisor of $q_{\pi_2} - 1$, and D be the largest integer such that $2^{D-1} \rho n' < \gamma$. Then, $(\text{Prove}_2, \text{Verify}_2)$ is knowledge sound with an extractor running in expected polynomial time and soundness error

$$\varepsilon_{snd,2} = \frac{2}{|\mathcal{C}_{\rho, \tilde{\eta}}|} + q_{\pi_2}^{-n'/2} + q_{\pi_2}^{-2\ell} + 2^{-128} + \varepsilon_{\text{M-SIS}}^{\text{sound}}$$

and zero-knowledge with loss $\varepsilon_{zk,2} = \varepsilon_{\text{M-LWE}}^{zk} + \text{negl}(\lambda)$. The term $\varepsilon_{\text{M-SIS}}^{\text{sound}}$ is the hardness bound of M-SIS $_{n', d_2, m_1 + m_2, q_{\pi_2}, \beta_{\pi_2}}$ where

$$\beta_{\pi_2} = 4\tilde{\eta} \sqrt{4c_{n'm_1}^2 \sigma_1^2 n'm_1 + \left(2c_{n'm_2} \sigma_2 \sqrt{n'm_2} + (2^D \tilde{\eta} + \gamma) \sqrt{n'd_2} \right)^2},$$

while $\varepsilon_{\text{M-LWE}}^{zk}$ is the hardness bound of M-LWE $_{n', m_2 - d_2 - \lfloor 256/n' \rfloor - \ell - 1, m_2, q_{\pi_2}, \mathcal{B}_1, \mathcal{B}_1}$ (with $\mathcal{B}_1$ the centered binomial distribution of parameter 1 over $\mathcal{R}'$).

The size of π_2 is computed from the parameters of the proof system introduced in Lemma A.2 by

$$\begin{aligned} |\pi_2| &= n'd_2 (\lceil \log_2 q_{\pi_2} \rceil - D + 2.25) + \left(\frac{256}{n'} + 2\ell + 1 \right) \lceil \log_2 q_{\pi_2} \rceil \\ &+ n' \lceil \log_2 (2\rho + 1) \rceil + n'm_1 \left(\frac{1}{2} + \log_2 \sigma_1 \right) \\ &+ n'(m_2 - d_2) \left(\frac{1}{2} + \log_2 \sigma_2 \right) + 256 \left(\frac{1}{2} + \log_2 \sigma_3 \right). \end{aligned}$$

A.4 Security Analysis

We now give the full security statement and security proof of anonymity and unforgeability of our anonymous credentials.

Theorem A.1. *The anonymous credentials of Section A.2 is anonymous based on the zero-knowledge property of Prove₂. More precisely, for any PPT adversary, it holds that $\text{Adv}_{\text{an}}[\mathcal{A}] \leq 2\varepsilon_{zk,2}$ where $\text{Adv}_{\text{an}}[\mathcal{A}]$ is the advantage of $\mathcal{A}$ in the anonymity game of Definition A.2 and $\varepsilon_{zk,2}$ is defined in Lemma A.2.*

Proof. The proof here simply consists simulating the proof π_2 in the Show interaction from the anonymity game in Definition A.2. In each experiment, i.e., for $b = 0$ and $b = 1$, we use the simulator of Lemma A.2 to generate π_2 and send it to the adversary. As we no longer resort to $\text{usk}_{j_b}, \mathbf{m}^{(j_b)}(\bar{T}), \text{cred}^{(j_b')}$, it holds that the view of the adversary does not depend on b . Hence, in this modified game, the adversary cannot distinguish the two experiments, and its advantage is therefore 0. As a result, we have $\text{Adv}_{\text{an}}[\mathcal{A}] \leq 2\varepsilon_{zk,2}$ by Lemma A.2. $\square$

Theorem A.2. *The anonymous credentials of Section A.2 is unforgeable based on the hardness of M-LWE, M-SIS, M-ISIS, the zero-knowledge and soundness properties of Prove₁ and Prove₂, and the EUF-CMA of the signature of Section 4.1. More precisely, the advantage of PPT adversary in breaking the unforgeability of the anonymous credentials is upper-bounded by*

$$\begin{aligned} \text{Adv}_{uf}[\mathcal{A}] \leq & \varepsilon_{\text{snd},2} + 2(\varepsilon_{zk,1} + \varepsilon_{zk,2} + \varepsilon'_{\text{M-LWE}} + \binom{n}{w} \varepsilon_{\text{M-ISIS}} \\ & + 2 \left(h^{\text{odk}_H} (C(\binom{n}{w} - Q) \varepsilon_{\text{M-SIS}}^{\bullet}) + mC^2 \varepsilon_{\text{M-LWE}} \right. \\ & \left. + \varepsilon_{\text{snd},1} + \frac{1+\varepsilon}{1-\varepsilon} \frac{4M_{1,1}M_{1,2}}{(1-p)^4} h^{\text{odk}_H} (CQ \varepsilon_{\text{M-SIS}}^{\bullet}) \right) + \text{negl}(\lambda) \end{aligned}$$

where $\varepsilon'_{\text{M-LWE}}, \varepsilon_{\text{M-ISIS}}, \varepsilon_{\text{M-LWE}}, \varepsilon_{\text{M-SIS}}^{\bullet}, \varepsilon_{\text{M-SIS}}^{\bullet}$ are the respective hardness bounds of $\text{M-LWE}_{n,d,2d,q,U(T_1)}, \text{M-ISIS}_{n,d,2d,q,\sqrt{2nd}}, \text{M-LWE}_{n,d,d,q,B_1}, \text{M-SIS}_{n,d,2d+3,q,\beta_{\bullet}}$ and $\text{M-SIS}_{n,d,2d+1,q,\beta_{\bullet}}$. The constant $C \approx 2$ is the one from Lemma 2.8, and the probability p corresponds to the tailcut probability used to set the Gaussian verification bounds. The function h^{odk_H} corresponds to the dk_H -th composition power of the function h defined by

$$h(x) = \varepsilon_{\text{M-LWE}} + \mu \left(2\varepsilon_{\text{M-LWE}} + \mu (\varepsilon_{\text{M-LWE}} + x)^{\frac{2\lambda-1}{2\lambda}} \right)^{\frac{2\lambda-1}{2\lambda}}.$$

The constant μ is defined by $\mu = \left(1 + \frac{\lambda(2\lambda-1)(\delta^2-1)^2}{(2-\delta^2)^{2\lambda+1}} \right)^{Q/2\lambda}$ with δ defined in Theorem 3.1 and Equation (5). Finally, we have

$$\begin{aligned} \beta_{\bullet} &= \sqrt{\left(\sqrt{B'_{1,1}{}^2 + B'_{1,2}{}^2} + \sqrt{nd}B_2 \right)^2 + B_3^2 + n + 1} \\ \beta_{\bullet} &= \sqrt{\left(2\sqrt{B'_{1,1}{}^2 + B'_{1,2}{}^2} + \sqrt{nd}\sqrt{4B_2^2 + n} \right)^2 + 4B_3^2} \end{aligned}$$

The values $\varepsilon_{snd,i}$ and $\varepsilon_{zk,i}$ are those from the security of Prove_i in Lemma A.1 and A.2.

Proof. The unforgeability proof essentially goes as follows. We first change the game by extracting the final **Show** proof π_2 . Then, we distinguish two types of forgeries. Either it is an impersonation forgery, in which case we make small changes to use the extracted proof π_2 to solve M-ISIS on the matrix $\mathbf{D}_s$. Or it is a credential forgery, in which case we basically follow the EUF-CMA security proof of Theorem 4.1 of the standard model signature with only minor tweaks to accomodate for the specificities of the issuance process. For simplicity, we assume that $\mathcal{I} = \emptyset$ in the issuances and showings and the message returned by the adversary, meaning no attributes is ever disclosed. Disclosing attributes will only improve the quality of the security proof, but will necessitate many notation subtleties. We thus work in the worst-case.

G_1 : Extracting the Show proof. We first change the unforgeability game of Definition A.3. At the end of the game, the challenger extracts the witness $\mathbf{w}^* = (t^*, \mathbf{v}_{1,1}^*, \mathbf{v}_{1,2}^*, \mathbf{v}_2^*, v_3^*, \mathbf{s}^*, \mathbf{m}^*)$ by Lemma A.2. In particular, the witness verifies

$$\begin{aligned} \mathbf{v}_{1,1}^* + \mathbf{A}\mathbf{v}_{1,2}^* + (q_L t^* \mathbf{G}_H - \mathbf{B})\mathbf{v}_2^* + \mathbf{a}_3 v_3^* - \mathbf{D}_s \mathbf{s}^* - \mathbf{D}\mathbf{m}^* &= \mathbf{u} \bmod q\mathcal{R}, \\ \|\mathbf{v}_{1,1}^*\|_2 \leq B'_{1,1}, \|\mathbf{v}_{1,2}^*\|_2 \leq B'_{1,2}, \|\mathbf{v}_2^*\|_2 \leq B_2, \|v_3^*\|_2 \leq B_3 \\ t^* \in \mathcal{T}_w, \mathbf{m}^* \in T_1^{m'}, \mathbf{s}^* \in T_1^{2d} \end{aligned}$$

By the special soundness in Lemma A.2, we then have

$$\text{Adv}_{uf}[\mathcal{A}] \leq \varepsilon_{snd,2} + \text{Adv}_1$$

Branching. We now introduce a branching in the security reduction. The challenger flips ρ_1 a coin and expects either an impersonation forgery or a credential forgery depending on the coin toss. We then have

$$\text{Adv}_1 \leq 2(\text{Adv}[\text{impersonation forgery}] + \text{Adv}[\text{credential forgery}])$$

and we treat both separately below. When $\rho_1 = 0$ (impersonation forgery), we denote the games by $G_{0,i}$, whereas when $\rho_1 = 1$ (credential forgery), we denote the games by $G_{1,i}$.

Impersonation Forgery. We perform a series of game hops to in the end reduce to M-ISIS using the extracted witness $\mathbf{w}^*$. We define $G_{0,0}$ as G_1 and change it below.

$G_{0,1}$. The challenger makes a guess on which honest user will be targeted by the impersonation forgery. It thus samples $j^+ \leftarrow U(\binom{n}{w})$, as the number of users requesting credentials is bounded by the number of possible tags, i.e., $|\mathcal{T}_w| = \binom{n}{w}$. It then changes the behavior of certain queries to $\mathcal{O}_{\text{Oblss}}$ and $\mathcal{O}_{\text{Show}}$. When $\mathcal{O}_{\text{Oblss}}(j^+, \mathbf{m})$ is queried, proceeds as usual except that if $j^+ \in \text{HU}$, it simulates the proof π_1 in the **Issue** protocol. And when $\mathcal{O}_{\text{Show}}(j', \mathbf{m}^{(j')}(\mathcal{I}))$ is queried, the challenger looks up the user index j that corresponds to this issuance index. The

challenger then proceeds as usual except if $j = j^+$ in which case it simulates the proof π_2 in the Show protocol. By Lemma A.1 and A.2, it holds that

$$\text{Adv}[\text{impersonation forgery}] \leq \varepsilon_{zk,1} + \varepsilon_{zk,2} + \text{Adv}_{0,1}$$

$G_{0,2}$. It then changes the way the key of the user j^+ is generated. Concretely, when $\mathcal{O}_{\text{HU}}(j^+)$ is queried, the challenger samples $\bar{\mathbf{b}} \leftarrow U(\mathcal{R}_q^d)$ and defines $\text{upk}_{j^+} = \bar{\mathbf{b}}$ which it sends to the adversary. Under the Knapsack version of M-LWE, $\mathbf{D}_s \mathbf{s}_{j^+} \bmod q\mathcal{R}$ is indistinguishable from uniform. So the adversary cannot notice. The only caveat is that $\mathcal{A}$ can potentially corrupt user j^+ , which means the guess was wrong. As a result, we also change the behavior of $\mathcal{O}_{\text{CU}}$ and the winning condition. Because the adversary must return an impersonation forgery, it holds that the extracted $\mathbf{s}^*$ verifies $\mathbf{D}_s \mathbf{s}^* = \text{upk}_{j^*}$ for some $j^* \in \text{HU}$. The challenger then aborts the game if $j^+ \neq j^*$ meaning the guess was wrong. Also, if $\mathcal{O}_{\text{CU}}(j^+)$ is queried, the challenger aborts as well but it wouldn't respect the new winning condition anyways as it would entail $j^+ \in \text{CU}$ and thus $j^+ \neq j^*$. As a result, we have

$$\text{Adv}_{0,1} \leq \varepsilon'_{\text{M-LWE}} + \binom{n}{w} \text{Adv}_{0,2}$$

where $\varepsilon'_{\text{M-LWE}}$ is the hardness bound of M-LWE $_{n,d,2d,q,U(T_1)}$ which is equivalent to the Knapsack version.

Reduction to M-ISIS. We now change the game so as to introduce an M-ISIS instance and bound $\text{Adv}_{0,2}$. The challenger receives the M-ISIS $_{n,d,2d,q,\sqrt{2nd}}$ instance $(\mathbf{D}_s, \bar{\mathbf{b}})$. It sets the public parameters using $\mathbf{D}_s$ instead of a freshly sampled one, and uses $\bar{\mathbf{b}}$ in the modified query $\mathcal{O}_{\text{HU}}(j^+)$. At the end of the game, it holds that $j^+ = j^*$ and therefore $\mathbf{D}_s \mathbf{s}^* = \text{upk}_{j^*} = \text{upk}_{j^+} = \bar{\mathbf{b}} \bmod q\mathcal{R}$. Also, it holds that $\mathbf{s}^* \in T_1^{2d}$ and therefore that $\|\mathbf{s}^*\|_2 \leq \sqrt{2nd}$, which solves M-ISIS. We then have

$$\text{Adv}_{0,2} \leq \varepsilon_{\text{M-ISIS}}$$

where $\varepsilon_{\text{M-ISIS}}$ is the hardness bound of M-ISIS $_{n,d,2d,q,\sqrt{2nd}}$. Overall, we have

$$\text{Adv}[\text{impersonation forgery}] \leq \varepsilon_{zk,1} + \varepsilon_{zk,2} + \varepsilon'_{\text{M-LWE}} + \binom{n}{w} \varepsilon_{\text{M-ISIS}}.$$

Credential Forgery. We now describe the branch $\rho_1 = 1$ corresponding to a credential forgery. As it follows almost exactly the proof of Theorem 4.1, we only describe the changes and refer to the games of the original proof when they are the same. We define $G_{1,0} = G_1$.

$G_{1,1}$ to $G_{1,2}$. We define these games by changing the issuance queries in $\mathcal{O}_{\text{ObtIss}}$ and $\mathcal{O}_{\text{Issue}}$ in the same way as G_1 up to G_2 from the proof of Theorem 4.1.

$G_{1,3}$. We follow G_3 from the proof of Theorem 4.1, except when $\rho = 2$ where, in addition to sampling i^+ and defining $t^+ = t_{i^+}$, we also extract the i^+ -th issuance proof π_1 , i.e., the one associated to the issuance that will use t^+ . The challenger then extracts $(\mathbf{r}_1^+, \mathbf{r}_2^+, \mathbf{s}^+, \mathbf{m}^+) \in T_1^{4d+m'}$ such that $\mathbf{c} = \mathbf{r}_1^+ + \mathbf{A}\mathbf{r}_2^+ + \mathbf{D}_s \mathbf{s}^+ + \mathbf{D}\mathbf{m}^+$. As a result, we have

$$\text{Adv}_{0,2}^{\mathbf{1}} = \text{Adv}_{0,3}^{\mathbf{1}}, \text{ and } \text{Adv}_{0,2}^{\mathbf{2}} \leq \varepsilon_{\text{snd},1} + \text{Adv}_{0,3}^{\mathbf{2}}$$

$G_{1,4}$. We follow G_4 from the proof of Theorem 4.1, except when $\rho = 2$ where the challenger samples $\mathbf{S}_1, \mathbf{S}_2$ from $\mathcal{B}_1^{d \times m}$ conditioned on $\|\mathbf{S}_i\|_{2,\mathbb{Z}} \leq \sqrt{nd/2}$. It then defines $[\mathbf{D}_s|\mathbf{D}] = \mathbf{S}_1 + \mathbf{A}\mathbf{S}_2 \bmod q\mathcal{R}$. As this represents an m -secret instance of M-LWE, the loss is updated to

$$\text{Adv}_{0,3}^{\bullet} = \text{Adv}_{0,4}^{\bullet}, \text{ and } \text{Adv}_{0,3}^{\Theta} \leq mC^2\varepsilon_{\text{M-LWE}} + \text{Adv}_{0,4}^{\Theta}$$

$G_{1,5}$ and $G_{1,6}$. We define these games by changing the generation of $\mathbf{u}$ in $\mathcal{O}_{\text{ObtLss}}$ and $\mathcal{O}_{\text{Issue}}$ in the same way as G_5 and G_6 from the proof of Theorem 4.1.

$G_{1,7}$. We follow G_7 from the proof of Theorem 4.1, except when $\rho = 2$ in the i^+ -th issuance query (the one linked to t^+). We perform the same rejection step but slightly tweak it to account for our new commitment structure. In this query, recall that as per $G_{1,3}$, the challenger extracted the commitment proof and has access to $(\mathbf{r}_1^+, \mathbf{r}_2^+, \mathbf{s}^+, \mathbf{m}^+)$ such that $\mathbf{c} = \mathbf{r}_1^+ + \mathbf{A}\mathbf{r}_2^+ + [\mathbf{D}_s|\mathbf{D}][\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T$. The challenger proceeds legitimately by sampling $(\mathbf{v}_{1,1}^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+)$ using TSampler . It then samples u_1, u_2 from $U([0, 1])$ and continues only if $u_1 \leq 1/M_{1,1}$, $u_2 \leq M_{1,2}$, and

$$\langle \tau(\mathbf{v}_{1,1}), \tau\left(\mathbf{S}_1 \begin{bmatrix} \mathbf{s}^+ \\ \mathbf{m}^+ \end{bmatrix} + \mathbf{r}_1^+\right) \rangle \geq 0, \text{ and } \langle \tau(\mathbf{v}_{1,2}), \tau\left(\mathbf{S}_2 \begin{bmatrix} \mathbf{s}^+ \\ \mathbf{m}^+ \end{bmatrix} + \mathbf{r}_2^+\right) \rangle$$

It aborts otherwise. If it did not abort, the challenger outputs $(t^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+)$ to pursue the i^+ issuance process. Again note that $\mathbf{v}_{1,1}, \mathbf{v}_{1,2}$ are only used to define $\mathbf{u}$ at this stage and not returned to $\mathcal{A}$. As in G_8 , we have

$$\text{Adv}_{0,6}^{\bullet} = \text{Adv}_{0,7}^{\bullet}, \text{ and } \text{Adv}_{0,6}^{\Theta} \leq (2M_{1,1})(2M_{1,2})\text{Adv}_{0,7}^{\Theta}$$

$G_{1,8}$. We follow G_8 from the proof of Theorem 4.1, except when $\rho = 2$ in the i^+ -th issuance query where we pursue the rejection sampling argument. Instead of sampling $(\mathbf{v}_{1,1}^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+)$ using TSampler , it defines $\mathbf{v}_{1,1}^+ = \mathbf{v}_{1,1} + \mathbf{S}_1[\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T + \mathbf{r}_1^+$, $\mathbf{v}_{1,2}^+ = \mathbf{v}_{1,2} + \mathbf{S}_2[\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T + \mathbf{r}_2^+$, $\mathbf{v}_2^+ = \mathbf{v}_2$ and $v_3^+ = v_3$. Notice that this satisfies the correct equation based on our definition of $\mathbf{u}$ and $[\mathbf{D}_s|\mathbf{D}]$. Indeed, we have

$$\begin{aligned} & \mathbf{v}_{1,1}^+ + \mathbf{A}\mathbf{v}_{1,2}^+ + (q_L t^+ \mathbf{G}_H - \mathbf{B})\mathbf{v}_2^+ + \mathbf{a}_3 v_3^+ \\ &= \mathbf{u} + \mathbf{S}_1[\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T + \mathbf{r}_1^+ + \mathbf{A}(\mathbf{S}_2[\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T + \mathbf{r}_2^+) \bmod q\mathcal{R} \\ &= \mathbf{u} + [\mathbf{D}_s|\mathbf{D}][\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T + \mathbf{r}_1^+ + \mathbf{A}\mathbf{r}_2^+ \bmod q\mathcal{R} \\ &= \mathbf{u} + \mathbf{c} \bmod q\mathcal{R} \end{aligned}$$

Before outputting them, the challenger samples $u_1, u_2 \leftarrow U([0, 1])$ and computes $\gamma_j = \langle \tau(\mathbf{v}_{1,j}^+), \tau(\mathbf{S}_j[\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T + \mathbf{r}_j^+) \rangle$ and continues only if

$$\gamma_1 \geq 0, \text{ and } u_1 \leq \frac{1}{M_{1,1}} \exp\left(\frac{\pi}{s_{1,1}^2} \left(\left\|\mathbf{S}_1 \begin{bmatrix} \mathbf{s}^+ \\ \mathbf{m}^+ \end{bmatrix} + \mathbf{r}_1^+\right\|_2^2 - 2\gamma_1\right)\right),$$

$$\text{and } \gamma_2 \geq 0, \text{ and } u_2 \leq \frac{1}{M_{1,2}} \exp \left(\frac{\pi}{s_3^2} \left(\left\| \mathbf{S}_2 \begin{bmatrix} \mathbf{s}^+ \\ \mathbf{m}^+ \end{bmatrix} + \mathbf{r}_2^+ \right\|_2^2 - 2\gamma_2 \right) \right).$$

and aborts otherwise. If it did not abort, sig_{i+} is outputted and is a valid signature only if the bounds are verified. Because we have $\|\mathbf{S}_j[\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T\|_2 \leq \sqrt{nd/2}\|\mathbf{s}^{+T}|\mathbf{m}^{+T}\|_2$ based on $G_{1,4}$, we then get $\|\mathbf{S}_j[\mathbf{s}^{+T}|\mathbf{m}^{+T}]^T + \mathbf{r}_j^+\|_2 \leq \sqrt{nd/2}\sqrt{nm} + \sqrt{nd}$. By the way the parameters $M_{1,j}$ are set, Lemma 2.10 shows that the distribution is the same. Hence, using the Gaussian tail bounds on $\mathbf{v}_{1,1}^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+$ after the rejection sampling passed, the challenger also aborts if the bounds are not verified. Because the bounds are verified each with probability $1 - p$, it yields

$$\text{Adv}_{1,7}^{\bullet} = \text{Adv}_{1,8}^{\bullet}, \text{ and } \text{Adv}_{1,7}^{\circ} = \frac{1}{(1-p)^4} \text{Adv}_{1,8}^{\circ}.$$

Rest of the Proof. The rest of the proof, i.e., the trapdoor switching argument and the reduction to M-SIS in both branches stays exactly the same. Instead of using the forgery in clear, we use the extracted witness from the final **Show** proof (see G_1) to test the tag guess and construct the M-SIS solutions. The only difference is that $\mathbf{v}_{1,1}^*, \mathbf{v}_{1,2}^*$ verify the bounds $B'_{1,1}, B'_{1,2}$ instead of $B_{1,1}, B_{1,2}$. Overall, we have

$$\begin{aligned} \text{Adv}[\text{credential forgery}] &\leq 2 \left(h^{\text{odk}_H} (C(\binom{n}{w}) - Q) \varepsilon_{\text{M-SIS}}^{\bullet} + mC^2 \varepsilon_{\text{M-LWE}} \right. \\ &\quad \left. + \varepsilon_{\text{snd},1} + \frac{1+\varepsilon}{1-\varepsilon} \frac{4M_{1,1}M_{1,2}}{(1-p)^4} h^{\text{odk}_H} (CQ \varepsilon_{\text{M-SIS}}^{\circ}) \right) + \text{negl}(\lambda) \end{aligned}$$

Combined with the impersonation forgery branch and the first hybrid change, we get the desired result. $\square$

A.5 Performance

To evaluate the performance and security of our construction, we essentially select the same parameters as the standard model signature of Section 4.1. Because only the values of $\alpha_{1,i}, M_{1,i}$ change, we refer to Table C.1. We choose $m' = 10$ as the number of attribute slots for a fair comparison with prior lattice anonymous credentials schemes. It then results in $\alpha_{1,1} = 6.02$, $\alpha_{1,2} = 3.05$, $M_{1,1} = 1.09$ and $M_{1,2} = 1.40$. This does not affect the performance and the other parameters but only (ever so slightly) the security for the unforgeability.

Then, based on the lifting strategy for the zero-knowledge arguments we described in Section A.3, the parameters of each proof system can be chosen in a very modular way without affecting the signature parameters. We provide parameter sets in Table C.2 and C.3 that satisfy all the conditions of Lemma A.1 and A.2, while providing sufficient security for the anonymity and unforgeability of the anonymous credentials. Said security is evaluated based on Theorems A.1 and A.2. Note that, as opposed to the evaluation of Section 4.1 where we used the Core-SVP cost model to assess the hardness of M-LWE and M-SIS, we here use

the more realistic cost model of $0.292 \cdot bikz + 16.4$, where $bikz$ is the BKZ block-size, for a fairer comparison with prior works. Using this new cost model with the parameters of Table C.1 yields 133 bits of security for the signature unforgeability. When combined with the other terms of Theorem A.2, our anonymous credentials reaches 125 bits of security for unforgeability, and 132 bits of security for anonymity. The final credential proof size, i.e, the size of π_2 , is 46.89 KB which is a 13% improvement over the anonymous credentials of [AGJ⁺24] with the sampler of [JS25b], which was the most compact to date.

B Round-Optimal Blind Signature

In this section, we give the full description of our blind signature scheme relying on our new standard model signature and sampler. The construction is based on that of [JS25a], which we revisit using our samplers, security proof argument for better compactness, as well as an improved parameter selection analysis on the decomposition bases introduced in [JS25a] to actually make it round optimal. Regarding the latter, we essentially observe that the mask for (v_2, v_3) does not impact the performance by much when choosing the decomposition base b_2 for optimal security. As a result, we do not need to transmit the tag in an artificial first round. Hence, our scheme is 2-round and still benefits from all the advantages of [JS25a] over prior state-of-the-art constructions (standard assumptions, computationally practical issuance), but in a more compact way. Our improved sampler opens for a better parameter selection, leading to significant gains. As reported in Section 4.3, we end up with a round-optimal blind signature that is 27% smaller than that of [JS25a] with the recent sampler of [JS25b]. Although our construction is fairly similar, we detail our construction and updated security proof for the sake of completeness and to ease future reference.

B.1 Syntax and Model

Blind signatures were introduced by Chaum [Cha82], and their security were formalized in, e.g., [JLO97]. At a high-level, we require the notion of blindness to ensure that a signature cannot be traced back to the specific issuance that emitted it. It then imposes to conceal the message during the issuance, or otherwise it could be used to break the blindness. This concealment then further constrains the standard unforgeability notion, which is why we now require *one-more unforgeability*. It captures the fact that a user cannot produce more blind signatures than they received from the signer, which then provides the security assurances we need for real-life uses.

Definition B.1. *A blind signature scheme is defined as a set of algorithms and protocols (Setup, KeyGen, Sign, Verify) where*

- *Setup: takes the security parameter λ and generates the public parameters pp (which includes λ)*
- *KeyGen: takes pp and generates the issuer’s key pair (pk, sk)*

- $\text{Sign}_{S,U}$: is a protocol between a signer S , holding $(\text{sk}, \text{pk}, \text{pp})$, and a user U holding (pk, pp) and message m . The user U either gets a blind signature bsig or $\perp$ if the protocol aborted, while S only gets a notification of whether the execution was successful.
- **Verify**: takes $\text{pk}, \text{pp}, \text{bsig}$ and the message m , and outputs 1 if bsig is a valid blind signature on m with respect to pk , and 0 otherwise.

A blind signature is called k -round if $\text{Sign}_{S,U}$ involves k exchanges between S and U . It is in particular round-optimal when $k = 2$. Let us now describe formally the security properties we expect for a blind signature scheme.

Anonymity. The anonymity ensures that a blind signature bsig cannot be traced back to the specific interaction with S that produced it. It in particular means the message m cannot be revealed to S in the execution of $\text{Sign}_{S,U}$. Otherwise, S could store the queried messages and the exchanged elements for each execution of $\text{Sign}_{S,U}$ and later pinpoint which interaction used the message m associated to the final blind signature bsig . This unlinkability property is crucial in the application of blind signatures to untraceable payments or ecash.

Definition B.2. A blind signature scheme is anonymous if, for any PPT adversary $\mathcal{A}$, $\text{Adv}_{\text{an}}[\mathcal{A}] = |\mathbb{P}[\text{Game}_{\text{an}-0}(\mathcal{A}) = 1] - \mathbb{P}[\text{Game}_{\text{an}-1}(\mathcal{A}) = 1]|$ is negligible in λ , where $\text{Game}_{\text{an}-b}(\mathcal{A})$ is defined below for $b \in \{0, 1\}$.

$\text{Game}_{\text{an}-b}(\mathcal{A})$:

- 1 $\text{pp} \leftarrow \text{Setup}(1^\lambda)$
- 2 $(\text{pk}, \text{sk}, m_0, m_1) \leftarrow \mathcal{A}(\text{pp})$
- 3 $\text{Execute}(\emptyset, \text{bsig}_0) \leftarrow \text{Sign}_{\mathcal{A},C}(\cdot, (\text{pk}, \text{pp}, m_b))$
- 4 $\text{Execute}(\emptyset, \text{bsig}_1) \leftarrow \text{Sign}_{\mathcal{A},C}(\cdot, (\text{pk}, \text{pp}, m_{1-b}))$
- 5 $b^* \leftarrow \mathcal{A}(\text{pp}, \text{bsig}_0, \text{bsig}_1)$
- 6 **return** b^*

One-More Unforgeability. The one-more unforgeability notion translates the standard unforgeability of digital signature to this specific context where the message cannot be known by the signer in the issuance. For digital signatures, the challenger receives a forgery and can check that the provided message differs from all the messages that were queried in the signing queries. Here, this check cannot be done as the messages are hidden during the signing queries. This specificity is circumvented in anonymous credentials in the model of [FHS19] by asking that the attributes are revealed at issuance. This is however only to make the security model simpler, and in many anonymous credentials use cases, this is actually a reasonable assumption. In blind signatures however, the whole point of the construction is to not link the signature to its issuance and thus hide the message in the latter. It then pushed towards an updated notion of one-more unforgeability, where we could detect a forgery has happened even without knowing the messages. Concretely, we simply ask that the adversary should not be able to produce more valid blind signatures than the number of queried blind

signatures. If the adversary manages that, it means it produced at least one forgery, i.e., at least one unlawfully obtained blind signature.

To formalize the one-more unforgeability notion, we define the $\mathcal{OSign}$ oracle where the challenger runs the signer's part of $\text{Sign}_{S,\mathcal{A}}((\text{sk}, \text{pk}, \text{pp}), \cdot)$ and the adversary plays the user.

Definition B.3. A blind signature scheme is one-more unforgeable if, for any PPT adversary $\mathcal{A}$, $\text{Adv}_{\text{omuf}}[\mathcal{A}] = \mathbb{P}[\text{Game}_{\text{uf}}(\mathcal{A}) = 1]$ is negligible in λ , where $\text{Game}_{\text{omuf}}(\mathcal{A})$ is defined below. The number of query to $\mathcal{OSign}$ in $\text{Game}_{\text{omuf}}(\mathcal{A})$ is N , which is bounded by Q .

Game_{omuf}($\mathcal{A}$):

- 1 $\text{pp} \leftarrow \text{Setup}(1^\lambda)$
- 2 $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(\text{pp})$
- 3 $(m_i^*, \text{bsig}_i^*)_{i \in [N+1]} \leftarrow \mathcal{A}^{\mathcal{OSign}(\text{pp}, \text{pk})}$.
- 4 **if** $\exists (i, j) \in [N+1], i \neq j$ and $m_i^* = m_j^*$, **return** 0
- 5 **if** $\exists i \in [N+1], \text{Verify}(\text{pk}, \text{pp}, \text{bsig}_i^*, m_i^*) = 0$, **return** 0
- 6 **return** 1

B.2 Our Anonymous Credentials Construction

We now give the full description of Setup , KeyGen , Sign , and Verify . They follow the same construction as [JS25a] with a few modifications. Based on our later parameter selection, we remove the commitment randomness for $(\mathbf{v}_2, v_3)$, which then removes the need to transmit the tag in advance. As a result, our scheme automatically becomes 2-round. The other modification comes from using the standard model signature of Section 4.1 or more precisely our samplers. At high level, it follows similar steps to the anonymous credentials, with optimizations for the context of blind signatures like the entropy recycling technique of [JS25a].

We define $\tilde{S}_\eta = \tau^{-1}(\{-\eta, \dots, \eta - 1\}^n)$. We also define the decomposition function High , Low , Decompose as $\text{High}(x, b) = 2\lfloor x/(2b) \rfloor + 1$, $\text{Low}(x, b) = x - b\text{High}(x, b) = (x \bmod^+ 2b) - b$, and $\text{Decompose}(x, b) = (\text{High}(x, b), \text{Low}(x, b))$. We recall from [JS25a] that for any $b \in \mathbb{N}^\times$, it holds that $\text{High}(U([-2b, 2b - 1], b)) = U(\{-1, 1\})$, $\text{Low}(U([-2b, 2b - 1], b)) = U([-b, b - 1])$, and that for any $y \in \mathbb{Z}$, $\text{Low}(y - U([-b, b - 1]), b) = U([-b, b - 1])$ and in particular perfectly hides the shift y .

Algorithm B.1: BS.Setup(λ)

1. Choose $n \geq \kappa$ to be powers of two.
2. Choose $d \in \mathbb{N}^\times$.
3. Choose $b_H \in \mathbb{N}^\times$ a small prime s.t. $b_H = 2\kappa + 1 \bmod 4\kappa$ and $b_H > \sqrt{\kappa}^\kappa$.
4. Choose $k_H, b_L, k_L \in \mathbb{N}^\times$
5. $q_L \leftarrow b_L^{k_L}$
6. $q_H \leftarrow b_H^{k_H}$
7. $q \leftarrow q_L q_H$
8. Choose $w \in \mathbb{N}^\times$ st $\binom{n}{w} \geq Q$.
9. $\mathcal{T}_w \leftarrow \{t \in T_1 : \|t\|_1 = w\}$.
10. $B_R \leftarrow \frac{3}{4}(\sqrt{nd} + \sqrt{ndk_H} + 6)$
11. For $x \in \{L, H\}$, define $\mathbf{G}_x = [1 \dots |b_x^{k_x-1}|] \otimes \mathbf{I}_d \in \mathcal{R}^{d \times dk_x}$

12. Choose $\varepsilon \in (0, 1)$ $\triangleright$ typically $\varepsilon \approx 2^{-40}$
13. $s_L \leftarrow b_L \cdot \sqrt{\ln(2ndk_L/\varepsilon)/\pi}$ $\triangleright s_L \approx b_L \eta_\varepsilon(\mathbb{Z}^{ndk_L}) \geq b_L \eta_{\varepsilon/dk_L}(\mathbb{Z}^n)$ by Lemma 2.4
14. $s_H \leftarrow b_H \cdot \sqrt{\ln(2ndk_H/\varepsilon)/\pi}$ $\triangleright s_H \approx b_H \eta_\varepsilon(\mathbb{Z}^{ndk_H}) \geq b_H \eta_{\varepsilon/dk_H}(\mathbb{Z}^n)$ by Lemma 2.4
15. $\eta \leftarrow \sqrt{\ln(2nd(k_L + k_H)/\varepsilon)/\pi}$ $\triangleright \eta \approx \eta_\varepsilon(\mathbb{Z}^{nd(k_L + k_H)})$ by Lemma 2.4
16. $s_1 \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$.
17. $s_2 \leftarrow s_L^2/\sqrt{s_L^2 - \eta^2}$.
18. $s_{1,1} \leftarrow \sqrt{s_L^4/(s_L^2 - \eta^2) \cdot (b_L^{2k_L} - 1)/(b_L^2 - 1) + 3s_H^4 B_R^2/(s_H^2 - \eta^2)}$.
19. $s_3 \leftarrow \sqrt{3s_H^2 B_R/\sqrt{s_H^2 - \eta^2}}$.
20. $s_4 \leftarrow \sqrt{3s_H^2/\sqrt{s_H^2 - \eta^2}}$.
21. Choose $M'_{1,1}, M'_{1,2} > 1$
22. $\alpha'_{1,i} \leftarrow \sqrt{\pi/\ln M'_{1,i}}$
23. $(b_{1,1}, b_{1,2}) \leftarrow (\lfloor s_{1,1}/(2\alpha'_{1,1}\sqrt{nd}) \rfloor, \lfloor s_3/(2\alpha'_{1,2}\sqrt{nd}) \rfloor)$
24. $(\alpha_{1,1}, \alpha_{1,2}) \leftarrow (s_{1,1}/(n\sqrt{d/2} + 2b_{1,1}\sqrt{nd}), s_3/(n\sqrt{d/2} + 2b_{1,2}\sqrt{nd}))$.
25. $M_{1,i} \leftarrow \exp(\pi/\alpha_{1,i}^2)$.
26. $\mathbf{d} \leftarrow U(\mathcal{R}_q^d)$
27. $\mathbf{A} \leftarrow U(\mathcal{R}_q^{d \times d})$.
28. $\mathbf{a}_3 \leftarrow U(\mathcal{R}_q^d)$.
29. $\mathbf{u} \leftarrow U(\mathcal{R}_q^d)$.
30. Choose a prime p and $d_e, \eta_e \in \mathbb{N}^\times$ and define $m_e \leftarrow 2d_e + 1$.
31. $\mathbf{A}_e \leftarrow U(\mathcal{R}_p^{m_e \times d_e})$
32. $\mathbf{b}_e \leftarrow U(\mathcal{R}_p^{m_e})$
33. **params** $\leftarrow (n, d, m', m, b_H, k_H, q_H, b_L, k_L, q_L, q, w, B_R, \varepsilon, s_L, s_H, \eta, s_1, s_2, s_3, s_4, s_{1,1}, p, m_e, d_e, \eta_e)$

Output: $\text{pp} = (\text{params}, \mathbf{A}, \mathbf{d}, \mathbf{a}_3, \mathbf{u}, \mathbf{A}_e, \mathbf{b}_e)$

Algorithm B.2: BS.KeyGen(pp)

1. $(\mathbf{B}, (\mathbf{R}_1, \mathbf{R}_2)) \leftarrow \text{KeyGen}(\text{pp})$

$\triangleright$ Algorithm 4.2

Output: $\text{pk} = \mathbf{B}, \text{sk} = (\mathbf{R}_1, \mathbf{R}_2)$

Algorithm B.3: BS.Sign $_{S,U}((\text{sk}, \text{pk}, \text{pp}, \text{st}); (\text{pk}, \text{pp}, m))$

User U .

1. $\mathbf{r}_{1,1,L} \leftarrow U(\tilde{\mathcal{S}}_{b_{1,1}}^d)$, $\mathbf{r}_{1,1,H} \leftarrow \tau^{-1}(U(\{-1, 1\}^{nd}))$.
2. $\mathbf{r}_{1,2,L} \leftarrow U(\tilde{\mathcal{S}}_{b_{1,2}}^d)$, $\mathbf{r}_{1,2,H} \leftarrow \tau^{-1}(U(\{-1, 1\}^{nd}))$.
3. $\mathbf{r}_{1,i} \leftarrow \mathbf{r}_{1,i,L} + b_{1,i}\mathbf{r}_{1,i,H}$
4. $\mathbf{c} \leftarrow \mathbf{r}_{1,1} + \mathbf{A}\mathbf{r}_{1,2} + \mathbf{d}m \bmod q\mathcal{R}$.
5. $\mathbf{r}_e \leftarrow \mathcal{B}_{\eta_e}^{m_e}$.
6. $\text{ct}_0 \leftarrow \mathbf{A}_e^T \mathbf{r}_e \bmod p\mathcal{R}$.
7. $\text{ct}_1 \leftarrow \mathbf{b}_e^T \mathbf{r}_e + \lfloor p/2 \rfloor m \bmod p\mathcal{R}$.
8. $\pi_1 \leftarrow \text{Prove}_1(\text{statement}_1, \mathbf{r}_{1,1}, \mathbf{r}_{1,2}, \mathbf{r}_e, m)$ $\triangleright$ See Sect. B.3
9. Send $(\mathbf{c}, \text{ct}_0, \text{ct}_1, \pi_1)$ to O .
- Signer S .
10. **if** $\text{Verify}_1(\pi_1, \text{statement}_1) = 0$, **abort**.
11. $t \leftarrow F(\text{st}) \in \mathcal{T}_w$. $\triangleright t \in T_1, \|t\|_1 = w$
12. $v_3 \leftarrow \mathcal{D}_{\mathcal{R}, s_4}$.
13. $(\mathbf{v}'_{1,1}, \mathbf{v}'_{1,2}, \mathbf{v}_2) \leftarrow \text{TSampler}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{A}, \mathbf{u} + \mathbf{c} - v_3\mathbf{a}_3 \bmod q\mathcal{R}, t\mathbf{I}_d, s_1, s_2, s_3, s_4, s_L, s_H)$
14. **if** $\|\mathbf{v}'_{1,1}\|_2 > B_{1,1} \vee \|\mathbf{v}'_{1,2}\|_2 > B_{1,2} \vee \|\mathbf{v}_2\|_2 > B_2 \vee \|v_3\|_2 > B_3$ **goto** 7).
15. Send $(t, \mathbf{v}'_{1,2}, \mathbf{v}_2, v_3)$ to U
16. $\text{st} \leftarrow \text{st} + 1$.
- User U .
17. $\mathbf{v}'_{1,1} \leftarrow \mathbf{u} + \mathbf{c} - \mathbf{A}\mathbf{v}'_{1,2} - (q_L\mathbf{G}_{H,t1} - \mathbf{B})\mathbf{v}_2 - \mathbf{a}_3v_3 \bmod q\mathcal{R}$
18. $\rho \leftarrow (\|\mathbf{v}'_{1,1}\|_2 \leq B_{1,1}) \wedge (\|\mathbf{v}'_{1,2}\|_2 \leq B_{1,2}) \wedge (\|\mathbf{v}_2\|_2 \leq B_2) \wedge (t \in \mathcal{T}_w)$
19. **if** $\rho = 0$, **abort**

20. $(\mathbf{w}_{1,i,H}, \mathbf{w}_{1,i,L}) \leftarrow \text{Decompose}(\mathbf{v}'_{1,i} - \mathbf{r}_{1,i,L}, b_{1,i}) - (\mathbf{r}_{1,i,H}, \mathbf{0})$ for $i \in \{1, 2\}$
21. $\pi_2 \leftarrow \text{Prove}_2(\text{statement}_2, t, \mathbf{w}_{1,1,H}, \mathbf{w}_{1,2,H}, \mathbf{v}_2, v_3)$ ▷ See Sect. B.3

Output: U gets $\text{bsig} = (\mathbf{w}_{L,1,1}, \mathbf{w}_{L,1,2}, \pi_2)$

Algorithm B.4: $\text{BS.Verify}((\text{pk}, \text{pp}, \text{bsig}, m)$

1. $\rho_1 \leftarrow (\mathbf{w}_{1,1,L} \in \tilde{S}_{b_{1,1}}^d)$.
2. $\rho_2 \leftarrow (\mathbf{w}_{1,2,L} \in \tilde{S}_{b_{1,2}}^d)$.
3. $\rho_3 \leftarrow \text{Verify}_2(\pi_2, \text{statement}_2)$.

Output: $\rho_1 \wedge \rho_2 \wedge \rho_3$ (1 if valid, 0 otherwise)

B.3 Zero-Knowledge Arguments

Akin our anonymous credentials, our blind signature scheme relies on two zero-knowledge arguments $(\text{Prove}_1, \text{Verify}_1)$ and $(\text{Prove}_2, \text{Verify}_2)$ which we instantiate with the framework of [LNP22]. The first argument proves the well-formedness of the commitment $\mathbf{c}$ and the ciphertext $(\text{ct}_0, \text{ct}_1)$. The user then proves knowledge of $(\mathbf{r}_{1,1}, \mathbf{r}_{1,2}, \mathbf{r}_e, m) \in \mathcal{R}^{2d+m_e+1}$ such that

$$\begin{aligned} \mathbf{r}_{1,1} + \mathbf{A}\mathbf{r}_{1,2} + \mathbf{d}m &= \mathbf{c} \bmod q\mathcal{R}, \\ \mathbf{A}_e^T \mathbf{r}_e &= \text{ct}_0 \bmod p\mathcal{R}, \text{ and } \mathbf{b}_e^T \mathbf{r}_e + \lfloor p/2 \rfloor m = \text{ct}_1 \bmod p\mathcal{R} \\ \|\mathbf{r}_{1,1}\|_2 &\leq 2b_{1,1}\sqrt{nd}, \|\mathbf{r}_{1,2}\|_2 \leq 2b_{1,2}\sqrt{nd}, \|\mathbf{r}_e\|_2 \leq B_{r,e}, \quad m \in T_1 \end{aligned}$$

where $B_{r,e} = 1.15\sqrt{nm_e\eta_e/2}$. The statement statement_1 contains all the public elements of the above relation, those not in red.

The second argument used to produce the final blind signature proves knowledge of the carries $\mathbf{w}_{1,i,H}$ and $t, \mathbf{v}_2, v_3$ but reveals m , as well as $\mathbf{w}_{1,i,L}$. Hence, the user proves knowledge of $(t, \mathbf{w}_{1,1,H}, \mathbf{w}_{1,2,H}, \mathbf{v}_2, v_3) \in \mathcal{R}^{d(2+k_H)+2}$ such that

$$\begin{aligned} b_{1,1}\mathbf{w}_{1,1,H} + b_{1,2}\mathbf{A}\mathbf{v}_{1,2,H} + (q_L t \mathbf{G}_H - \mathbf{B})\mathbf{v}_2 + \mathbf{a}_3 v_3 &= \mathbf{u}' \bmod q\mathcal{R}, \\ \|\mathbf{w}_{1,1,H}\|_2 &\leq B'_{1,1}, \|\mathbf{w}_{1,2,H}\|_2 \leq B'_{1,2}, \|\mathbf{v}_2\|_2 \leq B_2, \|v_3\|_2 \leq B_3 \\ t \in T_1, \quad \|t\|_2 &= \sqrt{w} \end{aligned}$$

where $\mathbf{u}' = \mathbf{u}' + \mathbf{d}m - \mathbf{w}_{1,1,L} - \mathbf{A}\mathbf{w}_{1,2,L} \bmod q\mathcal{R}$, and $B'_{1,i} = B_{1,i}/b_{1,i} + 3\sqrt{nd}$. Indeed, if we define $\mathbf{w}_{1,i} = \mathbf{v}'_{1,i} - \mathbf{r}_{1,i,L}$, we have $\|\mathbf{w}_{1,i,H}\|_2 \leq \|\mathbf{w}_{1,i} - \mathbf{w}_{1,i,L}\|_2/b_{1,i} + \|\mathbf{r}_{1,i,H}\|_2 \leq (B_{1,i} + 2b_{1,i}\sqrt{nd})/b_{1,i} + \sqrt{nd}$. The statement statement_2 contains all the public elements of the above relation, as before.

We instantiate $(\text{Prove}_1, \text{Verify}_1)$ and $(\text{Prove}_2, \text{Verify}_1)$ in the same way we did for our anonymous credentials in Section A.3. In particular, because our sampler requires a smooth modulus q , we use the lifting approach described in the latter. Note that for the issuance relation, the verifiable encryption modulo p can also be lifted modulo q_π using the same strategy. As p will be much smaller than q , it incurs no noticeable overhead on the approximate range proof. Let us now give the security result for the two proof systems.

Lemma B.1 (Security of Prove_1). *Let $\mathcal{R}'$ be the power-of-two cyclotomic ring of degree $n' \leq n$ and let $k' = n/n'$. We denote by τ the coefficient embedding*

in $\mathcal{R}'$. Let $\rho, \tilde{\eta}$ define the challenge space $\mathcal{C}_{\rho, \tilde{\eta}} = \{c \in \tau^{-1}([- \rho, \rho]^{n'}) : c^* = c \wedge \sqrt[64]{\|c^{64}\|_1} \leq \tilde{\eta}\}$. Let M_1, M_2, M_3 be in $(1, \infty)$, and let $\alpha_j = \sqrt{\pi / \ln M_j}$ for $j \in [3]$. Define the witness $\mathbf{s}_1 = (\mathbf{r}_{1,1}, \mathbf{r}_{1,2}, \mathbf{r}_e, m, \mathbf{b})$ of dimension $m_1 = k'(2d + m_e + 1) + 1$ when embedded in $\mathcal{R}'$. Then, define $B = \sqrt{4(b_{1,1}^2 + b_{1,2}^2)nd + B_{r,e}^2 + n + 1}$. We then define $B_{\mathbf{j},\mathbf{c}} = \sqrt{nd}(2b_{1,1}/q + (1 + 2b_{1,2}nd + n)/2)$, and $B_{\mathbf{j},\mathbf{ct}} = \frac{\sqrt{n}}{2}(1 + 1/p + \sqrt{d_e + 1} + B_{r,e}\sqrt{nm_e(d_e + 1)})$. We then define $\sigma_1 = \alpha_1 \tilde{\eta} B$, $\sigma_2 = \alpha_2 \tilde{\eta} \sqrt{n'm_2}$, and $\sigma_3 = \alpha_3 \sqrt{337} \sqrt{B^2 + B_{\mathbf{j},\mathbf{c}}^2 + B_{\mathbf{j},\mathbf{ct}}^2}$. We then define $B_{arp} = \frac{2}{\sqrt{26}} c_{256} \sigma_3 \sqrt{256}$. Then, let $d_2, m_2, \ell, q_{\pi_1}$ be positive integers such that

$$\begin{aligned} q_{\pi_1} & \text{ prime s.t. } q_{\pi_1} = 5 \bmod 8, \\ q_{\pi_1} & > \max(4b_{1,1}^2 nd, 4b_{1,2}^2 nd, B_{r,e}^2) \\ q_{\pi_1} & > B_{arp}^2 - \min(4b_{1,1}^2 nd, 4b_{1,2}^2 nd, B_{r,e}^2) \\ q_{\pi_1} & > \max(41n'm_1 B_{arp}, B_{arp}^2 + B_{arp} \sqrt{n}) \\ q_{\pi_1} & > p(1 + B_{r,e} \sqrt{nm_e}/2 + B_{arp}) \\ q_{\pi_1} & > 2b_{1,1} \sqrt{nd} + q \left(\frac{1 + 2b_{1,2}nd + n}{2} + B_{arp} \right), \end{aligned}$$

We also let γ be an even divisor of $q_{\pi_1} - 1$, and D be the largest integer such that $2^{D-1}pn' < \gamma$. Then, $(\text{Prove}_1, \text{Verify}_1)$ is knowledge sound with an extractor running in expected polynomial time and soundness error

$$\varepsilon_{snd,1} = \frac{2}{|\mathcal{C}_{\rho, \tilde{\eta}}|} + q_{\pi_1}^{-n'/2} + q_{\pi_1}^{-2\ell} + 2^{-128} + \varepsilon_{\text{M-SIS}}^{\text{sound}}$$

and zero-knowledge with loss $\varepsilon_{zk,1} = \varepsilon_{\text{M-LWE}}^{\text{zk}} + \text{negl}(\lambda)$. The term $\varepsilon_{\text{M-SIS}}^{\text{sound}}$ is the hardness bound of $\text{M-SIS}_{n', d_2, m_1 + m_2, q_{\pi_1}, \beta_{\pi_1}}$ where

$$\beta_{\pi_1} = 4\tilde{\eta} \sqrt{4c_{n'm_1}^2 \sigma_1^2 n'm_1 + \left(2c_{n'm_2} \sigma_2 \sqrt{n'm_2} + (2^D \tilde{\eta} + \gamma) \sqrt{n'd_2}\right)^2},$$

while $\varepsilon_{\text{M-LWE}}^{\text{zk}}$ is the hardness bound of $\text{M-LWE}_{n', m_2 - d_2 - \lfloor 256/n' \rfloor - \ell - 1, m_2, q_{\pi_1}, \mathcal{B}_1, \mathcal{B}_1}$ (with $\mathcal{B}_1$ the centered binomial distribution of parameter 1 over $\mathcal{R}'$).

The size of π_1 is computed from the parameters of the proof system introduced in Lemma B.1 by

$$\begin{aligned} |\pi_1| &= n'd_2 (\lceil \log_2 q_{\pi_1} \rceil - D + 2.25) + \left(\frac{256}{n'} + 2\ell + 1 \right) \lceil \log_2 q_{\pi_1} \rceil \\ &+ n' \lceil \log_2 (2\rho + 1) \rceil + n'm_1 \left(\frac{1}{2} + \log_2 \sigma_1 \right) \\ &+ n'(m_2 - d_2) \left(\frac{1}{2} + \log_2 \sigma_2 \right) + 256 \left(\frac{1}{2} + \log_2 \sigma_3 \right). \end{aligned}$$

Lemma B.2 (Security of Prove₂). Let $\mathcal{R}'$ be the power-of-two cyclotomic ring of degree $n' \leq n$ and let $k' = n/n'$. We denote by τ the coefficient embedding

in $\mathcal{R}'$. Let $\rho, \tilde{\eta}$ define the challenge space $\mathcal{C}_{\rho, \tilde{\eta}} = \{c \in \tau^{-1}([-\rho, \rho]^{n'}) : c^* = c \wedge \sqrt[64]{\|c^{64}\|_1} \leq \tilde{\eta}\}$. Let M_1, M_2, M_3 be in $(1, \infty)$, and let $\alpha_j = \sqrt{\pi/\ln M_j}$ for $j \in [3]$. Define the witness $\mathbf{s}_1 = (t, \mathbf{w}_{1,1,H}, \mathbf{w}_{1,2,H}, \mathbf{v}_2, v_3, \mathbf{b})$ of dimension $m_1 = k'(2 + d(2 + k_H)) + 1$ when embedded in $\mathcal{R}'$. Then, define the following bound on the witness as $B = \sqrt{w + B_{1,1}'^2 + B_{1,2}'^2 + B_2^2 + B_3^2 + 1}$ and

$$B_j = \frac{B_{1,1}'}{q} + \sqrt{nd} \left(\frac{1 + \sqrt{nd}B_{1,2}' + \sqrt{ndk_H}B_2 + \sqrt{n}B_3}{2} + wB_2 \frac{1 - 1/q_H}{b_H - 1} \right)$$

We then define $\sigma_1 = \alpha_1 \tilde{\eta} B$, $\sigma_2 = \alpha_2 \tilde{\eta} \sqrt{n'm_2}$, and $\sigma_3 = \alpha_3 \sqrt{337} \sqrt{B^2 + B_j^2}$. We then define $B_{arp} = \frac{2}{\sqrt{26}} c_{256} \sigma_3 \sqrt{256}$. Then, let $d_2, m_2, \ell, q_{\pi_2}$ be positive integers such that

$$\begin{aligned} q_{\pi_2} & \text{ prime s.t. } q_{\pi_2} = 5 \bmod 8, \\ q_{\pi_2} & > \max(B_{1,1}'^2, B_{1,2}'^2, B_2^2, B_3^2, w) \\ q_{\pi_2} & > B_{arp}^2 - \min(B_{1,1}'^2, B_{1,2}'^2, B_2^2, B_3^2, w) \\ q_{\pi_2} & > \max(41n'm_1 B_{arp}, w + \sqrt{nw}) \\ q_{\pi_2} & > B_{1,1}' + q \left(\frac{1 + \sqrt{nd}B_{1,2}' + \sqrt{ndk_H}B_2 + \sqrt{n}B_3}{2} + wB_2 + B_{arp} \right), \end{aligned}$$

We also let γ be an even divisor of $q_{\pi_2} - 1$, and D be the largest integer such that $2^{D-1}pn' < \gamma$. Then, $(\text{Prove}_2, \text{Verify}_2)$ is knowledge sound with an extractor running in expected polynomial time and soundness error

$$\varepsilon_{snd,2} = \frac{2}{|\mathcal{C}_{\rho, \tilde{\eta}}|} + q_{\pi_2}^{-n'/2} + q_{\pi_2}^{-2\ell} + 2^{-128} + \varepsilon_{\text{M-SIS}}^{\text{sound}}$$

and zero-knowledge with loss $\varepsilon_{zk,2} = \varepsilon_{\text{M-LWE}}^{\text{zk}} + \text{negl}(\lambda)$. The term $\varepsilon_{\text{M-SIS}}^{\text{sound}}$ is the hardness bound of M-SIS $_{n', d_2, m_1+m_2, q_{\pi_2}, \beta_{\pi_2}}$ where

$$\beta_{\pi_2} = 4\tilde{\eta} \sqrt{4c_{n'm_1}^2 \sigma_1^2 n'm_1 + \left(2c_{n'm_2} \sigma_2 \sqrt{n'm_2} + (2^D \tilde{\eta} + \gamma) \sqrt{n'd_2} \right)^2},$$

while $\varepsilon_{\text{M-LWE}}^{\text{zk}}$ is the hardness bound of M-LWE $_{n', m_2-d_2-\lfloor 256/n' \rfloor - \ell - 1, m_2, q_{\pi_2}, \mathcal{B}_1, \mathcal{B}_1}$ (with $\mathcal{B}_1$ the centered binomial distribution of parameter 1 over $\mathcal{R}'$).

The size of π_2 is computed from the parameters of the proof system introduced in Lemma B.2 by

$$\begin{aligned} |\pi_2| &= n'd_2 (\lceil \log_2 q_{\pi_2} \rceil - D + 2.25) + \left(\frac{256}{n'} + 2\ell + 1 \right) \lceil \log_2 q_{\pi_2} \rceil \\ &+ n' \lceil \log_2(2\rho + 1) \rceil + n'm_1 \left(\frac{1}{2} + \log_2 \sigma_1 \right) \\ &+ n'(m_2 - d_2) \left(\frac{1}{2} + \log_2 \sigma_2 \right) + 256 \left(\frac{1}{2} + \log_2 \sigma_3 \right). \end{aligned}$$

B.4 Security Analysis

We now give the full security statement and security proof of anonymity and one-more unforgeability of our blind signature scheme.

Theorem B.1. *The blind signature of Section B.2 is anonymous based on the zero-knowledge property of Prove₁ and Prove₂ and M-LWE. More precisely, for any PPT adversary, it holds that $\text{Adv}_{an}[\mathcal{A}] \leq 4(\varepsilon_{zk,1} + \varepsilon_{zk,2} + \varepsilon_{\text{M-LWE}}^{(e)} + \varepsilon_{\text{M-LWE}}^{(2)})$ where $\varepsilon_{zk,1}, \varepsilon_{zk,2}$ are defined in Lemma B.1 and B.2, and where $\varepsilon_{\text{M-LWE}}^{(e)}$ and $\varepsilon_{\text{M-LWE}}^{(2)}$ are the hardness bounds of $\text{M-LWE}_{n,d_e,m_e,p,\mathcal{B}_{\eta_e}}$ and $\text{M-LWE}_{n,d,d,q,U(T_1)}$ respectively.*

Proof. The proof follows in a similar way as that of [JS25a]. It essentially consists in simulating π_1 and π_2 in the two executions of $\text{Sign}_{\mathcal{A},\mathcal{C}}$ from the blindness game, simulating the ciphertexts in both executions, simulating the commitments, and finally simulating the $\mathbf{w}_{1,i,L}$ using the fact that $\mathbf{r}_{1,i,L}$ act as perfect masks. We let G_0 be the original blindness game. We let Adv_i denote the advantage of $\mathcal{A}$ in game G_i .

G_1 . In this game, and for each experiment for $b \in \{0,1\}$, the challenger calls the simulator of Prove₁ and Prove₂ as per Lemma B.1 and B.2. Based on the zero-knowledge property of the latter lemmas, we get

$$|\text{Adv}_1 - \text{Adv}_0| \leq 4(\varepsilon_{zk,1} + \varepsilon_{zk,2}),$$

where $\varepsilon_{zk,1}$ and $\varepsilon_{zk,2}$ are defined in Lemma B.1 and B.2.

G_2 . We now simulate the ciphertext. For each experiment ($b \in \{0,1\}$) and each execution of Sign, the challenger computes ct_0 and ct_1 as follows. It samples $\mathbf{u}_0 \leftarrow U(\mathcal{R}_p^{d_e})$ and $u_1 \leftarrow U(\mathcal{R}_p)$ and defines $\text{ct}_0 = \mathbf{u}_0$ and $\text{ct}_1 = u_1 + \lfloor p/2 \rfloor \cdot m \bmod p\mathcal{R}$ (where m is either m_b or m_{1-b}). Because $\mathbf{r}_e$ is not used in the zero-knowledge proof π_1 , this change can be done unbeknownst to the adversary under M-LWE. Indeed, $([\mathbf{A}_e|\mathbf{b}_e]^T, [\mathbf{A}_e|\mathbf{b}_e]^T \mathbf{r}_e \bmod p\mathcal{R})$ is an instance of the knapsack version of M-LWE, which is equivalent to $\text{M-LWE}_{n,d_e,m_e,p,\mathcal{B}_{\eta_e}}$ as $m_e = 2d_e + 1$. As a result, we have

$$|\text{Adv}_2 - \text{Adv}_1| \leq 4\varepsilon_{\text{M-LWE}}^{(e)},$$

where $\varepsilon_{\text{M-LWE}}^{(e)}$ is the hardness bound of $\text{M-LWE}_{n,d_e,m_e,p,\mathcal{B}_{\eta_e}}$.

G_3 . Because u_1 is a perfect mask for $\lfloor p/2 \rfloor m$, the challenger samples $u_1 \leftarrow U(\mathcal{R}_p)$ and sets $\text{ct}_1 = u_1$ in each execution of Sign in each experiment. We then have

$$\text{Adv}_3 = \text{Adv}_2.$$

G_4 . We now simulate the commitments by using its hiding property. For each execution of Sign of each experiment, the challenger samples $\mathbf{r}_{1,1,L}$ and $\mathbf{r}_{1,2,L}$ as usual but also samples $\mathbf{b} \leftarrow U(\mathcal{R}_q^d)$. It then sets $\mathbf{c} = \mathbf{r}_{1,1,L} + \mathbf{A}\mathbf{r}_{1,2,L} + b_{1,1}\mathbf{b} + \mathbf{d}m \bmod q\mathcal{R}$. We now argue that G_4 is indistinguishable from G_3 under $\text{M-LWE}_{n,d,d,q,U(T_1)}$. Indeed, let us write $b_{1,1}\mathbf{r}_{1,1,H} + b_{1,2}\mathbf{A}\mathbf{r}_{1,2,H} \bmod q\mathcal{R}$ as

$b_{1,1}(\mathbf{r}_{1,1,H} + \mathbf{A}'\mathbf{r}_{1,2,H}) \bmod q\mathcal{R}$ where $\mathbf{A}' = b_{1,2}b_{1,1}^{-1}\mathbf{A} \bmod q\mathcal{R}$ with the inverse taken in $\mathbb{Z}_q$. Because $\mathbf{A}$ is uniform, $\mathbf{A}'$ is also uniform provided that $b_{1,1}, b_{1,2}$ are invertible modulo q , which can be easily enforced in the parameter selection. As a result, under $\text{M-LWE}_{n,d,d,q,U(\widetilde{T}_1)}$, G_4 and G_3 are indistinguishable, where $\widetilde{T}_1 = \tau^{-1}(\{-1, 1\}^n)$. The latter assumption is equivalent to $\text{M-LWE}_{n,d,d,q,U(T_1)}$ using the bijection $x \in \{0, 1\} \mapsto 2x - 1 \in \{-1, 1\}$. In the end, we have

$$|\text{Adv}_4 - \text{Adv}_3| \leq 4\varepsilon_{\text{M-LWE}}^{(2)},$$

where $\varepsilon_{\text{M-LWE}}^{(2)}$ is the hardness bound of $\text{M-LWE}_{n,d,d,q,U(T_1)}$.

G_5 . Because $b_{1,1}$ is chosen invertible modulo q , it holds that $b_{1,1}\mathbf{b}$ follows the uniform distribution. The challenger can therefore generate the commitments of each execution of **Sign** in each experiment as $\mathbf{c} \leftarrow U(\mathcal{R}_q^d)$. We then directly have

$$\text{Adv}_5 = \text{Adv}_4.$$

G_6 . Because $\mathbf{r}_{1,1,L}$ and $\mathbf{r}_{1,2,L}$ are now only used in the definition of $\mathbf{w}_{1,1,L}$ and $\mathbf{w}_{1,2,L}$ respectively, we simulate the latter. For that, we simply use the fact that $\text{Low}(y - U([-b, b - 1]), b) = U([-b, b - 1])$ for any $y \in \mathbb{Z}$ and any $b \in \mathbb{N}^\times$. As a result, the challenger can sample $\mathbf{w}_{1,1,L}$ and $\mathbf{w}_{1,2,L}$ uniformly in $\widetilde{S}_{b_{1,1}}^d$ and $\widetilde{S}_{b_{1,2}}^d$ respectively without changing the distribution of the adversary's view. Hence, we get

$$\text{Adv}_6 = \text{Adv}_5.$$

Combining all the above equations yields the result. $\square$

Theorem B.2. *The blind signature of Section B.2 is one-more unforgeable based on the hardness of M-LWE, M-SIS, the soundness properties of Prove_1 and Prove_2 , and the EUF-CMA of the signature of Section 4.1. More precisely, the advantage of PPT adversary in breaking the one-more unforgeability of the blind signature is upper-bounded by*

$$\begin{aligned} \text{Adv}_{\text{omuf}}[\mathcal{A}] &\leq \varepsilon_{\text{M-LWE}}^{(e)} + \varepsilon_{\text{snd},2} \\ &\quad + 2 \left(h^{\circ dk_H} (C \binom{n}{w} - Q) \varepsilon_{\text{M-SIS}}^{\mathbf{1}} + C^2 \varepsilon_{\text{M-LWE}} \right. \\ &\quad \left. + \varepsilon_{\text{snd},1} + \frac{1+\varepsilon}{1-\varepsilon} \frac{4M_{1,1}M_{1,2}}{(1-p)^4} h^{\circ dk_H} (CQ \varepsilon_{\text{M-SIS}}^{\mathbf{2}}) \right) + \text{negl}(\lambda) \end{aligned}$$

with $\varepsilon_{\text{M-LWE}}^{(e)}$, $\varepsilon_{\text{M-LWE}}$, $\varepsilon_{\text{M-SIS}}^{\mathbf{1}}$, $\varepsilon_{\text{M-SIS}}^{\mathbf{2}}$ the hardness bounds of $\text{M-LWE}_{n,d_e,m_e,p,\mathcal{B}_{\eta_e}}$, $\text{M-LWE}_{n,d,d,q,\mathcal{B}_1}$, $\text{M-SIS}_{n,d,2d+2,q,\beta_\bullet}$ and $\text{M-SIS}_{n,d,2d,q,\beta_\bullet}$ respectively. The constant $C \approx 2$ is the one from Lemma 2.8, and the probability p corresponds to the tailcut probability used to set the Gaussian verification bounds. The function $h^{\circ dk_H}$ corresponds to the dk_H -th composition power of the function h defined by

$$h(x) = \varepsilon_{\text{M-LWE}} + \mu \left(2\varepsilon_{\text{M-LWE}} + \mu (\varepsilon_{\text{M-LWE}} + x)^{\frac{2\lambda-1}{2\lambda}} \right)^{\frac{2\lambda-1}{2\lambda}}.$$

We define μ by $\mu = \left(1 + \frac{\lambda(2\lambda-1)(\delta^2-1)^2}{(2-\delta^2)^{2\lambda+1}}\right)^{Q/2\lambda}$ with δ defined in Theorem 3.1 and Equation (5). Finally, defining $B'_{1,i} = b_{1,i}B'_{1,i} + b_{1,i}\sqrt{nd} = B_{1,i} + 4b_{1,i}\sqrt{nd}$, and $B = \sqrt{B_{1,1}^2 + B_{1,2}^2}$ and $B'' = \sqrt{B_{1,1}'^2 + B_{1,2}'^2}$ for clarity, we have

$$\beta_{\bullet} = \sqrt{\left(B'' + \sqrt{nd}B_2\right)^2 + B_3^2 + n + 1}$$

$$\beta_{\bullet} = \sqrt{\left(B'' + B + 2\sqrt{b_{1,1}^2 + b_{1,2}^2}\sqrt{nd} + \sqrt{nd}\sqrt{4B_2^2 + n}\right)^2 + 4B_3^2}$$

The values $\varepsilon_{\text{snd},i}$ are those from the security of Prove_i in Lemma B.1 and B.2.

Proof. We give the updated security proof following the blueprint of [JS25a] and adapt it to our sampler and specific construction. Overall, the security proof is very similar to that our anonymous credentials in Section A except for the fact that we do not need to tackle impersonation forgeries, but we need to deal with the subtleties of *one-more* unforgeability. We define G_0 to be the original one-more unforgeability game, and later denote by Adv_i the advantage of a PPT adversary $\mathcal{A}$ in game G_i . Hence $\text{Adv}_0 = \text{Adv}_{\text{omuf}}[\mathcal{A}]$. As many of the changes have already been described in the proof of Theorems 4.1 and A.2, we will refer to these proofs to avoid repetitions.

G_1 . We start by reprogramming the encryption-ot-the-sky public key to hide a secret key allowing the challenger to decrypt ciphertexts. Hence, when generating the public parameters, the challenger samples $\mathbf{A}_e$ from $U(\mathcal{R}_p^{m_e \times d_e})$ as usual, but now samples $(\mathbf{s}_e, \mathbf{e}_e) \leftarrow \mathcal{B}_{\eta_e}^{d_e + m_e}$ and sets $\mathbf{b}_e = \mathbf{A}_e \mathbf{s}_e + \mathbf{e}_e \bmod p\mathcal{R}$. Distinguishing the view of the adversary between G_0 and G_1 then naturally comes down to solving M-LWE $_{n,d_e,m_e,p,\mathcal{B}_{\eta_e}}$. We then have

$$\text{Adv}_0 \leq \text{Adv}_1 + \varepsilon_{\text{M-LWE}}^{(e)},$$

where $\varepsilon_{\text{M-LWE}}^{(e)}$ is the hardness bound of M-LWE $_{n,d_e,m_e,p,\mathcal{B}_{\eta_e}}$.

G_2 . In this game, the challenger generates all the tags $\{t_i; i \in [Q]\}$ that will be used in the at most Q queries to $\mathcal{OSign}$. For each such query, the challenger receives a ciphertext $(\text{ct}_{0,i}, \text{ct}_{1,i})$ in the first message. For each one, it computes $u_i = \text{ct}_{1,i} - \mathbf{s}_e^T \text{ct}_{0,i} \bmod p\mathcal{R}$, and defines $m'_i \in T_1$ coefficient-wise: for all $j \in [0, n-1]$, if $\tau_j(u_i)$ is closer to 0 than $\lfloor p/2 \rfloor$, it defines $\tau_j(m'_i) = 0$, and $\tau_j(m'_i) = 1$ otherwise. The challenger then stores (m'_i, t_i) in a table $\mathbb{T}$.

At the end of the game, the challenger gets a one-more forgery, that is $((m_i, \mathbf{w}_{1,1,L,i}, \mathbf{w}_{1,2,L,i}, \pi_{2,i}))_{i \in [N+1]}$ where N is the number of $\mathcal{OSign}$ queries, all the blind signatures verify, and all the m_i are distinct. The challenger then computes $j = \min\{i \in [N+1] : (m_i, \cdot) \notin \mathbb{T}\}$. Because all the m_i 's are distinct, and $\mathbb{T}$ contains at most N entries, such a j always exist. The challenger then extracts $\pi_{2,j}$ and gets $\mathbf{w}_{1,1,H}^*, \mathbf{w}_{1,2,H}^*, \mathbf{v}_2^*, v_3^*, t^*$ such that

$$b_{1,1}\mathbf{w}_{1,1,H}^* + b_{1,2}\mathbf{A}\mathbf{w}_{1,2,H}^* + (q_L t^* \mathbf{G}_H - B)\mathbf{v}_2^* + \mathbf{a}_3 v_3^*$$

$$= \mathbf{u} + \mathbf{d}m_j - \mathbf{w}_{1,1,L,j} - \mathbf{A}\mathbf{w}_{1,2,L,j} \bmod q\mathcal{R}$$

$$t^* \in \mathcal{T}_w, \quad \|\mathbf{w}_{1,1,H}^*\|_2 \leq B'_{1,1}, \quad \|\mathbf{w}_{1,2,H}^*\|_2 \leq B'_{1,2}, \quad \|\mathbf{v}_2^*\|_2 \leq B_2, \quad \|v_3^*\|_2 \leq B_3.$$

By Lemma B.2, it holds that

$$\text{Adv}_1 \leq \text{Adv}_2 + \varepsilon_{\text{snd},2}$$

where $\varepsilon_{\text{snd},2}$ is the soundness error defined in said lemma.

G₃. We now introduce the same branching as in the game G_2 from the proof of Theorem 4.1.

G₄. Here we follow the game $G_{1,3}$ from the proof of Theorem A.2. When we are in the branch $\rho = 2$ and in the i^+ -th query, the challenger extracts the proof π_{1,i^+} provided in the first message of said $\mathcal{OSign}$ query, i.e., the one that will use the tag guess $t^+ = t_{i^+}$. More concretely, the challenger extracts $(\mathbf{r}_{1,1}^+, \mathbf{r}_{1,2}^+, \mathbf{r}_e^+, m^+)$ such that $\mathbf{c} = \mathbf{r}_{1,1}^+ + \mathbf{A}\mathbf{r}_{1,2}^+ + \mathbf{d}m^+ \bmod q\mathcal{R}$, $\text{ct}_{0,i^+} = \mathbf{A}_e^T \mathbf{r}_e^+ \bmod p\mathcal{R}$, $\text{ct}_{1,i^+} = \mathbf{b}_e^T \mathbf{r}_e^+ + \lfloor p/2 \rfloor m^+ \bmod p\mathcal{R}$, and with

$$m^+ \in T_1, \quad \|\mathbf{r}_{1,1}^+\|_2 \leq 2b_{1,1}\sqrt{nd}, \quad \|\mathbf{r}_{1,2}^+\|_2 \leq 2b_{1,2}\sqrt{nd}, \quad \|\mathbf{r}_e^+\|_2 \leq B_{r,e}$$

Recall that m'_{i^+} is obtained by decrypting $(\text{ct}_{0,i^+}, \text{ct}_{1,i^+})$. By the equations verified by $\mathbf{r}_e^+$ and m^+ , it holds that $\text{ct}_{1,i^+} - \mathbf{s}_e^T \text{ct}_{0,i^+} = \mathbf{e}_e^T \mathbf{r}_e^+ + \lfloor p/2 \rfloor m^+ \bmod p\mathcal{R}$. Yet, we have $\|\mathbf{e}_e^T \mathbf{r}_e^+\|_\infty \leq \|\mathbf{e}_e\|_2 \|\mathbf{r}_e^+\|_2 \leq B_{r,e}^2$. As we choose $p > 4B_{r,e}^2$, we get that decoding u_{i^+} indeed gives m^+ . In other words, we have $m'_{i^+} = m^+$.

In the end, we have

$$\text{Adv}_3^{\bullet} = \text{Adv}_4^{\bullet}, \text{ and } \text{Adv}_3^{\circ} \leq \text{Adv}_4^{\circ} + \varepsilon_{\text{snd},1}$$

where $\varepsilon_{\text{snd},1}$ is the soundness error of Prove_1 from Lemma B.1.

G₅ to G₇. We here follow the games G_4 to G_6 of the proof of Theorem 4.1.

G₈ to G₉. We here follow the games G_7 and G_8 from the proof of Theorem 4.1 (or rather $G_{1,7}$ and $G_{1,8}$ from that of Theorem A.2) except that the rejection sampling and the definition of $\mathbf{v}_{1,1}^+, \mathbf{v}_{1,2}^+, \mathbf{v}_2^+, v_3^+$ instead features $\mathbf{s}_i m^+ + \mathbf{r}_{1,i}^+$.

G_{0,0} to G_{dk_H-1,9}. These games follow the corresponding ones from the proof of Theorem 4.1, with the trapdoor switching argument

G₁₀. We here follow the game G_9 from the proof of Theorem 4.1 by verifying the tag guess (i.e., comparing t^+ with the extracted tag t^*).

G₁₁. We now enforce another bound on the extracted forgery following G_{10} from the proof of Theorem 4.1. From the extracted proof $\pi_{2,j}$ in the j -th blind signature, we got $(\mathbf{w}_{1,1,H}^*, \mathbf{w}_{1,2,H}^*, \mathbf{v}_2^*, v_3^*, t^*)$ as described in G_2 above. In branch $\rho = 1$, the challenger aborts if $\|-\mathbf{R}\mathbf{v}_2^*\|_2 > \sqrt{nd}\|\mathbf{v}_2^*\|_2$ where $\mathbf{R} = [\mathbf{R}_1^T | \mathbf{R}_2^T]^T$. By Lemma 2.8, the bound is verified with probability $1/C$ with $C \approx 2$. In branch $\rho = 2$, we define $\Delta \mathbf{v}_{2m} = [(\mathbf{v}_2^* - \mathbf{v}_2^+)^T | m_j - m^+]^T$, and the challenger aborts if $\|[\mathbf{R}|\mathbf{s}]\Delta \mathbf{v}_{2m}\|_2 > \sqrt{nd}\|\Delta \mathbf{v}_{2m}\|_2$. This holds with probability $1/C$ using the same result. We then have

$$\text{Adv}_{10}^{\bullet} = C\text{Adv}_{11}^{\bullet}, \text{ and } \text{Adv}_{10}^{\circ} = C\text{Adv}_{11}^{\circ}$$

Exploiting the One-More Forgery. We finally bound Adv_{11} in a similar way as in the proof of Theorem 4.1, except with small subtleties due to the extraction and recomposition of the low and high parts.

For $i \in \{1, 2\}$, we then define $\mathbf{v}_{1,i}^* = \mathbf{w}_{1,i,L,j} + b_{1,i}\mathbf{w}_{1,i,H}^*$. By the bounds verified by $\mathbf{w}_{1,i,H}^*$ and because $\mathbf{w}_{1,i,L,j}$ lies in $\tilde{S}_{b_{1,i}}^d$, we then get $\|\mathbf{v}_{1,i}^*\|_2 \leq b_{1,i}B'_{1,i} + b_{1,i}\sqrt{nd} = B_{1,i} + 4b_{1,i}\sqrt{nd} = B''_{1,i}$. Notice that by the equations verified by the extracted witness, we have

$$\mathbf{v}_{1,1}^* + \mathbf{A}\mathbf{v}_{1,2}^* + (q_L t^* \mathbf{G}_H - \mathbf{B})\mathbf{v}_2^* + \mathbf{a}_3 v_3^* = \mathbf{u} + \mathbf{d}m_j \bmod q\mathcal{R}$$

We can use the same methodology as in the proof of Theorem 4.1 in each branch, and plugging the new bounds to derive the result. Indeed, in branch $\rho = 1$, we construct the M-SIS solution for $[\mathbf{I}_d | \mathbf{A} | \mathbf{a}_3 | \mathbf{d} | \mathbf{u}]$ as

$$\mathbf{x}^* = \begin{bmatrix} \begin{bmatrix} \mathbf{v}_{1,1}^* \\ \mathbf{v}_{1,2}^* \end{bmatrix} - \begin{bmatrix} \mathbf{R}_1 \\ \mathbf{R}_2 \end{bmatrix} \mathbf{v}_2^* \\ v_3^* \\ -m_j \\ -1 \end{bmatrix},$$

which is non-zero and bounded by $\sqrt{(B'' + \sqrt{nd}B_2)^2 + B_3^2 + n + 1} = \beta_{\bullet}$ where $B'' = \sqrt{B_{1,1}''^2 + B_{1,2}''^2}$. For branch $\rho = 2$, we can construct the M-SIS solution for $[\mathbf{I}_d | \mathbf{A} | \mathbf{a}_3]$ as

$$\mathbf{x}^* = \begin{bmatrix} \begin{bmatrix} \mathbf{v}_{1,1}^* - \mathbf{v}_{1,1}^+ + \mathbf{r}_{1,1}^+ \\ \mathbf{v}_{1,2}^* - \mathbf{v}_{1,2}^+ + \mathbf{r}_{1,2}^+ \end{bmatrix} - \begin{bmatrix} \mathbf{R}_1 \mathbf{s}_1 \\ \mathbf{R}_2 \mathbf{s}_2 \end{bmatrix} \Delta \mathbf{v}_{2m} \\ v_3^* - v_3^+ \end{bmatrix},$$

Because we defined j so that $m_j \neq m'_i$ for all $i \in [N]$, we in particular have $m_j \neq m_{i+} = m^+$. As a result, the unpredictability argument still holds and shows $\mathbf{x}^*$ is non-zero with overwhelming probability. Then, based on the bounds on $\mathbf{v}_{1,i}^*$, the bound verified by $\mathbf{r}_{1,i}^+$ extracted in G_4 , that $\|\mathbf{v}_{1,i}^+\|_2 \leq B_{1,i}$, and the bound enforced in G_{11} , we obtain that $\|\mathbf{x}^*\|_2$ can be bounded by

$$\sqrt{(B'' + B + 2\sqrt{b_{1,1}^2 + b_{1,2}^2}\sqrt{nd} + \sqrt{nd}\sqrt{4B_2^2 + n})^2 + 4B_3^2} = \beta_{\bullet},$$

as desired. We then end up with

$$\begin{aligned} \text{Adv}_0 &\leq \varepsilon_{\text{M-LWE}}^{(e)} + \varepsilon_{\text{snd},2} + 2 \left(h^{\text{odk}_H} (C((\binom{n}{w}) - Q)\varepsilon_{\text{M-SIS}}^{\bullet}) + C^2 \varepsilon_{\text{M-LWE}} \right. \\ &\quad \left. + \varepsilon_{\text{snd},1} + \frac{1+\varepsilon}{1-\varepsilon} \frac{4M_{1,1}M_{1,2}}{(1-p)^4} h^{\text{odk}_H} (CQ\varepsilon_{\text{M-SIS}}^{\bullet}) \right) + \text{negl}(\lambda) \end{aligned}$$

□

B.5 Performance

As opposed to [JS25a] which sets $b_{1,1}, b_{1,2}$ (as well as b_2, b_3 in their case), and then computes the Gaussian parameters for the sampler so that the rejection sampling in the security proof is correctly parameterized, we adopt a different view. For compactness and security, one needs to choose $b_{1,1}, b_{1,2}$ the largest possible so that the original sampling parameters $s_{1,1}, s_3$ already satisfy the rejection sampling condition for the desired repetition rates $M_{1,1}, M_{1,2}$. Doing so homogenizes the bounds $B'_{1,i} = B_{1,i}/b_{1,i} + 3\sqrt{nd}$ with B_2 . As a result, the witness in the zero-knowledge proof π_2 has roughly homogeneous norms which is optimal for the framework of [LNP22]. In our parameter selection described in Algorithm B.1, we therefore fix $M_{1,1}, M_{1,2}$ and choose $b_{1,1}, b_{1,2}$ accordingly.

We observe that the bases taken in [JS25a] were already close to optimal, because they used a gadget with a large base. In our case, we have two well distinct bases $b_{1,1}, b_{1,2}$, and we also observe that masking the low bits of $\mathbf{v}_2$ and v_3 is actually no longer optimal. In their case, they used $b_2 = 8$ which was already very small. In ours, the optimal value for b_2 is strictly smaller than 1, which essentially concludes that one should choose $b_2 = 0$ for optimality.

Then, evaluating the performance of our scheme can be done in a similar way as our anonymous credentials, i.e., by choosing roughly the same parameters as for the standard model signature (although we need to account for the bound overhead $B''_{1,i}$ in the M-SIS hardness), and then choosing the encryption and proof parameters somewhat independently. We then assess the security using Theorems B.1 and B.2, while estimating the hardness bounds of the various M-LWE and M-SIS assumptions using the lattice estimator [APS15].

The improved compactness provided by our sampler allows us to explore different parameter choices which were out of reach from [JS25a, JS25b]. In particular, the reduced truncation error allows for choosing $k_H = 2$ and $d = 4$ as in our anonymous credentials. In previous works, reaching $k_H = 2$ was only achievable using larger gadget bases, which then required to compensate the loss in security by choosing $d = 5$. On the contrary, if we were to force $d = 4$ in the blind signature of [JS25a, JS25b], reaching sufficient security would require working with a smaller gadget base, leading to similar parameters as the anonymous credentials case, i.e., $k_H = 3$. Overall, our sampler allows for reducing the dimension and thus diminish the impact of the zero-knowledge proofs. For similar security, we then reach blind signatures of 26.19 KB, which are 27% smaller, while keeping a practical issuance proof, a security on standard assumptions, and making the scheme round-optimal due to a different parameter selection for the decomposition bases. We give in Tables C.4, C.5, C.6, and C.7 a full parameter set for our blind signature scheme.

C Parameter Sets for Section 4.1, A, and B

Symbol	Description	Value
Signature Parameters		
λ	Security parameter	128
n	Signature ring degree	256
d	Module rank	4
q	Modulus	$405769 \approx 2^{18.63}$
k_L, k_H	Low and High Gadget lengths	2, 2
b_L, b_H	Low and High Gadget bases	49, 13
q_L, q_H	Low and High Moduli	2401, 169
ε	Smoothing loss for samplers	2^{-40}
s_L	Low Gadget sampling width	165.972
s_H	High Gadget sampling width	44.033
s_1	Gaussian width 1 of sampler	4779.581
s_2	Gaussian width 2 of sampler	166.008
s_3	Gaussian width 3 of sampler ($\mathbf{v}_{1,2}$)	4776.697
s_4	Gaussian width 4 of sampler ($\mathbf{v}_2, \mathbf{v}_3$)	76.499
$s_{1,1}$	Final Gaussian width of $\mathbf{v}_{1,1}$	9434.635
w	Hamming weight of tags	5
κ	Number of splitting factors of b_H	2
Q	Maximal number of signature queries	2^{32}
α_1, α_2	Rejection sampling slack (sec. proof)	26.06, 13.19
$M_{1,1}, M_{1,2}$	Rejection sampling repetition rate (sec. proof)	1.005, 1.018
$B_{1,1}$	Verification bound of $\mathbf{v}_{1,1}$	134719.51
$B_{1,2}$	Verification bound of $\mathbf{v}_{1,2}$	68207.66
B_2	Verification bound of $\mathbf{v}_2$	1496.27
B_3	Verification bound of $\mathbf{v}_3$	606.01
Security Estimates		
BKZ [Ⓛ]	Required BKZ blocksize for M-SIS [Ⓛ]	650
BKZ [Ⓢ]	Required BKZ blocksize for M-SIS [Ⓢ]	558
BKZ	Required BKZ blocksize for M-LWE	463
$\varepsilon_{\text{M-SIS}}^{\text{Ⓛ}}$	Hardness bound for M-SIS [Ⓛ]	$2^{-190.11}$
$\varepsilon_{\text{M-SIS}}^{\text{Ⓢ}}$	Hardness bound for M-SIS [Ⓢ]	$2^{-163.20}$
$\varepsilon_{\text{M-LWE}}$	Hardness bound for M-LWE	$2^{-135.41}$
λ_{uf}	Reached unforgeability bit security (Thm. 4.1)	119
Efficiency Estimates		
$ \text{pk} $	Size of public key $\mathbf{B}$ (+seed)	19488 B
$ \text{sk} $	Size of secret key $\mathbf{R}_1, \mathbf{R}_2$	4096 B
$ \text{sig} $	Size of signature $(t, \mathbf{v}_{1,2}, \mathbf{v}_2, v_3)$	3607 B

Table C.1. Example parameter set for the standard model signature. The hardness bounds are estimated in the Core-SVP model, i.e., obtained from the BKZ blocksize B with sieving SVP oracle as $2^{-B \log_2(\sqrt{3/2})} \approx 2^{-0.292B}$ [BDGL16].

Symbol	Description	Value
Proof System Parameters		
λ	Security parameter	128
n'	Proof system ring degree	64
k'	Subring embedding dimension	4
d_2	Module rank	16
q_{π_1}	Proof system modulus	228628724346101
ℓ	Soundness amplification dimension	2
m_1	Witness dimension	105
m_2	Dimension of ABDLOP randomness	56
χ	Distribution of ABDLOP randomness	$\mathcal{B}_1$
ρ	Infinity norm of challenges	8
$\tilde{\eta}$	Manhattan-like norm of challenges	93
$(\alpha_1, \alpha_2, \alpha_3)$	Rejection sampling slacks	$(2.13, 2.13, 2.13)$
(M_1, M_2, M_3)	Rejection sampling repetition rates	$(2, 2, 2)$
σ_1	First rejection sampling width	16230.421
σ_2	Second rejection sampling width	11853.024
σ_3	Third rejection sampling width	3522387.005
γ	Mask commitment compression parameter	544402
D	Witness commitment compression parameter	11
B	Witness bound	81.976
B_j	Lifting term bound	90128.000
Security Estimates		
$\text{BKZ}_{\text{M-SIS}}^{\text{sound}}$	Required BKZ blocksize for M-SIS, Lem. A.1	415
$\text{BKZ}_{\text{M-LWE}}^{\text{zk}}$	Required BKZ blocksize for M-LWE, Lem. A.1	382
$\varepsilon_{\text{M-SIS}}^{\text{sound}}$	Hardness bound for M-SIS, Lem. A.1	$2^{-137.77}$
$\varepsilon_{\text{M-LWE}}^{\text{zk}}$	Hardness bound for M-LWE, Lem. A.1	$2^{-128.12}$
$\varepsilon_{\text{snd},1}$	Soundness error	$2^{-128.79}$
$\varepsilon_{\text{zk},1}$	Zero-Knowledge security bound	$2^{-128.12}$
Efficiency Estimates		
$ \pi_1 $	Proof size	25.28 KB

Table C.2. Suggested parameter set for the anonymous credentials' issuance proof (Prove₁).

Symbol	Description	Value
Proof System Parameters		
λ	Security parameter	128
n'	Proof system ring degree	64
k'	Subring embedding dimension	4
d_2	Module rank	20
q_{π_2}	Proof system modulus	36893488147419103069
ℓ	Soundness amplification dimension	1
m_1	Witness dimension	149
m_2	Dimension of ABDLOP randomness	72
χ	Distribution of ABDLOP randomness	$\mathcal{B}_1$
ρ	Infinity norm of challenges	8
$\tilde{\eta}$	Manhattan-like norm of challenges	93
$(\alpha_1, \alpha_2, \alpha_3)$	Rejection sampling slacks	$(2.13, 2.13, 2.13)$
(M_1, M_2, M_3)	Rejection sampling repetition rates	$(2, 2, 2)$
σ_1	First rejection sampling width	29907260.158
σ_2	Second rejection sampling width	13440.066
σ_3	Third rejection sampling width	1417550244.794
γ	Mask commitment compression parameter	1129442642
D	Witness commitment compression parameter	22
B	Witness bound	151053.737
B_j	Lifting term bound	36270833.765
Security Estimates		
$\text{BKZ}_{\text{M-SIS}}^{\text{sound}}$	Required BKZ blocksize for M-SIS, Lem. A.1	387
$\text{BKZ}_{\text{M-LWE}}^{\text{zk}}$	Required BKZ blocksize for M-LWE, Lem. A.1	402
$\varepsilon_{\text{M-SIS}}^{\text{sound}}$	Hardness bound for M-SIS, Lem. A.1	$2^{-129.59}$
$\varepsilon_{\text{M-LWE}}^{\text{zk}}$	Hardness bound for M-LWE, Lem. A.1	$2^{-133.97}$
$\varepsilon_{\text{snd},2}$	Soundness error	$2^{-127.78}$
$\varepsilon_{\text{zk},2}$	Zero-Knowledge security bound	$2^{-133.97}$
Efficiency Estimates		
$ \pi_2 $	Proof size	46.89 KB

Table C.3. Suggested parameter set for the anonymous credentials' show proof (Show_2).

Symbol	Description	Value
Signature Parameters		
λ	Security parameter	128
n	Signature ring degree	256
d	Module rank	4
q	Modulus	$670761 \approx 2^{19.36}$
k_L, k_H	Low and High Gadget lengths	2, 2
b_L, b_H	Low and High Gadget bases	63, 13
q_L, q_H	Low and High Moduli	3969, 169
$b_{1,1}, b_{1,2}$	Decomposition bases	148, 50
ε	Smoothing loss for samplers	2^{-40}
s_L	Low Gadget sampling width	213.393
s_H	High Gadget sampling width	44.033
s_1	Gaussian width 1 of sampler	4781.463
s_2	Gaussian width 2 of sampler	213.420
s_3	Gaussian width 3 of sampler ($\mathbf{v}_{1,2}$)	4776.697
s_4	Gaussian width 4 of sampler ($\mathbf{v}_2, \mathbf{v}_3$)	76.499
$s_{1,1}$	Final Gaussian width of $\mathbf{v}_{1,1}$	14270.366
w	Hamming weight of tags	5
Q	Maximal number of signature queries	2^{32}
α_1, α_2	Rejection sampling slack (sec. proof)	1.45, 1.34
$M_{1,1}, M_{1,2}$	Rejection sampling repetition rate (sec. proof)	4.45, 5.74
$B_{1,1}$	Verification bound of $\mathbf{v}_{1,1}$	238869.41
$B_{1,2}$	Verification bound of $\mathbf{v}_{1,2}$	79956.39
$B'_{1,1}$	Proof bound of $\mathbf{w}_{1,1,H}$	1709.98
$B'_{1,2}$	Proof bound of $\mathbf{w}_{1,2,H}$	1695.13
B_2	Verification bound of $\mathbf{v}_2$	1681.46
B_3	Verification bound of $\mathbf{v}_3$	802.98
Security Estimates		
BKZ ^①	Required BKZ blocksize for M-SIS ^①	614
BKZ ^②	Required BKZ blocksize for M-SIS ^②	532
BKZ	Required BKZ blocksize for M-LWE (key)	442
BKZ ⁽²⁾	Required BKZ blocksize for M-LWE (cmt)	406
$\varepsilon_{\text{M-SIS}}^{\text{①}}$	Hardness bound for M-SIS ^①	$2^{-195.98}$
$\varepsilon_{\text{M-SIS}}^{\text{②}}$	Hardness bound for M-SIS ^②	$2^{-172.00}$
$\varepsilon_{\text{M-LWE}}$	Hardness bound for M-LWE (key)	$2^{-145.67}$
$\varepsilon_{\text{M-LWE}}^{(2)}$	Hardness bound for M-LWE (cmt)	$2^{-135.14}$
Efficiency Estimates		
$ \text{sig} $	Size of partial signature ($t, \mathbf{v}_{1,2}, \mathbf{v}_2, \mathbf{v}_3$)	3607 B
$ \mathbf{w}_L $	Size of revealed low bits ($\mathbf{w}_{1,1,L}, \mathbf{w}_{1,2,L}$)	2048 B

Table C.4. Example parameter set for the signature blind signature.

Symbol	Description	Value
Signature Parameters		
λ	Security parameter	128
n	Signature ring degree	256
d_e	Module rank	3
p	Modulus	$4993 \approx 2^{12.3}$
m_e	Number of samples	7
η_e	Binomial parameter	1
$B_{r,e}$	Binomial tail bound	34.42
$p/B_{r,e}^2$	Decryption correctness gap	4.21
Security Estimates		
$\text{BKZ}^{(e)}$	Required BKZ blocksize for M-LWE ^(e) (IND-CPA)	501
$\varepsilon_{\text{M-LWE}}^{(e)}$	Hardness bound for M-LWE ^(e)	$2^{-162.93}$
Efficiency Estimates		
$ \text{ct} $	Size of ciphertext	1664 B

Table C.5. Suggested parameter set for the blind signature (encryption to the sky).

Symbol	Description	Value
Proof System Parameters		
λ	Security parameter	128
n'	Proof system ring degree	64
k'	Subring embedding dimension	4
d_2	Module rank	22
q_{π_1}	Proof system modulus	144115188075855677
ℓ	Soundness amplification dimension	2
m_1	Witness dimension	68
m_2	Dimension of ABDLOP randomness	69
χ	Distribution of ABDLOP randomness	$\mathcal{B}_1$
ρ	Infinity norm of challenges	8
$\tilde{\eta}$	Manhattan-like norm of challenges	93
$(\alpha_1, \alpha_2, \alpha_3)$	Rejection sampling slacks	(3.01, 3.01, 3.01)
(M_1, M_2, M_3)	Rejection sampling repetition rates	$(\sqrt{2}, \sqrt{2}, \sqrt{2})$
σ_1	First rejection sampling width	2799456.780
σ_2	Second rejection sampling width	18606.928
σ_3	Third rejection sampling width	90793047.629
γ	Mask commitment compression parameter	596173316
D	Witness commitment compression parameter	21
B	Witness bound	9998.011
$B_{j,ct}$	Ciphertext lifting bound	23339.686
$B_{j,c}$	Commitment lifting bound	1642512.014
Security Estimates		
BKZ_{M-SIS}^{sound}	Required BKZ blocksize for M-SIS, Lem. B.1	389
BKZ_{M-LWE}^{zk}	Required BKZ blocksize for M-LWE, Lem. B.1	394
$\varepsilon_{M-SIS}^{sound}$	Hardness bound for M-SIS, Lem. B.1	$2^{-130.17}$
ε_{M-LWE}^{zk}	Hardness bound for M-LWE, Lem. B.1	$2^{-131.63}$
$\varepsilon_{snd,1}$	Soundness error	$2^{-128.32}$
$\varepsilon_{zk,1}$	Zero-Knowledge security bound	$2^{-131.63}$
Efficiency Estimates		
$ \pi_1 $	Proof size	28.50 KB

Table C.6. Suggested parameter set for the blind signature’s issuance proof (Prove_1).

Symbol	Description	Value
Proof System Parameters		
λ	Security parameter	128
n'	Proof system ring degree	64
k'	Subring embedding dimension	4
d_2	Module rank	20
q_{π_2}	Proof system modulus	549755813821
ℓ	Soundness amplification dimension	2
m_1	Witness dimension	77
m_2	Dimension of ABDLOP randomness	55
χ	Distribution of ABDLOP randomness	$\mathcal{B}_1$
ρ	Infinity norm of challenges	8
$\tilde{\eta}$	Manhattan-like norm of challenges	93
$(\alpha_1, \alpha_2, \alpha_3)$	Rejection sampling slacks	(3.01, 3.01, 3.01)
(M_1, M_2, M_3)	Rejection sampling repetition rates	$(\sqrt{2}, \sqrt{2}, \sqrt{2})$
σ_1	First rejection sampling width	852492.156
σ_2	Second rejection sampling width	16612.366
σ_3	Third rejection sampling width	168276.083
γ	Mask commitment compression parameter	565940
D	Witness commitment compression parameter	11
B	Witness bound	3044.600
B_j	Lifting term bound	2313277.726
Security Estimates		
$\text{BKZ}_{\text{M-SIS}}^{\text{sound}}$	Required BKZ blocksize for M-SIS, Lem. A.1	382
$\text{BKZ}_{\text{M-LWE}}^{\text{zk}}$	Required BKZ blocksize for M-LWE, Lem. A.1	396
$\varepsilon_{\text{M-SIS}}^{\text{sound}}$	Hardness bound for M-SIS, Lem. A.1	$2^{-128.12}$
$\varepsilon_{\text{M-LWE}}^{\text{zk}}$	Hardness bound for M-LWE, Lem. A.1	$2^{-132.22}$
$\varepsilon_{\text{snd},2}$	Soundness error	$2^{-127.42}$
$\varepsilon_{\text{zk},2}$	Zero-Knowledge security bound	$2^{-132.22}$
Efficiency Estimates		
$ \pi_2 $	Proof size	24.19 KB

Table C.7. Suggested parameter set for the blind signature proof (Prove₂).